

Integrum Energy Infrastructure Ltd.

Industry Assessment of the Indian Commercial & Industrial Renewable Energy Solutions Market

Final Report

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1 Brief overview of Global and Indian macroeconomic landscape

1.1 Economic Indicators

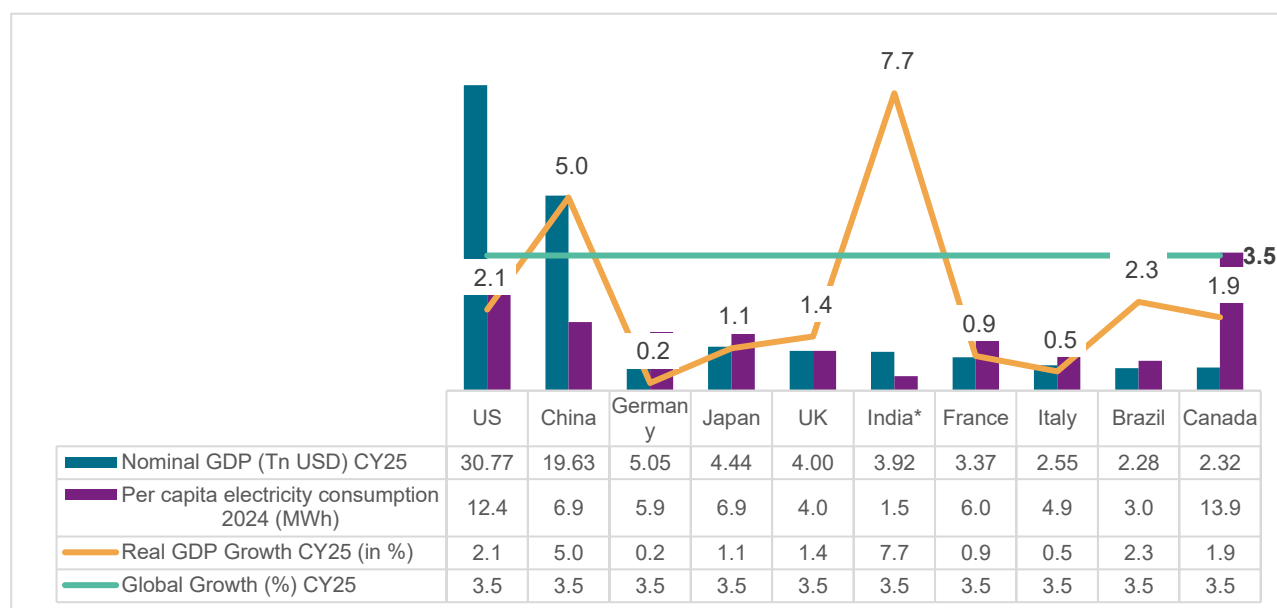
Macro backdrop India's growth remains resilient

India's economic performance over recent years has been marked by remarkable resilience, even as global headwinds such as trade disruptions, policy shifts in advanced economies, and volatile capital flows have challenged growth worldwide. In fiscal year 2026, India's real gross domestic product (GDP) is estimated to grow at 7.70%, outpacing most major economies and maintaining its status as the fastest-growing large economy. This robust growth is underpinned by strong domestic demand, prudent fiscal and monetary policies, and ongoing structural reforms in manufacturing, labour, and trade. Inflation has remained subdued, giving policymakers room to support growth, while foreign direct investment and capital markets have shown buoyancy. Despite some moderation in export growth and investment activity, India's outlook remains stable, demonstrating its capacity to adapt and thrive amid persistent global uncertainties.

On 27th February 2026, India's Ministry of Statistics and Programme Implementation (MoSPI) released the New Series of annual and quarterly national accounts estimates with base year 2022–23, which replaced the previous series with base year of 2011–12. As per June 2026 press release of MoSPI, India's real GDP is estimated to rise to Rs. 323.12 trillion in fiscal 2026, up from Rs. 299.89 trillion in fiscal 2025 and Rs. 280.00 trillion in fiscal 2024. This implies 7.70% growth in fiscal 2026, accelerating from 7.10% in fiscal 2025, a reflection of an economy still fundamentally powered by domestic consumption.

India's economy ranks sixth in the world and is projected to become the fourth-largest economy in fiscal 2027 by nominal GDP as per International Monetary Fund's World Economic Outlook (IMF – WEO) April 2026. Further, IMF's WEO (July 2026) estimates 7.70% real growth in fiscal 2026 which has improved by 0.1% as per WEO (April 2026) outlook and 6.40% in fiscal 2027 which has been reduced by 0.1% from 6.50% as per WEO (April 2026) outlook, which is the highest among the top 10 economies. The IMF also expects India to become the world's third-largest economy by fiscal 2031, overtaking Japan and Germany. India's trajectory is not only strong in absolute terms, it is outperforming at scale, but even as large economies also face slower trend growth.

Over the last decade (excluding the COVID shock), India has consistently delivered one of the strongest growth profiles among major economies. While as per WEO July 2026 the world is expected to grow at around 3.00% in CY2026 and at 3.40% in CY2027 respectively, India is expected to grow at more than double the global rate, even with a moderation to ~6.40% thereafter. India's growth is supported by domestic demand, ongoing formalisation, and rising consumption, while many advanced economies face ageing demographics and slower productivity gains.

Figure 1: Comparison of India's economy with other major nations


Note: India GDP data as of June 2026 as per NSO for Financial Year 2026,

Source: World Economic Outlook Database (April 2026) by IMF; International Energy Agency (IEA), Central Electricity Authority of India (CEA), World bank, Crisil Intelligence

External environment global growth steady, risks still elevated

In April 2026 WEO report, the IMF expects global growth to remain at 3.1% in 2026 and 3.2% in CY2027 however as per the WEO Report July 2026, global growth reduced by 0.1% to 3.0% for 2026 and increased by 0.2% to 3.4% for CY2027. The outlook still embeds headwinds from the war in West Asia, uncertainty and rising protectionism, even as the assessed tariff shock appears smaller than first announced. For India, this matters because a stable global baseline reduces downside risk, but persistent geopolitical volatility and protectionist policy shifts remain key watchpoints.

Table 1: Real GDP annual growth forecast of major economies (figures in %)

Country	CY25(A)	CY26(E)#	CY27(E)#	CY28(E)	CY29(E)	CY30(E)	CY31(E)
Brazil	2.3	2.4	2.2	2.4	2.5	2.5	2.5
Canada	1.7	1.1	1.7	1.7	1.7	1.7	1.8
China	5.0	4.6	4.1	4.0	3.7	3.3	3.3
France	0.9	0.6	0.9	1.2	1.2	1.1	1.1
Germany	0.2	0.7	1.0	1.2	0.9	0.7	0.6
India*	7.7	6.4	6.7	6.5	6.5	6.5	6.5
Italy	0.5	0.5	0.5	0.8	0.7	0.7	0.7
Japan	1.2	0.6	0.7	0.6	0.6	0.6	0.6
United Kingdom	1.3	1.0	1.3	1.6	1.6	1.5	1.4
United States	2.1	2.3	2.2	2.1	1.9	1.8	1.8
World	3.4	3.0	3.4	3.2	3.2	3.1	3.1

*For India financial Year (April-March) CY 25 is Fiscal 2026; # As per WEO July 2026

Source: World Economic Outlook Database (April and July 2026) by IMF; Crisil Intelligence

1.1.1 Key Projections

Macro conditions are supportive in fiscal 2026, but the inflation cycle is expected to turn in fiscal 2027 - keeping policy and markets focused on the next leg of normalisation

Crisil Intelligence expects real GDP growth to slow at 6.6% next fiscal 2027, around the estimated potential growth rate of 7% as per the Economic Survey for 2025-26. Growth will be supported by robust private consumption and a mild pickup in private investment. The positive effects on private consumption from fiscal support (via income tax cuts, GST cuts, unconditional cash transfers by states), lower interest rates and benign inflation will spill over into the next fiscal.

Crisil Intelligence expects the investment momentum to continue. Private investment sentiment is improving, with a recovery in private capex underway, especially in sectors benefiting from policy incentives and recent trade agreements. The central government's capex thrust will continue, with budgetary or effective capex budget to grow 22.1% in next fiscal 2027 compared with 6% in the revised estimates for this fiscal 2026.

As no durable agreement have been formalised between US and Iran, the West Asia conflict has resumed, and this has increased the concerns and sharply reduced shipping through the Strait of Hormuz which has pushed the oil prices. This forecast assumes a normal southwest monsoon and benign crude oil prices. Reports of possible developing of El Niño conditions forming beyond August 2026, according to the National Oceanic and Atmospheric Administration, remain a monitorable for the rural economy.

Crisil Intelligence forecasts current account deficit (CAD) at 1.5% of GDP in fiscal 2027. Crisil Intelligence expects retail inflation to rise to around 5.1% in fiscal 2027 from 2.0% in fiscal 2026 due to the crude prices easing from recent peak. With the West Asia conflict prolonged beyond April 2026 and energy prices remained elevated, the INR breached the 96 per dollar mark against the USD in May 2026, hitting record lows due to intense volatility from geopolitical tensions and rising energy prices. The benign current account deficit (CAD; 1.50% as per Crisil forecast) and India's robust macroeconomic fundamentals will also provide support. The rupee will likely weaken mildly to ~INR 93.5 per dollar by March 2027, with expectations of a manageable CAD next fiscal keeping pressure on the rupee in check.

Table 2: Crisil's key projections

Particulars	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27P
GDP growth (%)	6.50%	3.90%	-5.80%	9.70%	7.60%	7.20%	7.10%	7.70%	6.60%
CPI (% , average)	3.40%	4.80%	6.20%	5.50%	6.70%	5.40%	4.60%	2.00%	5.10%
CAD/GDP (%)	-2.10%	0.90%	-0.90%	-1.20%	-2.00%	-0.70%	-0.60%	-0.60%	-1.50%
FAD/GDP (%)	3.40%	4.60%	9.20%	6.70%	6.40%	5.60%	4.80%	4.40%	4.30%
Exchange rate (INR /\$ March-end)	69.50	74.40	72.80	76.20	82.30	83.00	86.60	92.80	~93.50
10-year G-sec yield (% , March-end)	7.50%	6.20%	6.20%	6.80%	7.40%	7.10%	6.70%	6.70%	7.00%

E: Estimated based on latest available information for the fiscal; P: Projected; CPI: Consumer Price Index-linked; CAD: Current account deficit; G-sec: Government security; FAD: Fiscal account deficit

1.1.2 Per capita electricity consumption

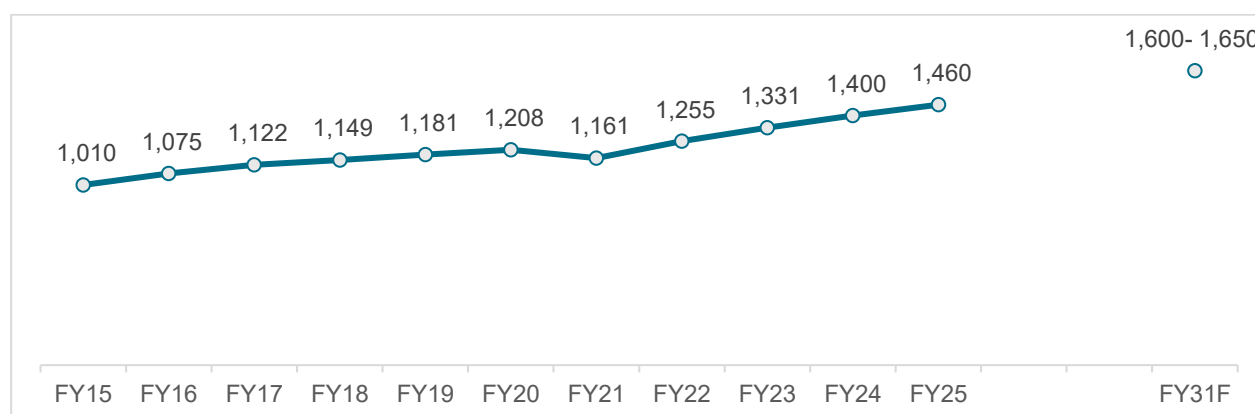
India's per capita electricity consumption: Steady growth with new drivers emerging

India's per capita electricity consumption has demonstrated a consistent upward trend over the past decade, increasing from 1,010 kWh in fiscal 2015 to a provisional 1,460 kWh in fiscal 2025: a CAGR of 3.7%. This growth has been fueled by robust economic expansion, higher domestic consumption, and

concerted efforts to extend electrification to rural and household sectors. The only notable interruption occurred during fiscal 2021, when the COVID-19 pandemic led to a temporary dip in demand, particularly from industrial and commercial users.

Looking forward, per capita electricity consumption is projected to continue rising at an annual rate of 2–3% through fiscal 2031. Key drivers include improved electricity access, rising household incomes, the growing adoption of electric vehicles, railway electrification, ongoing rural electrification, and increased use of consumer durables. As a result, Crisil Intelligence anticipates that per capita electricity consumption will reach approximately 1,600–1,650 kWh by fiscal 2031, reflecting India's ongoing economic development and the broadening of electricity use across sectors.

Figure 2: Per capita electricity consumption (in kWh)



F: Forecast

Source: Central Electricity Authority of India (CEA), Crisil Intelligence

India's per capita electricity consumption (kWh) is one of the lowest in the world. This persistent gap is a strong signal that electricity demand can compound well beyond the forecast period, supported by ongoing economic expansion and a structurally more electricity-intensive energy system as electrification deepens across transport (EVs), rail, and increasing industrial processes all occurring in a world where overall electricity demand is already on an upward trajectory.

1.2 Geopolitical uncertainties and impact on energy

Geopolitical developments and uncertainties have emerged as a significant factor influencing global energy markets, affecting fuel prices, supply chain stability, and investment decisions across the energy sector. Ongoing conflicts and regional tensions have increased volatility in crude oil and natural gas markets, particularly for import-dependent economies such as India. These uncertainties highlight the importance of strengthening energy security through diversification of energy sources and accelerating the transition towards domestically generated RE. The RE sector is expected to play an increasingly important role in enhancing India's long-term energy resilience and reducing exposure to external supply shocks.

- 1. Russia–Ukraine Conflict** - The ongoing Russia–Ukraine conflict has significantly affected the global energy trade flows since 2022, with India emerging as one of the largest importers of discounted Russian crude oil. While this has helped Indian refiners optimise procurement costs, the conflict continues to expose the country to global oil price volatility and supply chain uncertainties. India's dependence on imports for majority of its crude oil requirements and natural gas consumption, sustained increases in international crude prices affected impact inflation, the current account balance, and overall energy security. Consequently, the prevailing geopolitical environment has strengthened

the strategic rationale for accelerating domestic RE deployment, thereby reducing dependence on imported fossil fuels and enhancing long-term energy resilience.

2. **US–Iran Conflict and the Strait of Hormuz** - The escalation of geopolitical tensions involving the United States, Israel, and Iran during 2026 has heightened concerns regarding the security of energy supplies transiting through the Strait of Hormuz, one of the world's most critical oil and LNG shipping routes. Disruptions to commercial shipping and supply flows through the strait have resulted in heightened volatility in global crude oil and LNG markets. As India imports a substantial share of its crude oil and LNG from the Gulf region, prolonged disruptions in this corridor have affected the energy availability, import costs, and overall energy security. These developments further reinforce the importance of diversifying India's energy mix through greater deployment of domestic RE capacity, energy storage systems, and emerging technologies such as green hydrogen to reduce exposure to external supply disruptions.

1.3 Aatmanirbhar Bharat Abhiyan

The five focus points of the *Aatmanirbhar Bharat Abhiyan* are economy, infrastructure, system, vibrant demography, and demand. Its five phases are:

- Phase I: Businesses including MSMEs; Phase II: Poor, including migrants and farmers; Phase III: Agriculture; Phase IV: New horizons of growth and Phase V: Government reforms and enablers

Table 3: Sector-wise focus of *Aatmanirbhar Bharat Vision*

Sector	Government spends	Key schemes
Renewable energy	~Rs. 1,300 billion	- Rs 45 billion Production Linked Incentive Scheme 'National Programme on High Efficiency Solar PV Modules'. This was further increased by Rs 195 billion in the budget for Fiscal 2023, taking it to Rs 240 billion; in Tranche I 8.7 GW and in Tranche II 39.6 GW capacity were allocated for domestic solar module manufacturing capacity under PLI. - PM Surya Ghar Muft Bijli Yojna: This scheme has a proposed outlay of Rs. 750 billion and aims to light up 10 million households (rooftop solar) by providing up to 300 units of free electricity every month till FY 2027 - Public procurement (Preference for 'Make in India') to provide for purchase preference (linked with local content) in respect of renewable energy (RE) sector - Implementation of Pradhan Mantri Kisan Urja Suraksha Utthan Mahabhiyan (PM KUSUM) scheme; MNRE, in November 2020, scaled up and expanded the PM KUSUM scheme to add 30.8 GW by 2022 with central financial support of Rs 344 billion. The scheme has been extended till March 31, 2027 - Approved Models & Manufacturers of Solar Photovoltaic Modules (Requirement for Compulsory Registration) Order, 2019 (ALMM List-I) - Approved Models and Manufacturers of Solar PV Cells to be effective from 1st June 2026 (ALMM List-II)

Sector	Government spends	Key schemes
		- Scheme of grid connected wind-solar hybrid power projects - Basic customs duty (BCD) of 20% on solar cells and 20% on solar modules, effective May 1, 2025. However, the Agricultural and Infrastructure Development Cess (AIDC) of 20% on modules and 7.5% on cells keep the effective BCD rate of solar modules at 40% and for cells 27.50%. Further, antidumping duty of upto 30% on imported solar cells from China have been proposed by DGTR for three years
Power distribution companies (discoms)	~Rs.970 billion	- Rs 1.35 trillion liquidity infusion for discoms via Power Finance Corporation/ Rural Electrification Corporation (PFC/ REC) against receivables - Rebate for payment to be received by generation companies (gencos) to be passed on to industrial customers - Revamped distribution sector scheme (RDSS) to help discoms improve their operational efficiencies and financial sustainability by providing result-linked financial assistance; outlay of Rs 3,037.58 billion over 5 years i.e., Fiscals 2022 to 2026. The outlay includes an estimated Government Budgetary Support (GBS) of Rs 976.31 billion.
New Energy	Rs. ~388 billion	- Rs 181 billion under PLI scheme for Advanced Chemistry Cell (ACC) Battery Storage in India launched in October to achieve 50 GWh manufacturing capacity - National Green Hydrogen Mission with Rs 197.44 Bn incentive including other targets such as 5 MMTPA target annually and associated RE capacity - SIGHT Programme-174.90 Bn incentive (split between Electrolyser and Green hydrogen) - Pilot projects - R&D - Others - BCD exemption on critical minerals (cobalt, lead, zinc, etc.), scrap of lithium-ion batteries proposed in Budget 2025-26
Nuclear energy	~Rs. 200 billion	- Nuclear energy mission announced in Budget 2025-26 100 GW of nuclear power capacity by 2047 - Budgetary allocation to support R&D for indigenous development of small modular reactors (SMR) - Private sector participation in the development of Bharat small reactors, R&D of SMR and newer technologies

Source: Official portal of the Government of India; various ministries, PIB press releases, Crisil Intelligence

India's RE initiatives are aligned with its climate commitments and is progressing towards net-zero emissions by 2070. The Government has introduced several measures to accelerate RE deployment, including permitting up to 100% Foreign Direct Investment (FDI) under the automatic route in the RE sector, providing Viability Gap Funding (VGF) support for Battery Energy Storage Systems (BESS), and implementing schemes to strengthen domestic manufacturing and grid integration. These policy interventions are expected to attract investments, enhance energy security, and support the country's transition towards a low carbon economy.

India's sustained economic growth, rising industrial activity and relatively low per-capita electricity consumption are expected to support structural growth in electricity demand, creating the need for significant investment in generation, transmission and renewable-energy infrastructure. As businesses scale their operations, power consumption particularly within the C&I segment will increase, creating greater demand for reliable, cost-effective, and sustainable energy solutions.

2 Indian Power Sector

2.1 Evolution of power sector and its structure in India

Governance framework - a shared mandate between the Centre and States

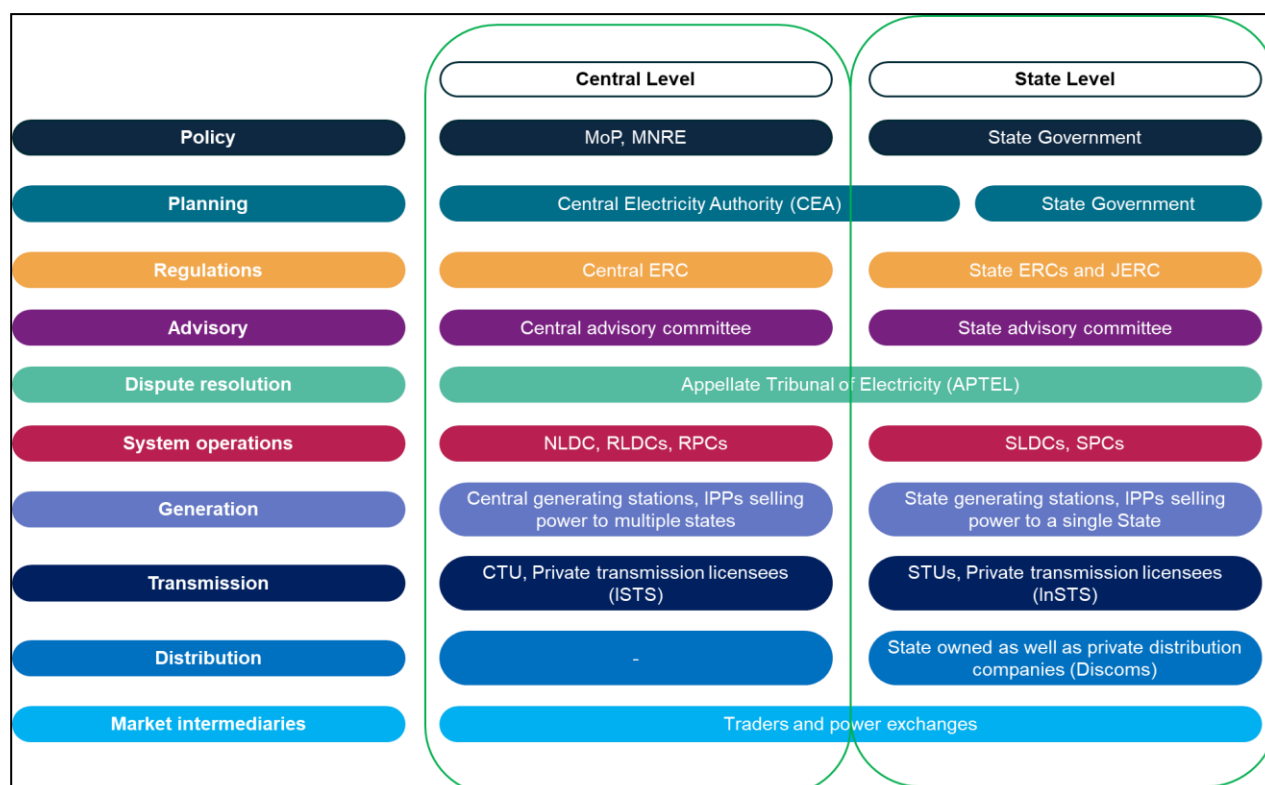
Electricity is a concurrent subject in India with the Ministry of Power (MoP) mainly being responsible for creating the overall policy framework for the power sector at national level while all state-level policies and issues come under the purview of the respective state governments.

To regulate electricity markets and strengthen sector discipline, India has established the Central Electricity Regulatory Commission (CERC). At the state level, State Electricity Regulatory Commissions (SERCs) have been set up by respective state governments. Together, these institutions are intended to regulate tariffs and market behaviour, encourage competition, and facilitate private investment, while ensuring that sector development aligns with policy priorities.

Roles across the value chain - states drive implementation; the centre enables scale and consistency

State governments have a major role in the creation of generation capacity, and in overseeing state-level transmission and distribution. The Central Government develop the national electricity policy, the national electricity plan, and other rules, guidelines, and schemes, typically in consultation with state governments and the CEA.

Figure 3: Institutional and structural framework



Note: MoP: Ministry of Power, MNRE: Ministry of New and Renewable Energy; ERC: Electricity Regulatory Commission, JERC: Joint Electricity Regulatory Commission, NLDC: National Load Despatch Centre, RLDC- Regional Load Despatch Centre, RPC: Regional Power Committee, SLDC- State Load Despatch Centre, SPC: State Power Committee, IPP: Independent Power

Producer, CTU- Central Transmission Utility; ISTS: Inter State Transmission System, STU- State Transmission Utility, InSTS: Intra State Transmission System

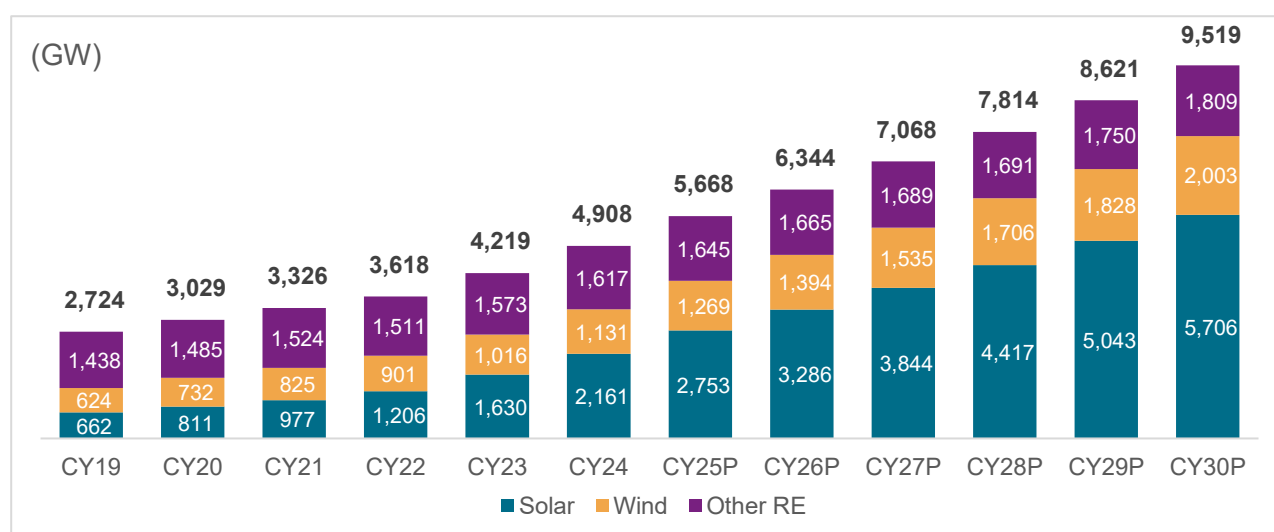
Source: Crisil Intelligence

The collaborative approach in India's power sector is designed to accelerate development while ensuring that electricity supply reaches all areas, consumer and stakeholder interests are protected, and planning reflects critical factors such as resource availability, technology options, generation economics, and energy security considerations. This Centre–State design is central to understanding how India scales capacity and reforms the sector, providing the institutional backbone for the transition to a larger, more reliable, and increasingly cleaner power system.

2.2 Overview and outlook for global power sector

India is the third largest producer and consumer of electricity along with installed capacity of renewable energy and it also ranks third in terms of hydrogen consumption. As per International Energy Agency (IEA), global RE capacity was 4,907.9 GW in CY 2024. Global installed solar capacity in utility expanded to 2,161 GW in 2024 from 662 GW in 2019, clocking a CAGR of 26.7%. Wind capacity increased to 1,131 GW in 2024 from 624 GW in 2019, at a CAGR of 12.6%. Solar accounts for ~44% and wind accounts for ~23% of the total RE capacity installed globally in 2024. Robust growth has been led by fall in costs, favourable government policies and strong thrust on avoiding carbon emissions. Total RE capacity is expected to grow at a CAGR of 11.7% to 9,519 GW between 2024 and 2030.

Figure 4: Source-wise installed RE capacity globally



Solar includes solar photovoltaic (PV)- Distributed, Utility, and concentrated solar. Wind includes offshore and onshore

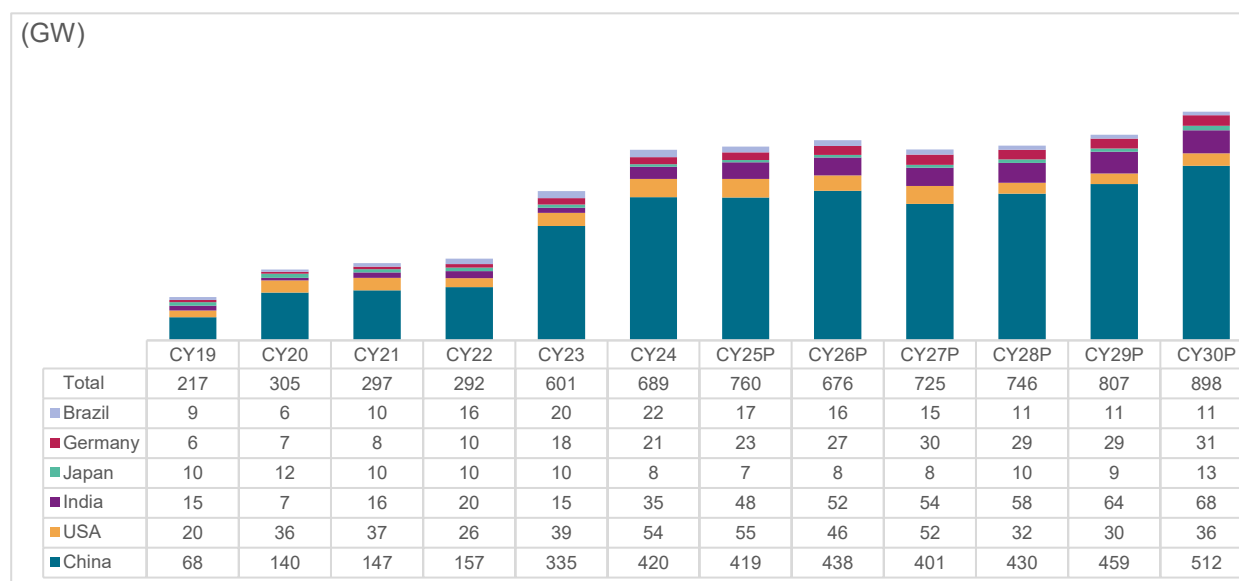
Source: IEA, Crisil Intelligence

Continuing the trend reported in recent years, 2024 saw the largest increase in RE capacity to date with 689 GW addition. Solar and Wind accounted for ~94% of global renewable power additions, as they are the most affordable options to add new capacity in almost every country in the world, and policies in more than 130 countries continue to support them.

Solar power alone accounted for over three-fourths of RE additions, with a record 531 GW added during the year, while 115 GW of wind energy was added. China and the US collectively account for 473 GW (or 81.7%) of RE capacity installed in 2024, while India ranks third in annual RE installation with 35 GW (6.0%) added in 2024. The global RE capacity addition is expected to reach 898 GW by 2030. Renewable energy

capacity in Asia Pacific (excluding China) is set to almost double over 2025-2030, expanding by 670 GW, the second-highest regional increase after China.

Figure 5: RE capacity additions in key countries



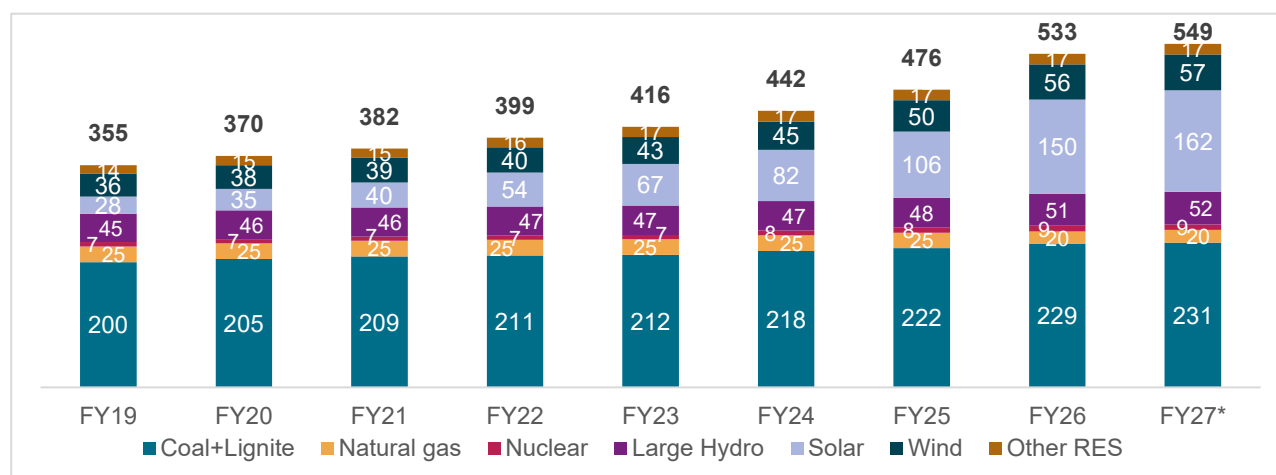
Source: IEA, Crisil Intelligence

2.3 Total generation capacity in India

Total generation capacity in India - scale-up continues, while the mix steadily transitions

India's power system has continued to expand in line with rising electrification and economic activity. India is also the 3rd largest producer and consumer of power. Total installed generation capacity stood at ~549 GW as of June 2026, with ~194 GW added during fiscal 2019 to fiscal 2027 (as of June 2026), translating into an overall capacity CAGR of 6.2% over the period. Even as the system grows, the fuel mix highlights a transition underway rather than complete. Coal and Lignite remains the single largest block of capacity at ~42.2%, reflecting their ongoing role in meeting baseload demand and supporting grid stability. At the same time, renewable energy (including large hydro) has expanded to ~289 GW as of June 2026, up from ~123 GW as of March 2019, taking renewables to 52.6% of total installed capacity. This shift has been led by solar, with installed solar capacity rising to ~162 GW from ~28 GW over the same period.

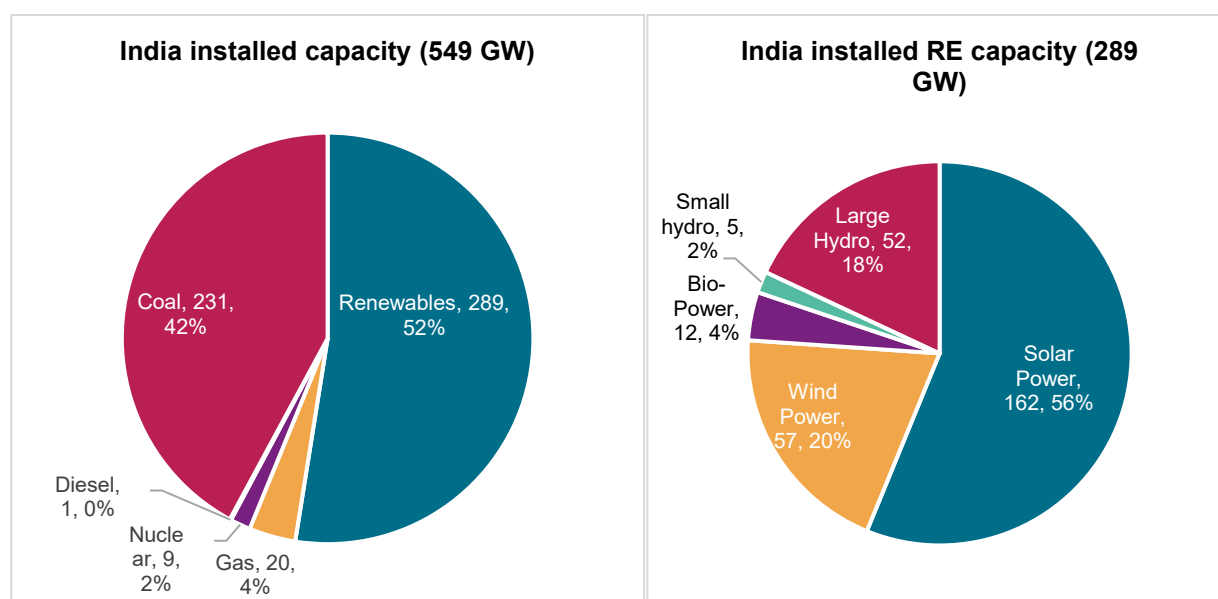
Figure 6: Historical Fuel-wise installed capacity (GW)



Thermal power includes coal, lignite, and gas; Other Renewable energy sources (RES) include biogas, bagasse, small hydro, and waste-to-energy

* As of June 2026 (fiscal 2027); Source: CEA, Crisil Intelligence

Figure 7: Details of source wise installed capacity as of June 2026 (GW, % share)



Source: CEA, Crisil Intelligence

2.4 India – Demand growth

Consumption growth - a stronger post-pandemic trajectory

India's electricity requirement has expanded steadily, rising at a CAGR of 4.3% between fiscals 2019 and 2026. Notably, the pace accelerated in the post-pandemic period, with requirement growth increasing to 6.0% between fiscals 2021 and 2026, reflecting the rebound in economic activity and structurally higher consumption.

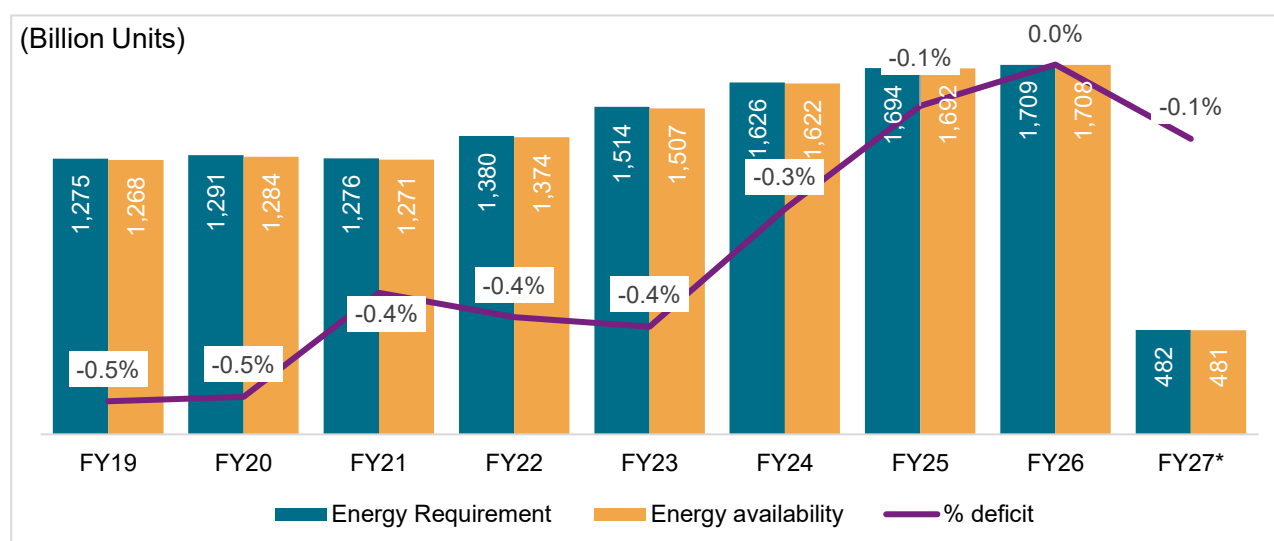
Supply response - availability keeps pace, pulling deficits to near-zero

Electricity availability has broadly matched the rise in requirement, growing at a comparable CAGR of 4.3% between fiscals 2019 and 2026, supported by strong capacity additions in both generation and transmission. As a result, India's energy deficit narrowed to 0.3% in fiscal 2024 and further to 0.1% in fiscal 2025, improving from 0.5% in fiscal 2020. In fiscal 2026 the energy deficit stands at 0.03%. This improvement coincided with capacity addition growth of 6.0% between fiscal 2019 and fiscal 2026, indicating that system expansion has largely kept shortages contained. However, the energy deficit slightly increased to 0.14% in fiscal 2027 (as of June 2026).

Transmission strengthening - easing congestion and improving regional balancing

A key factor behind the rapid reduction in deficit levels has been the strengthening of inter-regional transmission capacity over the past five years. Improved transfer capability reduced supply constraints caused by congestion and limited corridor availability, supporting better balancing of demand and supply across regions, and contributing to the decline in the deficit.

Figure 8: Aggregate power demand supply



*As on June 2026 (fiscal 2027); Source: CEA, Crisil Intelligence

State-level picture - broad adequacy, with pockets of marginal deficit

Electricity requirement and supply across key states Tamil Nadu, Maharashtra, Rajasthan, Telangana, Gujarat, Uttar Pradesh, and Karnataka remain critical to assessing India's power-sector dynamics. Collectively, these states account for over 55% of India's total electricity requirement and include several of the country's largest manufacturing, commercial, and services hubs.

These states are also central to India's energy transition, given their substantial renewable-energy potential and capacity additions particularly solar in Rajasthan and Gujarat, and wind and solar in Tamil Nadu, Karnataka and Maharashtra. Their large and growing renewable-energy penetration increases the importance of transmission availability, grid balancing, energy storage, and flexible conventional generation in maintaining supply reliability. At the all-India level, the deficit remained low at ~0.03% in fiscal 2026 which slightly increased to ~0.14% in fiscal 2027 (as of June 2026).

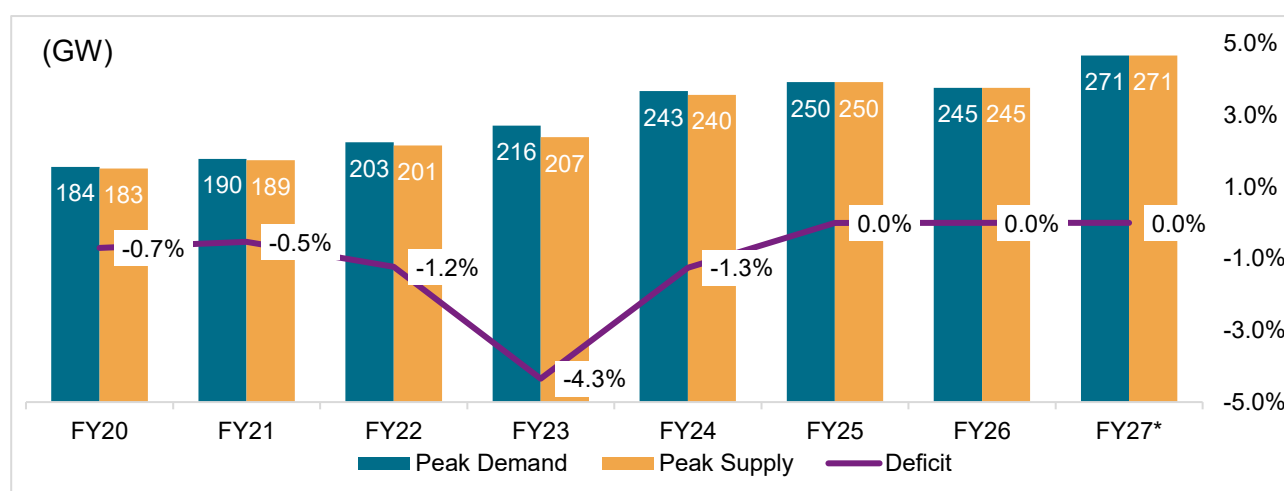
A structural driver behind the higher base demand is the expansion of access: India has achieved ~99.9% household electrification, as per the Saubhagya dashboard. While supply has largely kept pace, marginal deficits persist in some states, reflecting peak-period pressures and regional supply constraints rather than

a broad-based energy shortage. These marginal deficits become most visible not in annual energy numbers, but during peak load periods where the system is increasingly being tested.

Peak demand - the binding constraint is shifting to extremes and weather volatility

India's peak electricity demand rose from 184 GW in fiscal 2020 to 250 GW in fiscal 2025, implying an average growth rate of 6.3%, however it moderated to 245 GW in fiscal 2026. The peak demand surged to ~271 GW in May 2026. The constant rise in peak demand can be attributed to economic growth, seasonal vagaries, and an increasing daily average temperature that India has experienced over the last decade. In fiscal 2026, power demand depreciated by 2% year-on-year due to seasonal vagaries despite a 7.7% year-on-year growth in GDP.

Figure 9: Peak power demand and supply



*As on June 2026 (fiscal 2027); Source: CEA, Crisil Intelligence

While India has largely achieved annual energy adequacy, as reflected in the low all-India energy deficit, this masks emerging challenges in meeting sharp, increasingly weather-linked peak demand at the regional and state levels. The growing concentration of demand in key industrial and urban centres, alongside the rising share of variable renewable energy, heightens the need for timely capacity additions, stronger inter-state transmission corridors, adequate storage, and flexible generation resources. Consequently, power-system planning is increasingly shifting from merely ensuring sufficient annual energy availability to maintaining reliable supply during peak periods through improved regional balancing, grid readiness, and system flexibility.

2.5 Long term drivers and constraints for growth demand

Demand drivers - growth, industrialisation, and electrification deepen electricity intensity

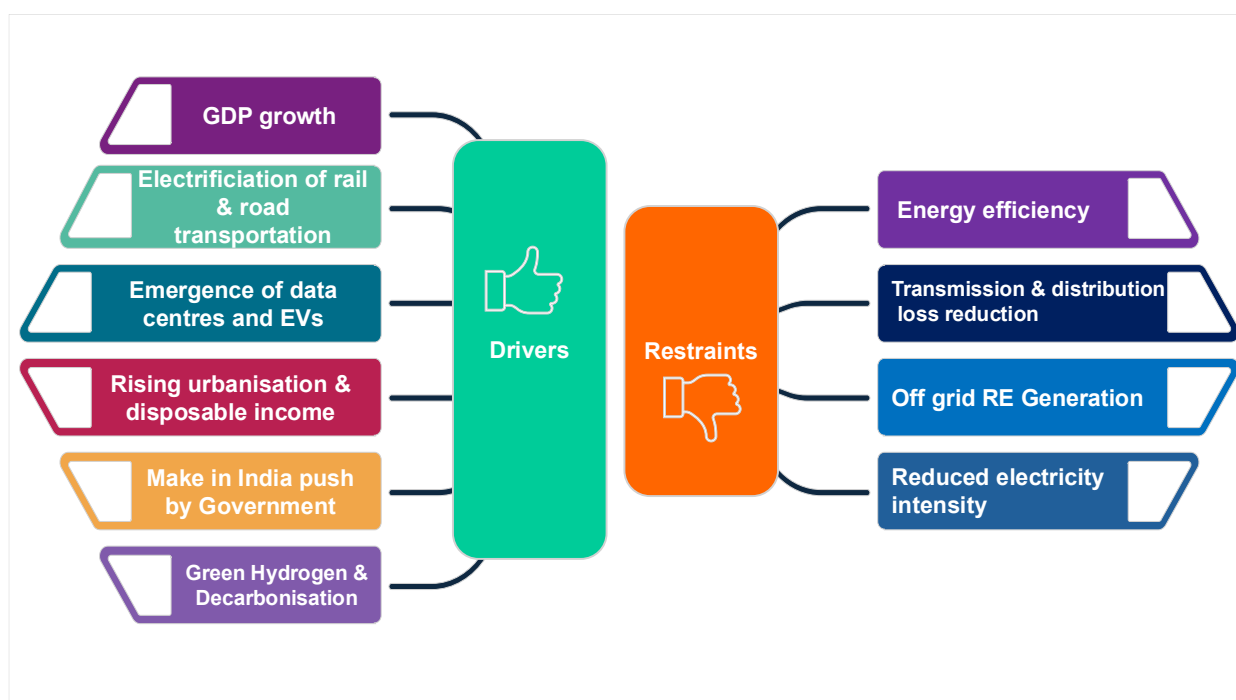
Power demand in India remains closely linked with macroeconomic performance, as higher output, incomes, and urbanisation translate into greater electricity consumption. India has been among the fastest-growing major economies, recording average GDP growth of 5.7% and a 5.9% increase in per capita income over the past decade. This macro momentum is expected to keep electricity demand on an upward trajectory.

A set of economy-wide and sector-specific developments further reinforce this trend: the broader investment impulse and reforms (including the Aatmanirbhar Bharat relief package), public capex through

the National Infrastructure Pipeline, logistics improvements from dedicated freight corridors, expansion of the services industry, rapid urbanisation, and stronger rural demand supported by farm income improvements and agriculture-related reforms.

Policy initiatives have complemented these fundamentals by expanding access and enabling industrial growth. Schemes such as 24x7 power for all and SAUBHAGYA have supported universal household electrification, while enabling infrastructure initiatives such as the Green Energy Corridor and the Green City scheme support higher integration of renewable power and cleaner urban growth. In parallel, manufacturing-focused measures including the PLI scheme and concessional corporate tax rates for “Made in India” manufacturing (15% for eligible new manufacturing companies) have helped create the conditions for large-scale industrial capacity addition, which is inherently power intensive.

Figure 10: Factors influencing power demand



Source: Crisil Intelligence

Next-wave use cases - transport, digital infrastructure, and new industrial loads

In addition to core economic drivers, incremental demand is expected from targeted railway electrification where the government’s goal of achieving 100% railway electrification by fiscal 2026 largely completed, with ~99.6% of electrification works completed as of March 2026. Similarly, over 1,018 km of under-construction metro rail projects are expected to add ~120–125 billion units (BU) of electricity demand by fiscal 2030, new economic loads will also contribute. Increased adoption of electric vehicles will require expanded charging infrastructure, adding ~40–50 BU of demand. Growth in digital infrastructure is another structural driver, with data centers expected to require ~2–3 GW of power by fiscal 2027. Alongside these, electrification of industrial processes and higher demand from key infrastructure and manufacturing sectors are likely to steadily raise the electricity intensity of GDP.

Constraints and offsets - efficiency gains and captive renewables moderate net demand growth

While demand drivers are strong, part of the growth will be moderated by structural improvements and self-supply. Increasing energy efficiency, a reduction in technical losses over time, and rising captive as well as off-grid renewable generation are expected to offset ~1% of overall electricity demand.

India's power market is positioned for sustained growth, with rising electrification, industrialisation, urbanisation and emerging loads such as data centers, EV and green hydrogen driving incremental electricity demand and, consequently, the need for additional generation capacity. While energy-efficiency improvements and the growing adoption of captive RE projects may moderate grid demand growth in certain segments, they are unlikely to materially alter the need for continued system-wide capacity additions.

The increasing share of geographically concentrated and variable RE, particularly solar and wind, also shifts the investment requirement beyond generation. It necessitates timely expansion of inter-state and intra-state transmission networks, power-evacuation infrastructure, energy storage and flexible generation resources. Power evacuation refers to the transmission infrastructure required to carry electricity generated at a power plant or renewable-energy park to the broader grid and demand centers; evacuation infrastructure includes associated transmission lines, substations, pooling stations and grid interconnection facilities.

In this context, the Green Energy Corridor programme assumes increasing importance, as it is intended to develop dedicated transmission capacity for integrating renewable generation from resource-rich regions with major load centers. Delays in the commissioning of such corridors or associated evacuation infrastructure could lead to renewable curtailment, constrain new project development and increase system-balancing costs. Therefore, coordinated planning of generation capacity, renewable-energy zones, transmission corridors, storage and demand centers will be critical to maintaining reliable, least-cost power supply.

2.6 Capacity addition outlook (fiscal 2027-2031)

A larger system with a greener mix and more flexibility

India's installed generation capacity has expanded materially over the last seven years, rising from 355 GW at the end of fiscal 2019 to ~533 GW in fiscal 2026 and 549 GW as of June 2026. This growth has been driven primarily by strong renewable additions (solar, wind, hybrid, and other renewable sources), even as additions in coal and other conventional fuels have moderated. Reflecting this shift, renewables (excluding large hydro) increased their share to ~42% in fiscal 2026, from ~22% in fiscal 2019, while coal-based capacity tapered to ~43% over the same period.

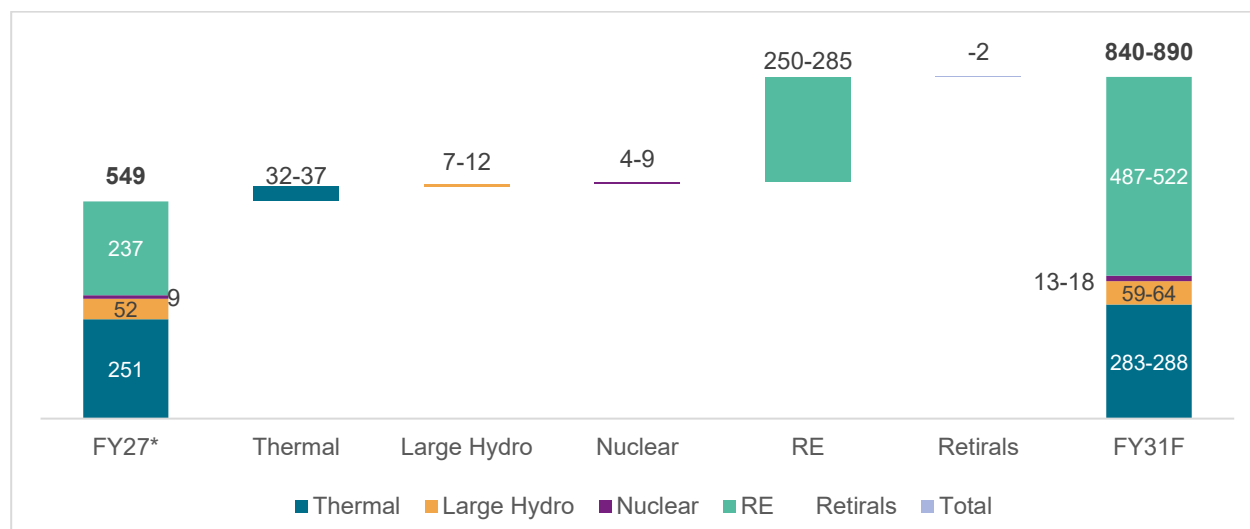
Looking ahead, capacity additions are expected to broaden beyond renewables alone, as the system prepares for higher peaks and the need for firm and flexible power. Conventional capacity additions are projected at ~32–37 GW during fiscals 2027–2031, supported by planned projects to meet peak demand and replace ageing coal plants. Nuclear additions of ~4–9 GW and hydro additions of ~7–12 GW are also expected during the period.

A defining feature of the next phase of capacity build-out is the acceleration in storage. Growing need for energy storage systems is expected to further drive the capacity additions of PSP and BESS over the next 5 years and is expected to reach 56-58 GW by fiscal 2031.

Between fiscal 2027- 2031, solar and wind capacity cumulatively is expected to add ~247-282 GW (~50-55 GW per annum), supported by government initiatives, favourable policies, competitive tariffs, innovative

tenders, solar parks, and enabling infrastructure such as green energy corridors. Other RE capacity (including small hydro, biomass and waste-to-energy) expected to add 2-3 GW over the same period. As a result, total installed capacity is expected to reach ~840–890 GW by fiscal 2031, with renewables (including large hydro) accounting for ~65% of the mix.

Figure 11: All India installed capacity addition by Fiscal 2031 (in GW)



*FY27 as of June 2026; RE includes solar, wind and other RE (small hydro, biomass, WTE)

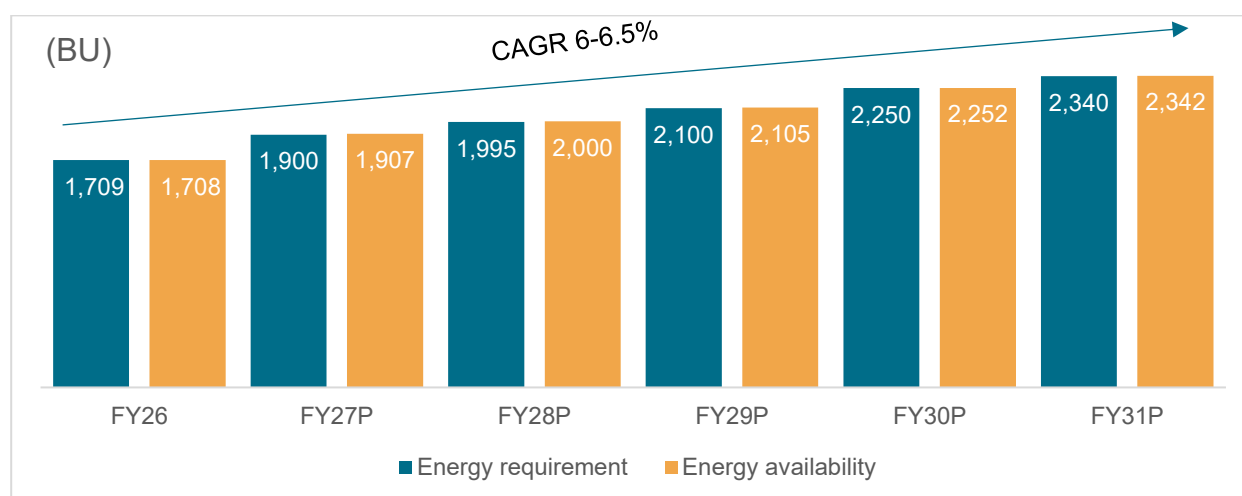
Source: CEA, Crisil Intelligence

2.7 Outlook for demand and supply (fiscal 2027-2031)

Steady energy growth, faster peak growth

After four years of elevated demand growth (fiscals 2022 to 2026) at ~5.5%, Crisil Intelligence expects power demand growth to normalise but remain strong, expanding at a CAGR of 6.0–6.50% over the next five years. This outlook is supported by infrastructure-linked capex, resilient economic fundamentals, green hydrogen production, adoption of EV, establishment of data centers and a broader power footprint enabled by strengthening transmission and distribution infrastructure. In parallel, ongoing reforms aimed at improving the health of distribution utilities are expected to enhance the quality and reliability of supply further supporting consumption growth. While energy demand is set to rise steadily, system planning is increasingly driven by peak demand which is growing faster and is more sensitive to weather and urbanisation trends.

Figure 12: India-Energy demand outlook (fiscals 2027-31)

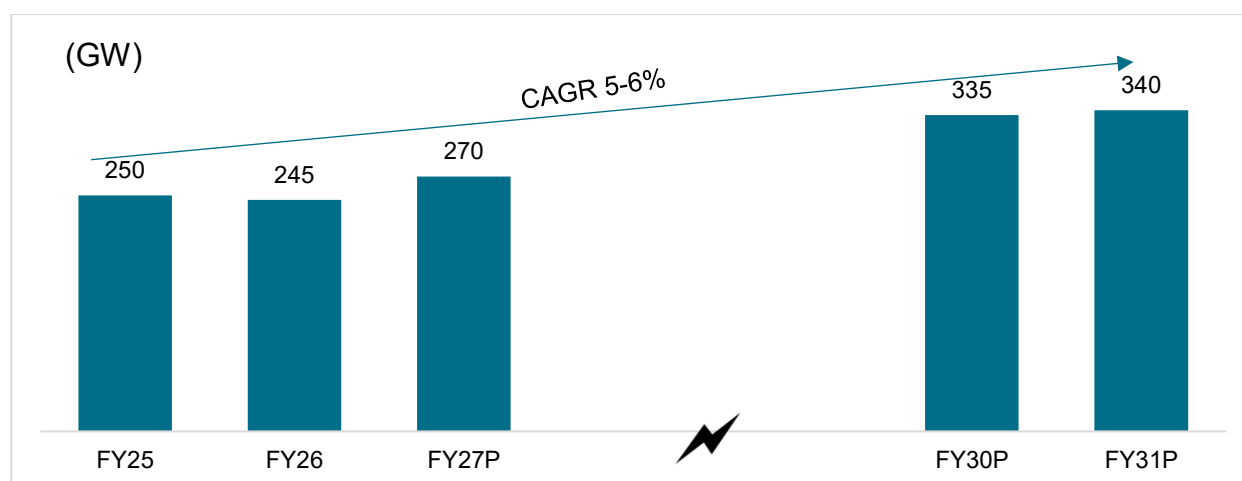


P: Projected, Source: CEA, Crisil Intelligence

India's power consumption is expected to increase sharply to approximately 3,776 TWh by 2041-42 as per CEA, primarily due to rapid economic and industrial growth.

Peak demand is expected to grow at an annual average CAGR of ~5-6% between fiscal 2026 and fiscal 2031, reaching 335-345 GW by fiscal 2031. Persistent high temperatures, rising urbanisation, economic growth, and the infrastructure push are expected to keep peak loads elevated. Despite India's significant strides toward becoming a power-surplus nation, peak power deficits cannot be completely ruled out, especially during high-demand summer months or in specific regions.

Figure 13: India-Peak demand to increase to cross 340 GW by fiscal 2031



P: Projected, Source: CEA, Crisil Intelligence

Meeting higher peaks reliably, especially with a rising share of variable renewables - requires not just generation and storage, but also a transmission backbone that can move power across regions and maintain grid stability.

India's rising power demand is expected to accelerate renewable energy deployment as the country seeks to meet incremental consumption through cost-effective, reliable, and sustainable sources while advancing its energy-transition and decarbonisation goals. Further, as renewable capacity becomes increasingly

concentrated in resource-rich regions, access to transmission capacity, substations, evacuation infrastructure and timely connectivity is becoming an increasingly important determinant of project viability.

3 Overview of Renewable Energy sector in India

3.1 Government of India initiatives for RE growth

Central Government's Decarbonisation Targets and Nationally Determined Contributions (NDCs) under the Paris Agreement

COP 26

The 2021 United Nations Climate Change Conference (COP 26) was a landmark event, as it saw several important decisions including:

- A commitment to phase down coal power and to accelerate the transition to clean energy.
- A commitment to reduce methane emissions by 30% by 2030.
- A commitment to provide \$100 billion per year in climate finance to developing countries.

India has submitted its updated first NDCs, working towards climate justice after COP26. Some of the key NDCs are:

- To reduce Emissions Intensity of its GDP by 45% by 2030, from the 2005 level.
- To achieve about 50% cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030, with the help of the transfer of technology and low-cost international finance including from the Green Climate Fund (GCF)
- To create an additional carbon sink of 2.5 to 3 billion tonnes of CO₂ equivalent through additional forest and tree cover by 2030.

India has achieved a reduction of 37.38% in the emissions intensity of its GDP in 2022 from the 2005 level, against the NDC goal of reducing emissions intensity by 45% by 2030. The share of cumulative installed electric power capacity from non-fossil fuel-based sources crossed 50% in June 2025, thereby achieving the NDC goal five years ahead the committed timeline. India has created an additional carbon sink of 2.44 billion tonnes of CO₂ equivalent during 2005 to 2022, against the NDC goal of creating an additional carbon sink of 2.5–3.0 billion tonnes of CO₂ equivalent by 2030 through additional forest and tree cover.

In March 2026, India submitted its updated NDC targets for the year, with quantitative targets to reduce the emissions intensity of GDP by 47% from the 2005 level, achieve 60% cumulative installed electricity capacity from non-fossil fuel-based energy resources, and create carbon sink of 3.5 to 4.0 billion tonnes of CO₂ equivalent through forest and tree cover as compared to the baseline year of 2005.

India's RE build-out is underpinned by a policy stack that does two things in parallel: creates demand for green power and reduces execution/operational risks for suppliers. The framework flows from compliance obligations (RPO/RCO) to compliance instruments (RECs), to operational protections (must-run), and finally to market-enabling rules for access, grid connectivity, and contracting.

Apart from the well-established mechanisms of RCO, Must Run status, REC framework, the government has rolled out several other policies, schemes, and guidelines to accelerate the adoption and implementation of renewable energy across the country. These initiatives span across various segments including solar, wind, energy storage, and distributed generation, and are aimed at addressing barriers related to financing, grid integration, land acquisition, etc.

3.2 Evolving state RE open access regulations helps in the adoption of RE by C&I consumers

Consumers opting to procure power under open access are liable to pay various charges to transmission and distribution utilities to use their network to wheel power from the third-party supplier to the consumption point. These charges include cross-subsidy surcharge, transmission and wheeling charges and losses, connectivity charges, and LDC charges, among others. Further, cross-subsidy surcharge is determined by respective state regulatory commissions to compensate distribution companies for loss of revenue on account of the shift of high paying C&I consumers. However, as per the National Tariff Policy 2016, cross-subsidy surcharge should not be so onerous to eliminate competition through open access route. Moreover, Cross Subsidy Surcharge (CSS) are not applicable to captive / Group Captive sales and are applicable only to third party non-captive sales. CSS is capped at 20% of the ACoS rate for industries. Further, Green Open Access Rules 2022 provide that AS will not be applicable to renewable energy if fixed charges are paid by such consumers. The Electricity (Promoting renewable energy through green energy open access) rules, 2022 state that cross-subsidy surcharge should not be increased by more than 50% for a period of 12 year from the date of project commissioning and full waiver on additional surcharge if fixed charges being paid by the consumer.

The expansion of the T&D networks, availability of more efficient systems to maintain grid stability, and increased operating efficiency of the utilities will free up network capacity to open access users. The increased operating efficiency and network availability would keep a rein on transmission and wheeling charges & losses.

Thus, the rationalisation of cross-subsidy and efficiency in grid operations would make open access routes more attractive for bulk consumers.

SERCs of different states have either exempted or provided concessions to RE projects from payment of various open access charges from time to time to promote the use of renewable energy among C&I consumers. For instance, Tamil Nadu and Rajasthan have provided 50% concession in the wheeling and transmission charges for renewable projects, Tamil Nadu has concessional cross-subsidy surcharges (at 60% and 70% of conventional power for third-party wind and solar OA, respectively). Many states have reduced banking provisions for renewable energy projects from an annual energy banking settlement to monthly energy banking.

Despite the discontinuation of open access charges exemptions/incentives for new projects, it is important to note that these benefits will be applicable until their expiry for the old projects commissioned during the period when such benefits were introduced. However, the landed cost of open access projects (for some states) without any exemptions/incentives could remain competitive as compared to utility tariff as the discom variable tariffs for C&I category are expected to continue to rise in future. There would be a sufficient margin between the levelised cost of RE and variable discom's C&I tariff to mitigate the risk of reducing/withdrawing the open access charges incentives. The savings will improve further if consumers opt for captive mode.

3.3 Growing prospects for open access sale

3.3.1 Introduction of green energy open access rules

The Electricity (Promoting renewable energy through green energy open access) rules, 2022 give consumers an option to draw green energy through open access whose contract demand is 100 kW or above and no limit for supply of power for captive consumers. It states that cross-subsidy surcharge should not be increased by more than 50% for a period of 12 years from the date of project commissioning and full waiver on additional surcharge if fixed charges being paid by the consumer. Banking is allowed on a monthly basis on payment of banking charges. The un-utilised surplus banked energy would be lapsed at the end of each banking cycle

and the RE generating station would get RECs to the extent of the lapsed banked energy. It also allows a consumer to purchase green energy by placing a requisition with their discom.

In addition, as per the rules, the electricity produced from offshore wind projects commissioned up to December 2032 will not attract additional surcharge for the energy supplied to open access consumers.

The rules provide long-term certainty of the open access charges for the stakeholders which will help in determining their returns from the project. Further, if an open access application is not approved within 15 days, it will be deemed approved. This will ensure timely execution of projects by minimising any risk of cost escalations. The demand for green energy open access from C&I consumers is likely to increase after implementation of these rules. However, discom cooperation, regulatory proactiveness in timely tariff orders and green energy tariffs are key monitorable.

3.3.2 Deviation in Green energy open access regulations adopted by the States from MoP rules

Most of the large industrial and RE rich states have notified green energy open access regulations, largely in line with the original Rules. However, there are a few deviations observed which are listed in the table below.

Table 4: Comparison of MoP's green open access rules with state notified RE open access regulations

	Eligibility	Banking	Cross subsidy surcharge for third party sale (waived for group captive)	Additional surcharge for third party sale (waived for group captive)	Standby charges	Compensation for unused banked power
MoP Rules	100 kW or above	Monthly; Banked quantum – 30% of monthly consumption for discom Banking charges at 8% of banked energy	Shall not increase by more than 50% in 12 years. CSS should not exceed 20% of ACoS	Not applicable if fixed charge being paid	Not applicable if notice is given 24 hrs. before the delivery of power. Should not exceed 25% of energy charges	Entitled to get REC for the lapsed banked energy
Andhra Pradesh	✓	✓	No cap on CSS	No AS exemption	✓ (20%)	Paid at 75% of the last discovered SECI tender for given RE source
Gujarat	✓	Banking charges of Rs. 1.5/kWh, subject to change half-yearly	✓	✓	✓ (10%)	✓
Karnataka	✓	Banking charges at	Cap upto 20% of	✓	✓ (125%)	✓

	Eligibility	Banking	Cross subsidy surcharge for third party sale (waived for group captive)	Additional surcharge for third party sale (waived for group captive)	Standby charges	Compensation for unused banked power
		8% of banked energy	applicable consumer tariff			
Madhya Pradesh	✓	✓	✓	✓	✓	✓
Maharashtra	✓	Banking charges at 8% of banked energy	No cap on CSS	✓	-	✓
Rajasthan	✓	Banking charges at 8% of banked energy	✓ Cap of 20% of ACOS	✓	✓ (25%)	Entitled to get REC for the lapsed banked energy
Telangana	✓	✓	✓	✓	✓	✓
Tamil Nadu	✓ (63 kVA)	Banking charges at 8% of banked energy	Incentives only for the intra-state transaction of RE power	No clarity	✓ (125%)	75% of APPC or preferential tariff whichever is lower

Source: MoP, Respective SERC Regulations, Crisil Intelligence

3.3.3 Summary of State RE policy

State	Existing benefits/Open Access (OA) provisions	Evolution
Tamil Nadu	50% concession on Transmission Charge / Wheeling (Distribution) Charge 100% concession on Cross Subsidy / Additional Surcharge, for Captive sale 40% (Wind) & 30% (Solar) concession on Cross Subsidy / Additional Surcharge, for non-Captive sale - Monthly banking facility; 12 months banking applicable for wind projects commissioned before Mar-2018	- No annual electricity banking facility available to solar projects commissioned after March 2020 - Prior projects not connected to high voltage grids also have to pay an additional wheeling charge plus additional wheeling losses - Resource charges of Rs.5 Mn/MW for all future wind projects and pending applications, with CTU connectivity
Gujarat	- CSS and AS not applicable for RE projects fulfilling captive power project criteria - No concession on transmission and wheeling charges and losses - Monthly banking with banked quantum limited to 30% of net drawl from discom in the month.	- Earlier Policy benefits only to solar projects commissioned by December 2020, wind projects commissioned by December 2023, Hybrid projects by June 19, 2023 - No concession in Wheeling Charges and Losses

State	Existing benefits/Open Access (OA) provisions	Evolution
Madhya Pradesh	100% exemption from payment of ED for 10 years from COD 50% in Wheeling Charges for all RE projects for 5 years from COD 100% concession on Cross Subsidy / Additional Surcharge, for Captive sale - No energy development cess payable on RE power upto 10 years from COD 20% exemption in registration fee - State levies an additional development fee of INR 0.10/ kWh on open access projects	
Rajasthan	100% concession on Cross Subsidy / Additional Surcharge, for Captive sale - For 5% of storage integration, 75% Waiver; For every additional 1% of BESS capacity added, the waiver increases by 1%; Once the BESS capacity reaches 30% of solar capacity, a 100% waiver on both Transmission and Wheeling charges for a period of 7 years.	

Source: State Policies, Industry, Crisil Intelligence

The States are slowly moving away from the benefits/waivers given to RE projects since the industry is now mature and new projects are not getting the concessions/waivers available to old projects commissioned before the implementation of new policies.

3.3.4 Captive/group captive policies and Electricity (Amendment) Rules, 2026

As per electricity Rules, 2005 a power plant could qualify as a captive generating plant (CGP) if the captive user holds at least 26% ownership. A group captive mode is where a developer develops a power plant for collective usage of many consumers. A power project is considered 'captive' if the consuming entity or entities consume at least 51% of the power generated and owns at least 26% of the equity.

On March 13, 2026, the Ministry of Power, Government of India, notified amendments to Rule 3 (Amendment) of the Electricity Rules, 2005. These amendments significantly ease compliance and resolve structural ambiguities for CGPs and group captive renewable projects. They modernise captive regulations for C&I consumers.

The amendment upholds the "twin qualifying criteria" (26% ownership and 51% consumption) under Rule 3 of the Electricity Rules, 2005.

- Captive users now explicitly include end-users consuming electricity directly or through energy storage systems.
- A captive user, its subsidiaries, and its holding company (plus other subsidiaries) are treated as a single entity, mitigating risks if a group member under-consumes
- In case of a power plant set up by a registered co-operative society, the eligibility conditions shall be satisfied collectively by the members of the co-operative society
- Captive user(s) must ensure that the eligibility conditions are complied with during the financial year. If minimum captive consumption requirement is not met during any period, the entire electricity generated

shall be treated as supply of electricity by a generating company and cross-subsidy surcharge and additional surcharge shall be levied on such consumption:

- Users can draw up to 100% of their proportional consumption share, and compliance can be assessed for specific units identified for captive use rather than the entire station.
- Where the ownership pattern of the power plant varies during the financial year, the proportionate consumption of each captive user shall be determined on the basis of the weighted average shareholding of such captive user during the financial year
- In case, a power plant set up by an association of persons, any consumption by an individual captive user in excess of 100% of its proportionate consumption shall be treated as supply of electricity by a generating company and cross-subsidy surcharge and additional surcharge shall be levied on such excess consumption.
- Verification to be conducted by state-designated nodal agencies for intra-state projects and by the NLDC for inter-state projects.
- An appeal against the verification shall lie before a Grievance Redressal Committee constituted by the Appropriate Government.
- Cross-subsidy and additional surcharges are suspended while captive status is pending verification.

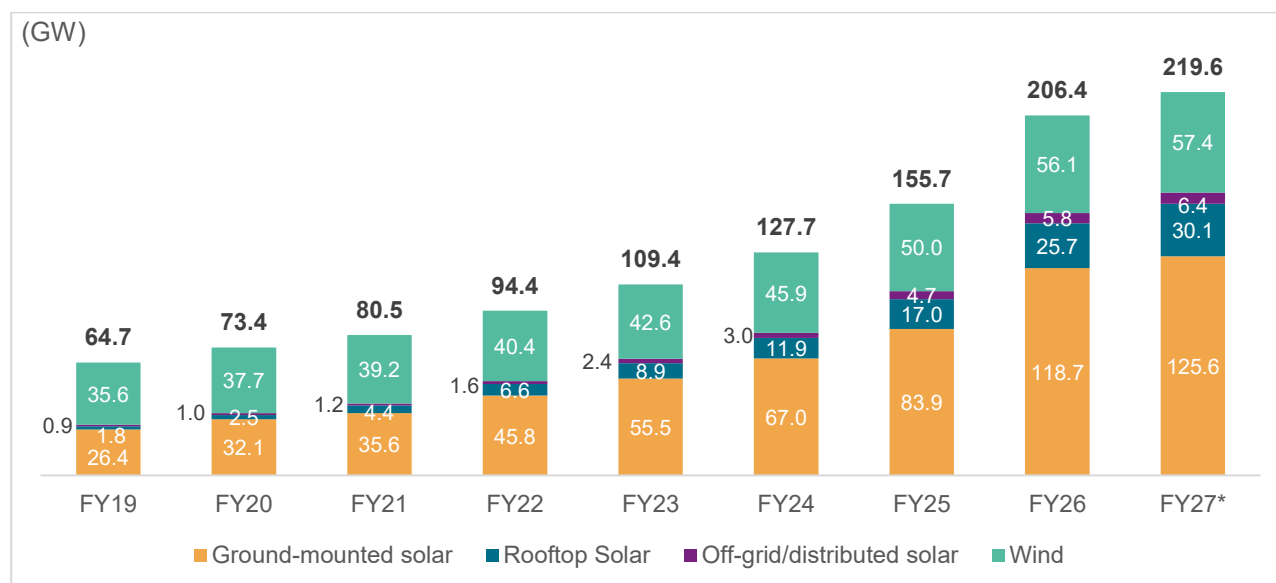
As far as benefits for captive users are concerned, as per section 42(2) of the Electricity Act, 2003, captive power plants, set up by end-users for their consumption, are exempted from payment of cross-subsidy surcharge and the additional surcharges, otherwise would be applicable for open access consumers. Thus, the central government's policy support has played a role in promoting the captive/group captive model, especially for C&I consumers.

3.4 Overview of solar and wind capacity additions

With the increased support of the Government and improved economics, the solar and wind sector has become attractive from an investor's perspective. India's renewable energy market is led by solar and wind, which are already charting a significant growth trajectory. During fiscals 2019-27 (as on June 2026), India added around ~ 155 GW of solar and wind capacities cumulatively. The installed capacity has grown from 64.7 GW in fiscal 2019 to 219.6 GW in fiscal 2027 (as on June 2026) at a CAGR of 18.4%.

Solar segment accounted for the majority of new capacity additions during this period, contributing ~133 GW, across utility scale ground-mounted, rooftop and off-grid/distributed installations. Among these segments, rooftop solar recorded the fastest growth, with installed capacity increasing from 1.8 GW in fiscal 2019 to 30.1 GW by June 2026, reflecting a CAGR of 47.5%. Distributed and off-grid solar capacity expanded from 0.9 GW to 6.4 GW over the same period, registering a CAGR of 30.8%. Utility-scale ground-mounted solar projects remained the largest segment, growing at a 24.0% CAGR to reach 125.6 GW of installed capacity as of June 2026.

Wind power also continued its growth trajectory, although at a more moderate pace. Installed wind capacity increased by approximately 22 GW between Fiscal 2019 and June 2026, reaching 57.4 GW, representing a CAGR of 6.8% over the period.

Figure 14: Historical annual RE Capacity additions in India


*FY 27 as of June 2026.

Source: CEA, MNRE, Crisil Intelligence

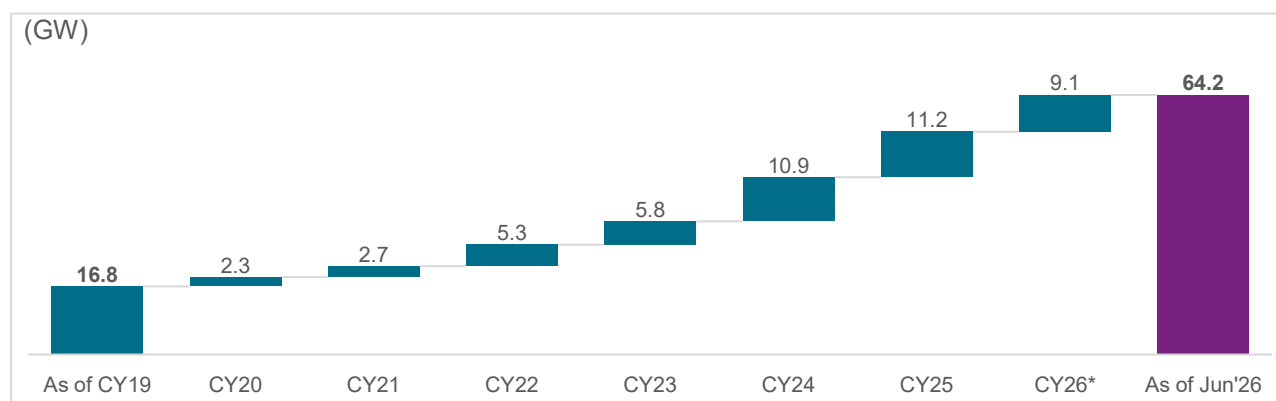
3.5 RE capacity addition by OA routes

The cumulative open access renewable C&I capacity including solar rooftop installation is ~64.2 GW as of June 2026, of which utility scale solar and wind OA project accounts for ~43.4 GW. The remaining 20.8 GW is from rooftop and behind-the-meter solar. The overall OA capacity recorded a CAGR of 22.9% between CY2019 and June 2026. India's RE OA market is estimated to be predominantly captive/group captive in structure, accounting for 65-75% of installed capacity, with 3rd party owned models accounting for remaining 25-35%.

Utility-scale open access solar emerged as the fastest-growing segment, with installed capacity increasing from approximately 4.1 GW in CY2019 to 26.6 GW by June 2026, representing a CAGR of 33.4%. Rooftop solar also witnessed robust growth during the period, expanding at a CAGR of 27.7%. In the wind segment, capacity additions accelerated from CY2023 onwards, with over 7 GW of new OA wind capacity installed between CY2023 and June 2026.

Among the renewable sources, solar holds the majority share at ~74% (including rooftop solar). This dominance is attributed to its advantages, such as higher accessibility, wider availability of resources, shorter project gestation periods, and reduced construction and operational risks.

The sustained expansion of OA and rooftop solar capacity underscores the increasing preference of C&I consumers to move beyond conventional grid-based power procurement toward direct RE sourcing models. This transition has been driven by a combination of factors, including cost competitiveness of RE power, rising corporate sustainability commitments, the need for long-term energy price visibility, and an evolving regulatory framework that increasingly facilitates direct procurement of RE. Consequently, the growth of OA, rooftop, and distributed RE solutions provides a more relevant indicator of the underlying opportunity within the C&I renewable energy market than overall renewable capacity additions alone.

Figure 15: C&I renewable annual capacity addition trend


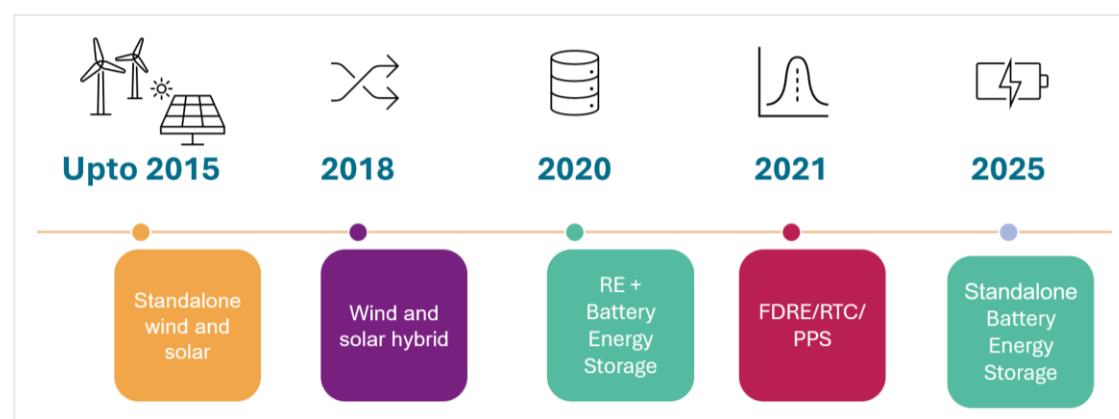
Source: Crisil Intelligence

Open access has become a key procurement route, enabling C&I consumers to source renewable energy directly from IPPs through long-term PPAs. The Green Energy Open Access Rules, 2022 lowered the eligibility threshold to 100 kW and streamlined approvals, broadening access. This model offers tariffs significantly below grid rates, supports ESG goals and provides long-term cost certainty. Rising electricity tariffs and supportive state policies continue to drive demands.

3.6 Standalone solar/wind vs hybrid RE systems

India's renewable energy market has progressively evolved from predominantly standalone, single technology generation projects towards integrated configurations designed to address the increasing requirements of grid integration and power supply reliability.

During the initial phase of RE deployment, solar and wind projects were largely developed and procured independently, with project economics and competitive tariffs being key considerations and heavy reliance on favourable banking provisions provided by the discoms. With increasing penetration of RE, the industry has progressively placed greater emphasis on the temporal alignment of generation with demand, utilisation of transmission infrastructure and management of generation variability with regulatory incentives applicable on RE also being substantially diluted parallelly. This has resulted in growing adoption of hybrid RE configurations, including combinations of solar and wind, and increasingly, integration with BESS to provide a more predictable and flexible power supply.

Figure 16: Evolution of RE project configuration in India


Source: Crisil Intelligence

Standalone Solar: Solar generation is constrained to daylight hours, with peak output typically occurring between 10 am and 2 pm. This limits availability during early morning, evening and night-time, when electricity demand may remain high. Consequently, standalone solar projects experience lower capacity utilisation factors (CUFs) and may require grid support or storage to meet round-the-clock demand.

Since C&I consumers typically have electricity demand extending beyond solar generation hours, this creates a temporal mismatch between the solar generation profile and the C&I consumption profile. Consequently, solar generation alone may not substantially meet the customer's electricity requirements across the day. This generation-load mismatch has become an important driver for the adoption of hybrid RE (with or without storage) configurations.

Standalone Wind: Wind generation is variable across seasons and depends on local wind patterns but provides a more consistent generation through the day on 24-hour basis. While it can operate throughout the day, output often peaks during night-time or early morning hours in many regions. However, wind availability is less predictable.

Given the seasonal variability in wind generation, the timing and magnitude may therefore not consistently coincide with the electricity consumption profile of C&I customers across months.

These limitations have supported the evolution towards hybrid RE configurations, particularly combinations of solar and wind, where the differing generation profiles of the two technologies can provide greater temporal complementarity. Wind and solar generation are complimentary in many ways – wind generation is tilted towards evening and night and solar generation is limited to daytime only, similarly wind generation is greater during monsoon months, and solar generation is least during the monsoon months. Combining the two can therefore broaden the RE generation profile and improve alignment between RE supply and the C&I consumer's load profile, potentially reducing periods of renewable shortfall and increasing the utilisation of contracted RE power. The addition of energy storage can further address residual generation-load mismatches by shifting available RE to periods of demand when generation from the underlying RE sources is insufficient. Storage being a cost, requirement of storage also comes down significantly from a standalone solar to a wind solar hybrid configuration.

Hybrid power reduces variability by pairing wind and solar energy

Wind-solar hybrid (WSH) project benefit from the complementary nature of solar and wind generation, providing a more stable and predictable energy supply throughout the day and seasons.

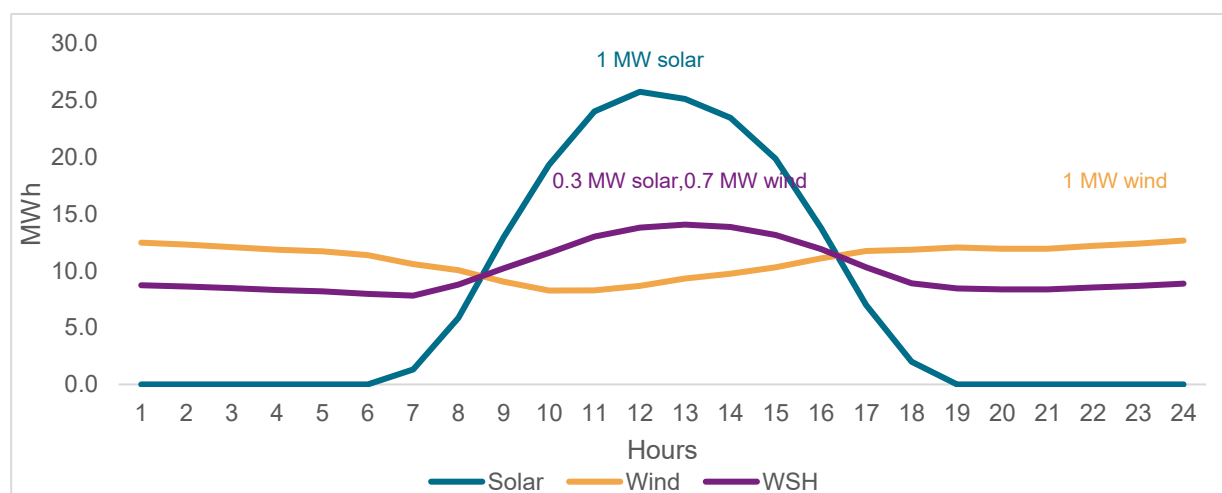
There are two primary types of hybrid projects: co-located and non-co-located. Co-located hybrid projects integrate wind and solar generation at a single site, sharing infrastructure such as land, transmission lines, and grid interconnection facilities, thereby optimising the utilisation of existing land, evacuation and transmission capacity, which help to reduce capital and operational costs. In contrast, non-co-located hybrid projects involve wind and solar components situated at different locations but operated in a coordinated manner to meet combined energy delivery commitments or grid requirements, often through centralised control systems.

The combination of wind and solar results in:

- **Higher capacity utilisation:** By capturing generation across different time periods and seasons, hybrid projects achieve higher overall CUFs compared to standalone solar or wind. The shared infrastructure including land, transmission connectivity and evacuation systems also improves asset utilisation and reduces per-unit costs.

- **More stable and predictable supply:** The blended output smooths variability across daily and seasonal cycles, improving grid integration and reducing reliance on backup power or storage. This enhanced predictability also reduces scheduling deviations and associated penalties under the Deviation Settlement Mechanism.
- **Optimised land and transmission use:** Co-locating wind and solar on the same site allows developers to maximise generation from a given land parcel and transmission capacity, improving project economics and reducing development timelines. Since wind and solar are combined in the same transmission network with overall evacuation being optimised in comparison to AC capacity of each, the overall transmission charges are substantially minimised as compared to standalone systems.
- **C&I customer energy replacement and power availability –** The ability of C&I customers to offtake green power are primarily determined by three factors. (1) allowable open access capacity, which is typically linked to contract demand, and therefore higher the ability to evacuate power in the same MW (as is the case with hybrid), higher will be the discom power replacement. (2) tightening banking requirements and revision in ToD profile by Discoms, is increasingly resulting in solar generation settlement being limited to solar hours only. Wind solar hybrid generation plant affords a much more uniform settlement across the time blocks.
- **Enhanced value for end users:** Can broaden RE generation window and improve the alignment between RE supply and C&I load profile, reduces reliance on grid power and exposure to grid tariff volatility and improves overall project bankability for both developers and offtakers.

Figure 17: Indicative hourly power output profile of solar, wind and WSH power projects



Source: Crisil Intelligence

Beyond basic hybrids, more sophisticated project structures are emerging to meet evolving grid and offtaker requirements

While WSH projects represent a significant improvement over standalone technologies, they still cannot fully guarantee power supply aligned with grid or consumer demand at all hours. With rising electricity consumption, demand patterns are becoming more complex, requiring supply solutions that can adapt to time-of-day and seasonal variations. This has led to the emergence of more complex project structures that integrate storage, demand-matching and dispatchability requirements with solar and wind have been introduced:

- **Firm and Dispatchable Renewable Energy (FDRE):** It is a delivery design construct rather than a single technology configuration. These projects are designed to supply power in alignment with the offtaker's varying demand curve throughout the day. They typically combine wind, solar and BESS to ensure that generation matches consumption patterns, effectively replicating the dispatchability characteristics of conventional thermal power while remaining fully renewable. SECI recently concluded an innovative tender for 1000 MW Round-the-Clock Thermal Mimic (RTC-TM) Power. The tender had supply commitments of 90% during the Peak Hours, 50-60% during the solar hours and 70% during the non-solar hours.
- **Round-the-Clock (RTC):** RTC projects are designed to deliver the contracted power continuously across 24-hour period, rather than simply generating RE power whenever wind/solar resources are available.
- **Peak Power Supply (PPS):** PPS tenders require high CUF during identified peak demand periods often both morning and evening peak while allowing relaxed CUF requirements during remaining hours. These projects are particularly valuable for grid operators seeking to displace expensive peaking thermal generation with renewable alternatives.

In effect, power procurement has evolved from simply buying the lowest-cost renewable energy to procuring consumption/demand aligned renewable capability with significantly reduced dependency on banking incentives. Complex projects translate grid needs firmness, dispatchability and peak support into bankable contracts and scalable pipelines. A similar transition is underway in the C&I segment, where customers are increasingly moving beyond procuring low-cost RE power towards securing RE supply profiles that align with their operational load patterns, reliability requirements, and sustainability objectives. The progression from standalone projects to hybrids to FDRE and RTC structures reflects a maturing market across both utility and C&I segments, where the value of renewable energy is increasingly defined not just by cost but by its ability to reliably serve load profiles.

Industries too are moving beyond basic RE procurement to hybrid and complex green power solutions

This trend is also increasingly visible in the C&I segment, where industries, data centers, commercial consumers, green hydrogen and other high-uptime facilities are seeking RTC or firm power commitments in line with tapering incentives for banking, transmission and wheeling charges and change in ToD blocks from Discoms. Corporate buyers are moving beyond simple solar procurement towards structured solutions that can meet baseload and peak requirements with green power, supporting both operational reliability and sustainability commitments. As corporate net-zero timelines tighten and RE100-type pledges become more common, demand for firm, dispatchable renewable energy in the C&I space is expected to grow significantly, creating new opportunities for Integrated energy players capable of delivering complex, integrated energy solutions.

The optimal solution depends on the operating hours, load factor, available rooftop or land area, grid tariffs, regulatory provisions, and the consumer's requirements for supply reliability. Therefore, the integrated service providers are increasingly playing an important role in offering customised design tailored to individual client's needs as expertise in the right configuration economics saves not just cost but provides significant predictability and viable options for DISCOM power replacement also.

Table 5: Technology configuration versus load profile

Source	Continuous process	Two-shift	Single-shift	Seasonal	Commercial daytime
Solar	Daytime peak; no night-time generation	Daytime generation	Strong alignment with daytime operations	Daytime generation	Strong alignment with daytime demand
Wind	Variable; often stronger during non-daytime hours	Variable profile with partial night-time alignment	Variable profile with relatively limited load matching	Variable and seasonal generation	Variable profile with relatively limited matching
Hybrid	Smoother profile with higher overall yield	Smoother profile with partial night-time coverage	Smoother daytime-oriented profile	Smoother seasonal generation profile	Smoother daytime-oriented profile
FDRE	Firm and dispatchable	Firm and dispatchable	Firm and dispatchable	Firm and dispatchable	Firm and dispatchable
Annual Consumption	Very high	High	Low	Very Low	Low
Best Match	Solar + wind /FDRE	Solar + wind / FDRE	Solar	Solar and/or wind	Solar

Source: Crisil Intelligence

3.7 RE tariff trend

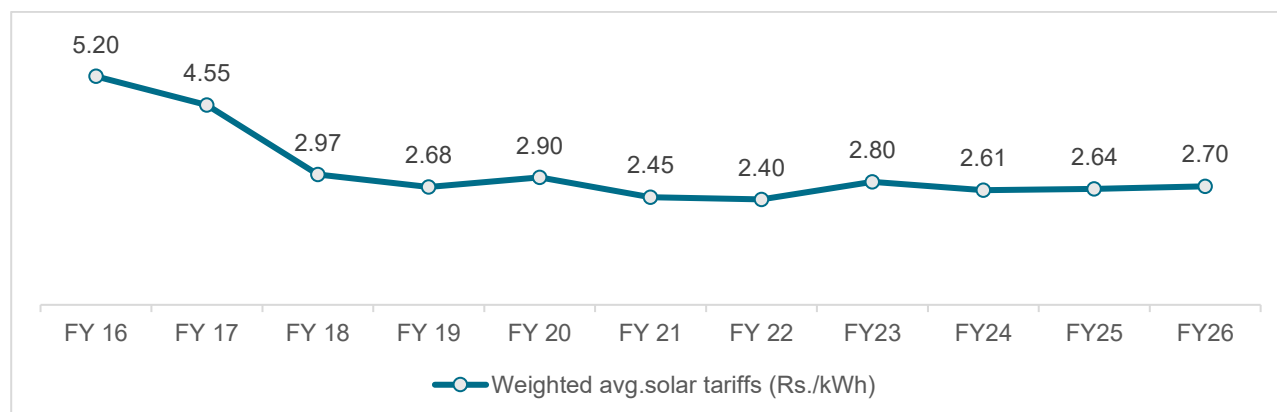
To promote the renewable energy sector, the government introduced feed-in tariffs (FiTs) in the early stages of RE development, when production was not economically feasible. The FiT mechanism guaranteed a fixed tariff to renewable energy producers for energy supplied to the grid. The tariff is fixed for a long period (15-20 years) to minimise the risk of renewable IPPs. For solar projects over 2009-2013, most states signed PPAs at FiTs determined by the state commission on a fixed regulated equity return of ~16%. For wind energy projects, states followed the FiT mechanism until March 2017. However, from fiscal 2018 onwards, the sector veered towards competitive bidding.

Solar tariffs showed a rapid decline from fiscal 2016 to fiscal 2020, with a rapid fall in component pricing, technological improvements in efficiency, and the government's policy push. While declining module prices contributed to a reduction in tariffs over fiscals 2017-2019, access to low-cost financing was the primary driver for the decline in tariffs over fiscals 2020-2022. Over the said period, global investments in the Indian renewable energy segment picked up via green bond issuance and external commercial borrowings, helping the lower cost of debt for the space. The participation of global players and entities with strong credit profiles (CPSUs) has helped tariffs remain in the Rs 2.4 - 2.6 per unit range even until fiscal 2022 when supply-side disruptions started to emerge. Weighted average tariff recorded a 16% uptick in fiscal 2023, when it rose to Rs 2.8 per unit before declining to Rs 2.61 per unit in fiscal 2024, a drop of close to 7%. Tariff drop in fiscal 2024 was majorly on account of price decline in upstream components, mainly solar cells, where prices fell

by ~47% in fiscal 2024. In fiscal 2025, the tariffs remained consistent with a weighted average tariff of Rs. 2.64 per unit.

With the applicability of ALMM List-II, solar modules will be required to use domestic cells and are therefore expected to be even more expensive, with prices at least 50% higher than those using imported cells. Crisil Intelligence estimates prices for imported cell-based modules to be in the range of \$0.14-0.16/Wp and for domestic cell-based modules in the range of \$0.24-0.26/Wp in fiscal 2027. The capex for solar plants using domestic cell-based modules is higher by Rs. 5-10 million per MW. As a result, the tariffs for the solar plants using domestic cell-based modules may see an upward trend.

Figure 18: Weighted average solar tariff trend

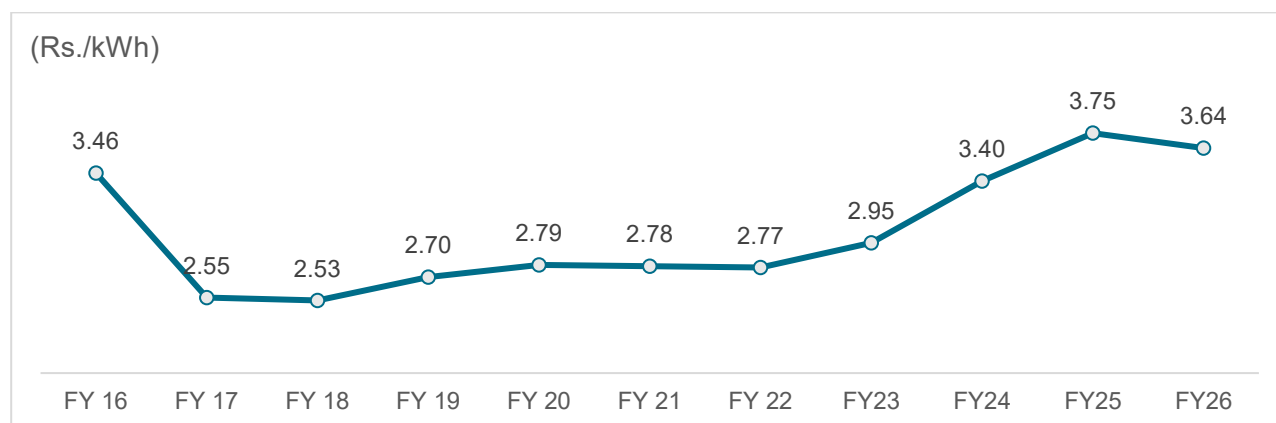


Source: Crisil Intelligence

For Wind projects, the discovered weighted average tariffs for competitively bid projects reached as low as Rs 2.53-2.55 /kWh in fiscal 2017 and 2018 as against Rs. 3.6-3.9/kWh tariff required for earning 10-13% equity IRRs. Crisil Intelligence believes that these projects were aggressively bid even when availability of developed land banks (availability of wind micro-siting data and proximity to the transmission) with high wind density sites were not tied up prior or finalised before bidding. This has caused execution challenges for several projects in the sector, for e.g., the SECI ISTS III projects were previously not able to acquire required wind sites in the preferred region of Gujarat. This followed the cancellation by some prominent tenders due to unviable tariffs. However, the tariff cap removal in March 2020 provided an opportunity for developers to factor in the added execution challenges along with concerns over project viability, leading to higher bid tariffs in successive auctions. Hence, wind sector power tariffs have risen steadily since 2022.

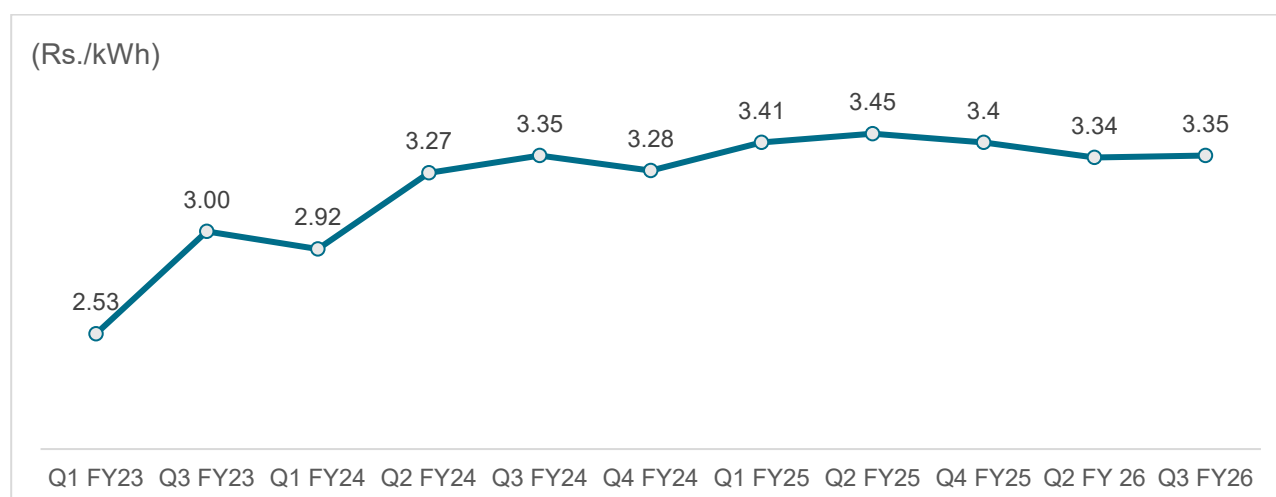
This experience has shown that the competitive renewable tariffs alone do not guarantee project viability. Timely access to suitable land, grid connectivity and evacuation infrastructure as well as effective management of approvals, supply chains, construction and commissioning determines whether contracted capacity becomes operational on schedule and delivers the expected savings.

Execution capability is therefore a critical differentiator among renewable energy solution partners. Developers with strong land-acquisition networks, secured connectivity, disciplined project management and a track record of timely delivery are better positioned to mitigate delays, cost overruns and curtailment risks. For C&I customers, evaluating a provider's execution readiness and operational credentials is as important as comparing headline tariffs.

Figure 19: Weighted average wind tariff trend


Source: Crisil Intelligence

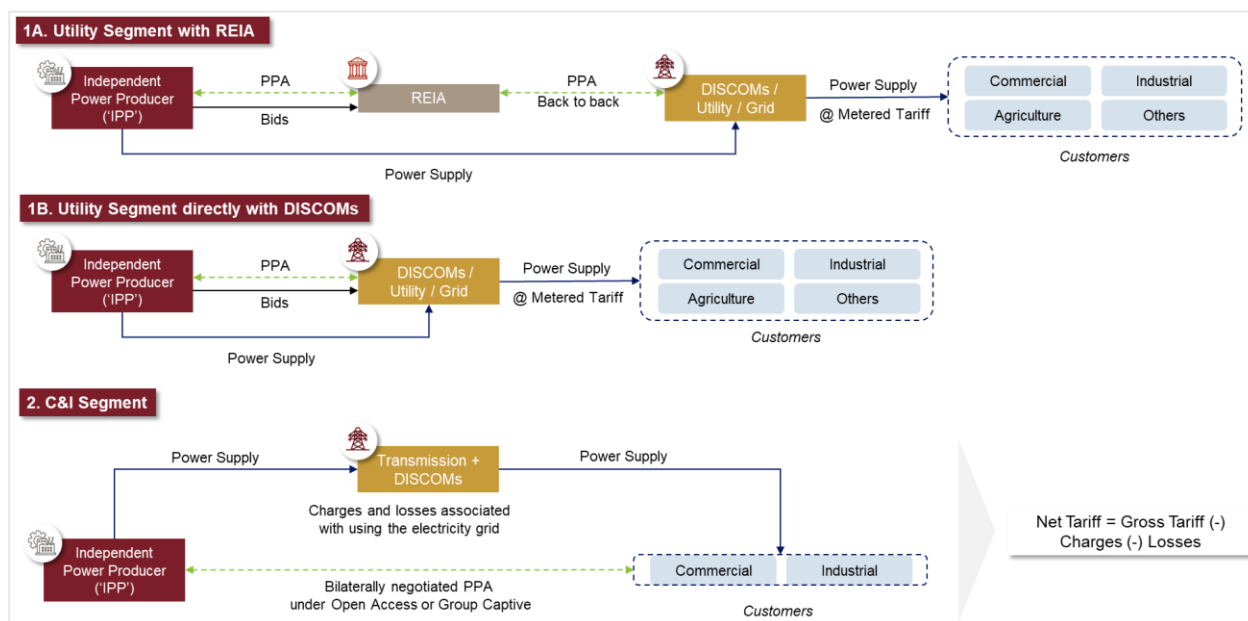
WSH tariffs are affected by concurrent trends from the solar and wind sectors. Like solar, the lowest-ever WSH tariff in India, Rs. 2.41/kWh was achieved in a December 2020 allotment. The recent upturn in wind and solar tariffs has also affected WSH tariffs. Recently, the WSH tariff has stayed above the threshold of Rs. 3/kWh, lower than vanilla wind tariffs but competitive with vanilla solar tariffs.

Figure 20: Recent trend in competitively bid out wind-solar hybrid tariff (weighted average tariffs)


*No WSH tender allocated in Q1 FY26 and Q4 FY26; Source: Crisil Intelligence

3.8 Power procurement options

C&I consumers have several options to procure power through discoms or open access. These options provide consumers in India with a range of choices to get power through the grid from Discoms or directly from generators, enabling them to optimise their power procurement cost and also meet their sustainability goals.

Figure 21: Power procurement options


Source: Industry, Crisil Intelligence

3.9 Capacity additions outlook on solar and wind

Solar: Strong growth trajectory and the drivers behind future capacity additions

Solar sector growth in India primarily has been spurred by robust government backing, demonstrated through an aggressive tendering strategy. Some of the key catalysts include technological advancements, affordable financing, supportive policies, thrust on go-green initiatives/sustainability targets, cost optimisation due to increased grid electricity tariffs, subsidy initiative (especially in rooftop solar) and various incentives such as ISTS charge waiver, banking facilities provided by DISCOMs.

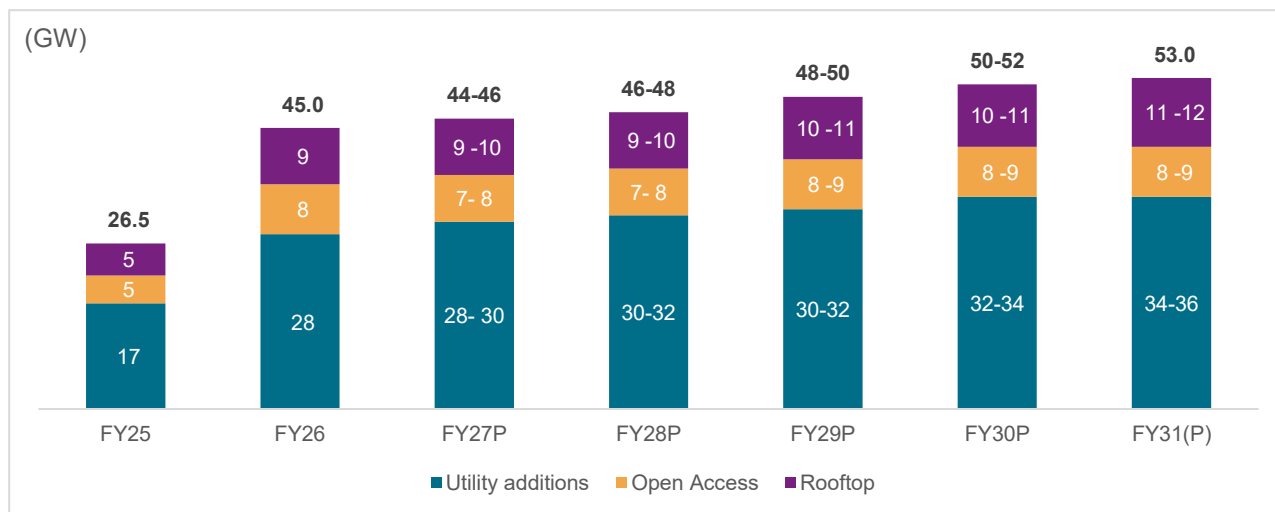
Crisil Intelligence expects 244-254 GW of solar capacity additions over fiscal 2027-2031. This will be driven by the additions under:

- **Other central schemes:** SECI has also started tendering projects outside the JNNISM Batch programme. It has initiated the ISTS scheme, wherein projects are planned for connection with the ISTS grid directly. Under this, SECI has already tendered and allocated ~38 GW (including hybrid).
- **State solar policies:** ~27 GW of projects are under construction and are expected to be commissioned over the fiscal 2027-2031 based on tendered capacities by states.
- **PSUs:** The CPSU programme under JNNISM has been extended to 12 GW in February 2019. The government is also encouraging cash-rich PSUs to set up renewable energy projects. NTPC has already commissioned a total of over ~3.7 GW of capacities in fiscal 2025. It has a target of installing ~60 GW of renewable energy capacities by fiscal 2032. Similarly, NHPC has an installed capacity of 262 MW and a pipeline of 1,290 MW, while the Indian Railways has committed to 20 GW of solar power by 2030. Other PSUs such as NLC, defence organisations, and governmental establishments are also expected to contribute to this addition.
- **Rooftop solar and behind the meter projects:** Crisil Intelligence expects 45-50 GW (18-20% in incremental solar capacity additions) of rooftop solar projects (under the capex and opex mode) to be

commissioned by fiscal 2031, led by PM Surya Ghar Yojana and industrial and commercial consumers under net/gross metering schemes of various states.

- **Open-access solar projects:** Crisil Intelligence expects 40-45 GW (16-18% in incremental solar capacity additions) of open-access solar projects (under the capex and opex mode) to be commissioned by fiscal 2031, led by green energy open access rules 2022, sustainability initiatives/RE 100 targets of the corporate consumers, better tariff structures and policies of states such as Uttar Pradesh and Karnataka, which are more long term in nature.

Figure 22: Year wise expected solar capacity addition



Source: Crisil Intelligence

However increasingly solar capacity addition will be driven in hybrid mode with another technology such as wind, storage, with standalone solar capacity addition tapering down substantially.

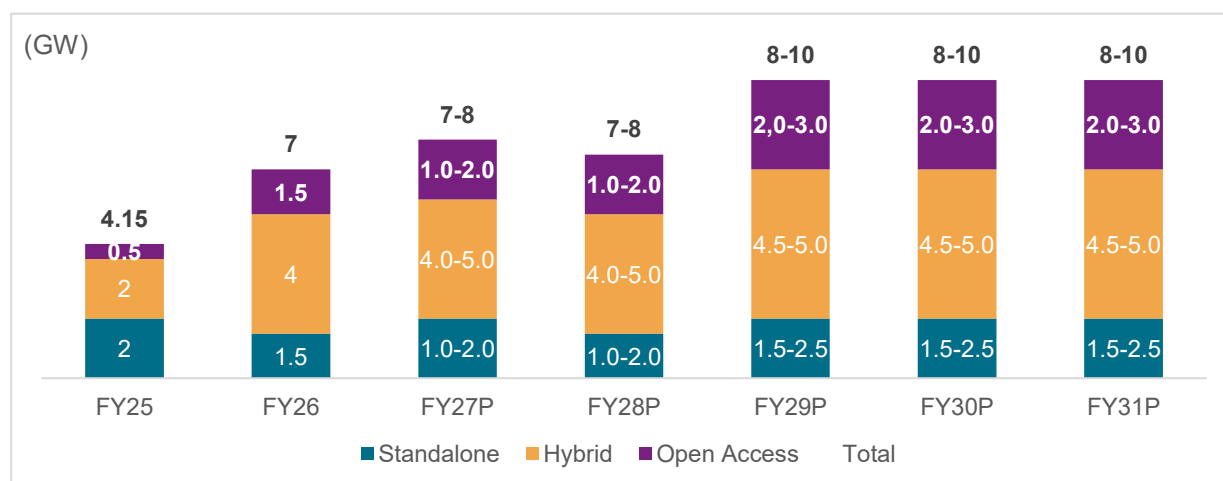
Wind: Hybrid and RTC power projects to emerge as key drivers for wind capacity addition

Crisil Intelligence expects wind capacity additions to grow over the next five years led by pipeline build-up under existing schemes and new tendering schemes, improvement in technology, thrust on green hydrogen, renewable generation obligation, and mixed resource models (RTC, hybrid, FDRE etc.). However, incremental challenges pertaining to wind-site/land availability, grid connectivity, and viability at low tariffs due to elevated capital cost pose challenges for the sector.

Led by India's ambitious clean energy targets declared under NDCs, focus on clean segments such as wind is expected to continue coupled with a healthy pipeline existing in the segment. The government policy to tender 10 GW wind capacity annually till fiscal 2028 will further boost the capacity additions. Central government allocations under relatively strong off-takers such as SECI and PTC, reduce risk and support developer interest. State allocation, on the other hand, has slowed as several states have instead signed power sale agreements (PSAs) with PTC and SECI for procurement of wind power to help fulfil their non-renewable purchase obligation targets.

Crisil Intelligence expects capacity additions of 35-50 GW over fiscals 2027-2031. Crisil Intelligence expects additions of 28-36.5 GW to come from competitively bid wind projects, while remaining 8-13 GW is expected from the open access segment. Mixed resource segment is estimated to see over half of total additions in competitively bid wind segment adding 21.5-25 GW over fiscal 2027-31.

Figure 23: Expected annual wind power capacity additions

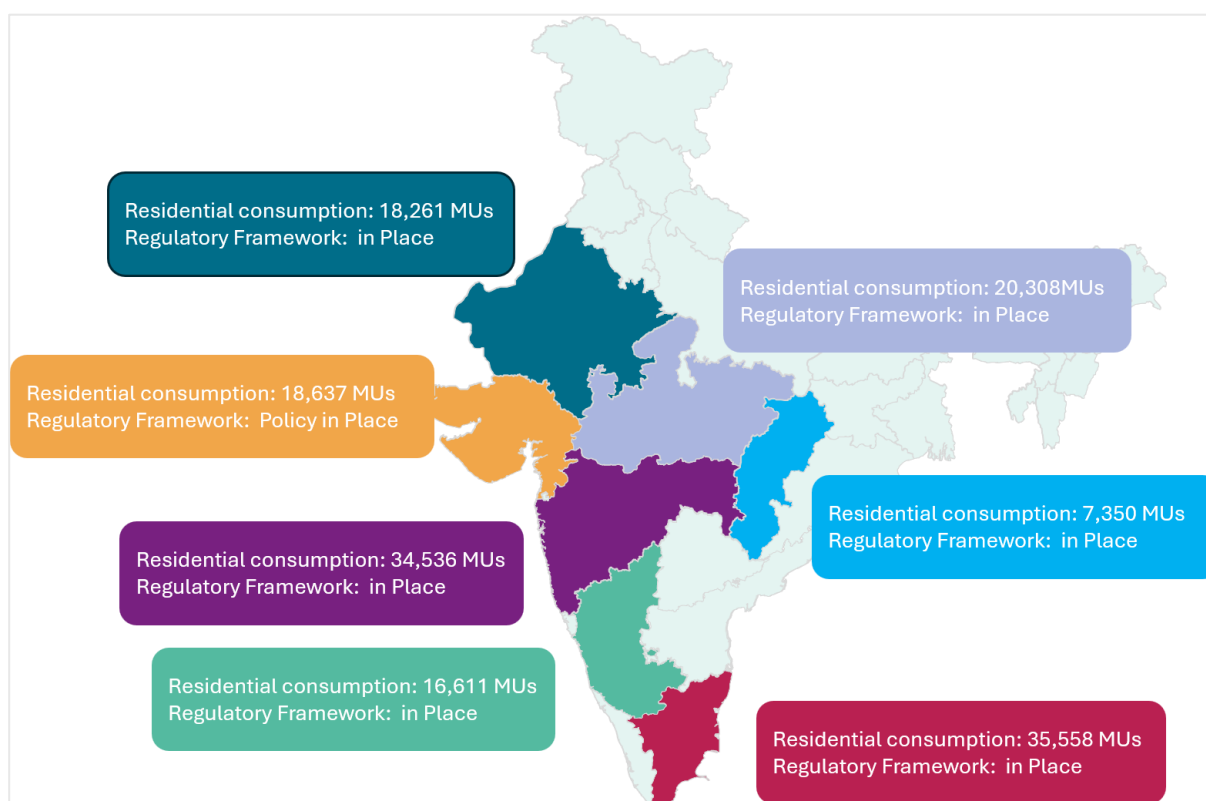


Source: Crisil Intelligence

3.10 Virtual net metering and residential solar rooftops

Virtual Net Metering (VNM) emerging as an adjacent opportunity and important regulatory mechanism for solar adoption among residential consumers, including the one with no access to suitable individual rooftop space, particularly residing in apartments, group housing societies and multi-story residential complexes. Under a VNM arrangement, electricity generated from a common or collectively owned rooftop solar system is exported to the distribution grid, while the corresponding energy credits are allocated across multiple participating electricity connections based on a pre-defined sharing ratio. This enables residents to receive the economic benefits of rooftop solar without requiring separate solar installation for every household. By broadening rooftop solar participation beyond independent houses, this can help address structural constraints such as limited roof availability, fragmented ownership and high-rise urban development.

Maharashtra leads with the highest residential consumption, driven by its large population and extensive urban centers. Tamil Nadu, Gujarat and Karnataka also show substantial residential demand, highlighting their advanced electrification and urban growth. The high and rising residential electricity consumption in key states creates a strong foundation for scaling up virtual net metering. With supportive policies, improved consumer awareness, and declining solar costs, the residential segment is poised to play a pivotal role in India's distributed energy transition.

Figure 24: Residential category consumption for FY2025


Source: India Climate and Energy Dashby by NITI Aayog, Crisil Intelligence

From a regulatory and market development perspective, VNM can complement India's broader push toward distributed renewable energy and residential solar deployment, including initiatives such as the PM Surya Ghar: Muft Bijli Yojana. State electricity regulators and distribution utilities have increasingly incorporated or considered mechanisms such as virtual and group net metering within rooftop solar regulations, although eligibility, capacity limits, credit allocation and settlement arrangements vary across jurisdictions. Wider adoption of VNM can reduce participating consumers' effective electricity expenditure and facilitate aggregation of household demand into larger, potentially more economical installations. This can also spur concepts like P2P trading and UPIization of electricity. Going forward, regulatory standardisation, simplified DISCOM procedures, digital allocation and settlement of energy credits, and broader availability of VNM across states could support residential solar penetration, particularly in India's densely populated urban residential market.

Table 6 : Status of Virtual Net Metering

Sr no	States	Year	Policy provisions	Incentives	Status
1	Maharashtra	2024	- Residential consumers in multi-story buildings/housing societies - Participating consumers can jointly establish common	Open access charges and losses are exempted till the roof top capacity reaches 5 GW	Operational

Sr no	States	Year	Policy provisions	Incentives	Status
			RE generating system		
2	Goa	2019	- Residential consumers, Group housing societies and government establishments - Capacity up to 500 kWp	Banking charge exemption	Operational
3	Telangana	2025	- Residential consumers, Group housing societies and government establishments, - Residential and Government consumers - should not exceed sum of contracted demand - C&I consumers - up to 80% of the sum of contracted demand	NA	Operational
4	Delhi	2019	- Residential consumers, Group housing societies and government establishments - RE generators registered under <i>Mukhya Mantri Kisaan Aay Bhadotari Yojna</i> - The generator should be less than 5 MW however for ground mounted RE systems can be upto 5 x sanctioned load or maximum of 10 MW	NA	Operational
5	Jharkhand	2024	Eligible consumers with no pending arrears	NA	Operational
6	Karnataka	2026	Domestic Consumers including residential apartments, group housing societies, institutions run or managed by charitable Institutions/organisations	NA	Notified/Operational

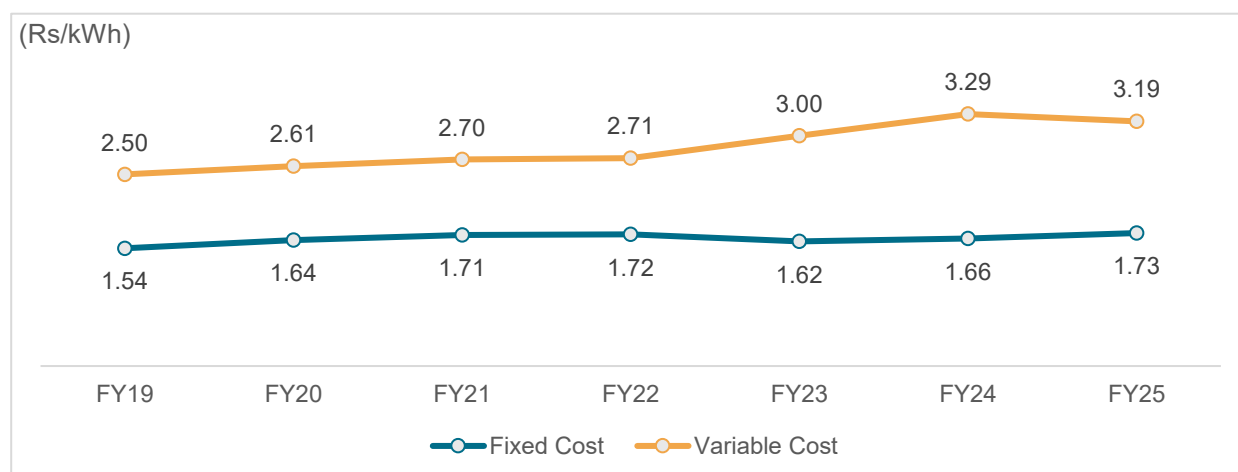
Sr no	States	Year	Policy provisions	Incentives	Status
			- Government buildings including schools, buildings belonging to local authorities - Only DSPV plant can be installed with minimum capacity of 5 kW and maximum shall not exceed the sanctioned load		
7	Chhattisgarh	2022	- Residential consumers, Group housing societies, and establishments of Government/Local Authorities - Any RE generator with minimum capacity of 1 kW and maximum of 500 kW - The capacity installed at eligible consumers premises shall not exceed the sanctioned load	Wheeling, cross subsidy, transmission & distribution and banking charges exemption	Operational

Source: SERC, Crisil Intelligence

3.11 Evolving Tariff Structures and Implications for Open Access

Tariff design for C&I consumers across Indian states has been shifting steadily towards cost recovery through fixed or demand charges, based on sanctioned load or contracted demand, alongside the conventional variable/energy charge linked to actual consumption. This shift allows distribution utilities to recover network, capacity and other fixed costs more reliably. The trend has gained relevance as C&I consumers increasingly turn to rooftop solar, captive generation and open-access procurement, reducing their DISCOM energy offtake while continuing to rely on the distribution network for connectivity, balancing and backup power. A higher fixed-charge share therefore helps utilities stabilise their revenues and offset energy sales from high-paying C&I categories.

Figure 25: Weighted average power purchase cost bifurcation for Discoms



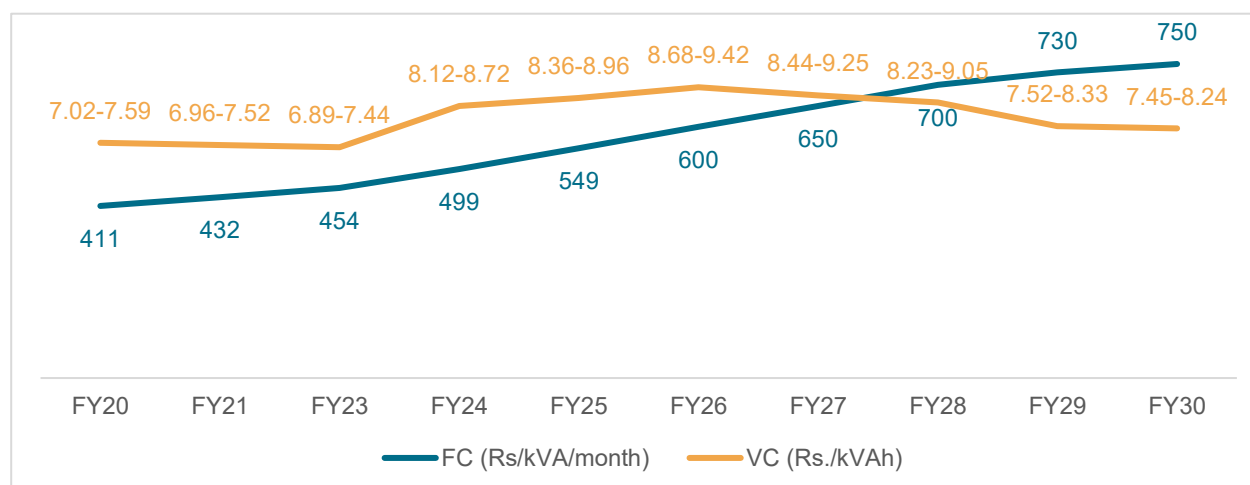
Source: NITI Aayog, Crisil Intelligence

The rising power procurement costs is contributing to an upward trend in the cost structure of Discoms. The chart above indicates that the fixed and variable components of power procurement cost have increased at a CAGR of 2.0% and 4.2%, respectively over fiscal 2019–2025. Power procurement constitutes 75-78% of the cost structure of Discoms. Higher procurement costs, together with transmission, distribution and other system costs, place upward pressure on the Average Cost of Supply (ACoS) and, over time, on retail tariffs, subject to regulatory approvals and subsidy considerations. Higher retail electricity tariffs can improve the relative economic attractiveness of renewable OA; however, the resulting customer savings need to be assessed on a case-to-case basis.

Almost all the Discoms in India use two-part industrial tariffs: an energy charge linked to consumption and a fixed or demand charge linked to sanctioned load, connected load, or billing demand. However, slowly, there has been a shift toward capacity-based recovery, particularly for larger LT and HT industrial consumers. However, this leads to increase in the effective cost per kWh when a manufacturing plant operates at low capacity.

For example, the fixed charge for MSEDCL HT Industrial consumers has increased from Rs. 411/kVA/month to Rs.750 /kVA/month i.e., an increase of 6.9% per annum (CAGR) from fiscal 2020 to fiscal 2030. However, variable component is moving more slowly and in its most recent tariff order the MERC has adopted a decreasing trajectory of variable tariff over the 5-year period. This pattern means the tariff mix is becoming more capacity-cost oriented over time and this makes open access more complex since the consumer pays a larger portion of the bill for being available at a certain capacity level (kVA or billing/contract demand). When OA replaces purchases from the distribution utility with power from a third-party generator/merchant, the consumer may still face charges that are driven by network use and capacity reservation.

Figure 26: Trend of approved HT Industrial category tariffs for MSEDCL



The VC range is for EHV and HT category consumers

Source: MERC, Crisil Intelligence

C&I consumers increasingly seek to reduce their delivered energy costs on overall basis while progressing toward their sustainability and decarbonisation goals. Rather than procuring renewable power or assets through multiple standalone vendors, they are looking for a credible energy partner that can deliver value, reliability, with end-to-end accountability over the life of these assets.

An integrated energy partner can manage the entire renewable energy journey from strategy, technology selection, site identification and approvals, land acquisition, procurement, financing, and execution to lifecycle O&M, asset management and performance optimisation. This enables customers to focus on their core businesses while an experienced specialist manages their energy requirements.

Customers also require long-term guidance on changing regulations, procurement structures, and market conditions, not merely a short-term project solution. Proactive regulatory and commercial support can help protect them from adverse policy changes, minimise compliance risks, and prevent value erosion over the contract or asset lifecycle.

Beyond project delivery, customers increasingly expect their energy partner to provide:

- ongoing asset operations and performance management.
- optimisation of renewable energy procurement and consumption.
- solutions to trade, reallocation, or monetise surplus renewable power, where regulations permit.
- insurance placement, claim support, and risk management.
- direct coordination with OEMs for warranties, spares, and technical issues.
- regulatory, scheduling, forecasting, and compliance assistance; and
- alternative technical or commercial solutions when operating conditions change.

The value proposition is therefore shifting from supplying a renewable energy asset or contract to managing the customer's energy outcome over the long term. A trusted, integrated partner provides a single point of accountability, reduces the need to coordinate with multiple external parties, and helps customers achieve lower costs, greater reliability, and sustained progress on their sustainability ambitions.

Additionally, RE OA consumers incur a separate set of network and regulatory charges, including transmission charges, wheeling charges, CSS (if applicable), banking charges, standby charges and applicable system-operation or scheduling charges. The Green Energy Open Access framework specifically recognises these components and seeks to make them transparent and predictable, while providing certain protections and concessions for green-energy consumers. For example, cross-subsidy surcharge (CSS) for green open access is subject to prescribed limits and cannot be increased during the specified protection period by more than 50% of the surcharge applicable when open access is granted; additional surcharge treatment is also subject to specific exemptions under the Green Energy Open Access Rules. Accordingly, while higher DISCOM procurement costs can support the underlying value proposition of renewable open access, the net customer savings will increasingly depend on the evolution of state-specific open-access charges and the ability to optimise the renewable configuration, captive/open-access structure, banking and energy profile.

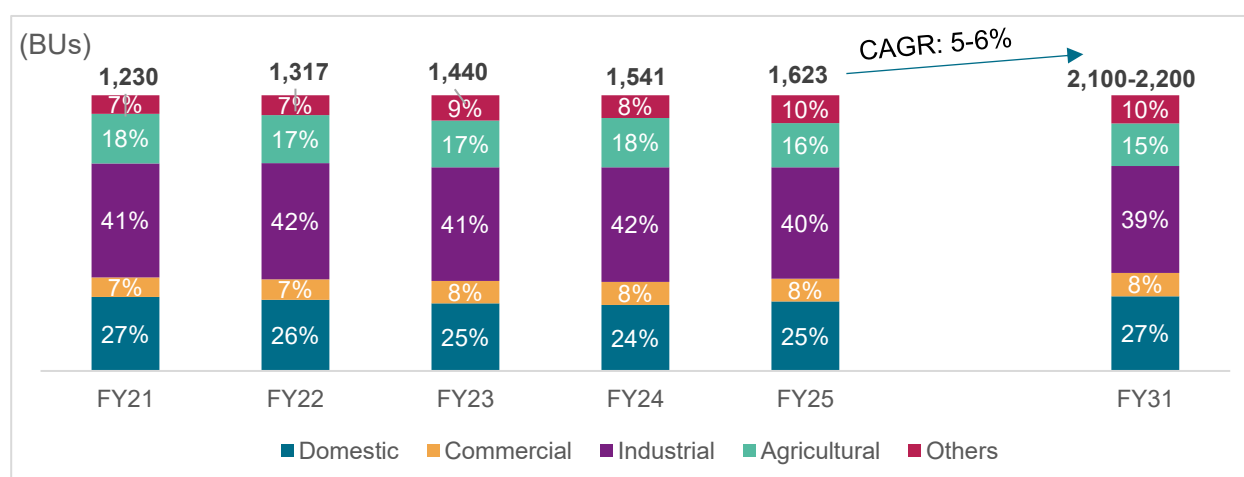
This increases the importance of integrated solution partners that optimise the complete procurement configuration rather than merely offering generation capacity. Such providers combine onsite solar, captive or third-party open access, wind-solar hybrids, storage; size these components against the customer's load profile; and select suitable locations and contractual structures. Their value lies in minimising total delivered cost while managing reliability, regulatory compliance, volume risk and decarbonisation outcomes not simply in supplying the lowest generation tariff.

4 Overview of open access/C&I market

4.1 C&I share in total energy requirement

The growth in energy demand can be attributed to the significant growth in domestic and industrial consumption. Over the period of fiscal 2015 to 2025, domestic electricity consumption grew at a CAGR of 6.5%, from 217 BUs to 409 BUs, while industrial consumption increased at a CAGR of 4.6%, from 418 BUs to 656 BUs. Industrial consumers constitute approximately 83% of the electrical consumption in the C&I segment, which in turn constitutes approximately 49% of total electricity consumption in India for fiscal 2025. This upward trend can be linked to factors such as a growing population, expanding economy, urbanisation, and industrialisation, Make in India and the PLI scheme. The establishment of special economic zones and industrial parks, along with increased access to grid-based electricity in rural areas, has also contributed to the rise in energy consumption. The demand in fiscal 2025 has grown by 7.1% from fiscal 2022 driven by increased industrial production, mining activity, and infrastructure development. Overall energy sales is expected to reach 2,100-2,220 BUs in 2032 i.e., growth at a CAGR of 5-6%. The C&I consumption will be around 47% of total sales.

Figure 27: Consumer segment-wise energy sales in India



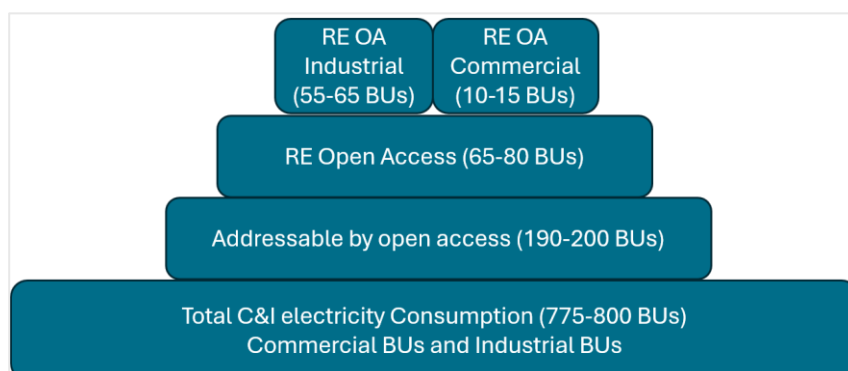
Source: NITI Aayog, Crisil Intelligence

India's C&I segment accounts for approximately 45–50% of the country's electricity consumption, making it the largest addressable market for renewable-energy procurement. Based on recent electricity-sales trends, C&I consumption is estimated at approximately 775–800 TWh in fiscal 2025, comprising:

- Industrial consumption: approximately 650–660 TWh
- Commercial consumption: approximately 130–140 TWh

The annual RE open-access supply to C&I consumers is estimated at approximately 60–80 TWh in fiscal 2025, equivalent to roughly 9–10% of total C&I electricity consumption. In C&I RE Open Access, Industrial category holds around 75-80% share whereas as commercial category share is about 20-25%.

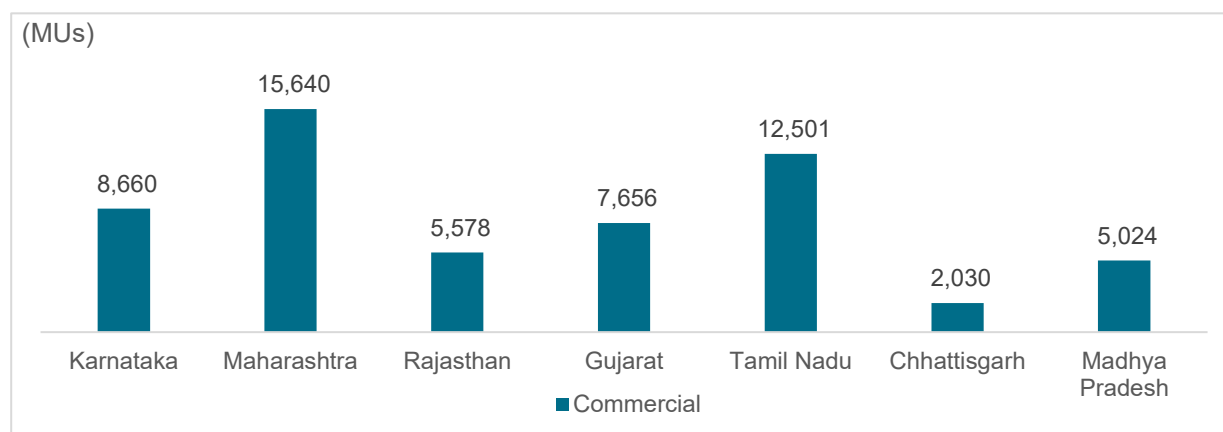
Figure 28: India RE C&I open access market fiscal 2025



Source: Crisil Intelligence

Commercial electricity consumption in India accounts for roughly 8% to 9% of the national total, driven heavily by urbanisation, commercial real estate, malls, and the services sector in major economic hubs.

Figure 29: Commercial category consumption for FY2024



Source: India Climate and Energy Dashby by NITI Aayog, Crisil Intelligence

C&I renewable energy consumption is projected to increase nearly fourfold, rising from approximately 60 TWh in fiscal 2023 to more than 250 TWh by fiscal 2030. This growth is expected to be driven by rising underlying demand and a higher share of renewables in the overall C&I energy mix, supported by greater policy clarity, improved financing availability, pressure on industries to reduce their scope I and Scope II emissions, and increased developer focus on the segment. In India, timely evacuation approvals and land acquisition remain critical to pipeline development, as they enable the infrastructure and site access required for effective project execution.

To assess state-level opportunities more effectively, a structured attractiveness index was developed based on various key parameters. Each parameter was assigned a specific weight, and benchmark criteria were established to evaluate the relative attractiveness of individual states.

- **Total C&I electricity consumption:** This parameter reflects C&I power demand in fiscal 2024. States with higher consumption such as Gujarat (107 BU), Maharashtra (93 BU), and Tamil Nadu (62 BU) offer greater potential for renewable energy adoption due to the scale of their demand, making them inherently attractive markets.

- **Policy and regulatory environment:** Policy stability and ecosystem readiness are critical to execution certainty. Karnataka and Tamil Nadu benefit from mature and stable policy frameworks, while Gujarat is viewed as supportive despite certain constraints, including provisions for electricity banking.
- **RE resource availability:** This parameter evaluates the quality of solar and wind resources, together with land availability. Karnataka, Gujarat, and Rajasthan benefit from strong solar irradiation and wind potential. Maharashtra, despite its excellent renewable resource base, is rated “moderate” because of its relatively high land costs.
- **Presence of Industries:** Maharashtra, Gujarat, Tamil Nadu, and Karnataka are among India’s most industrialised states, supported by strong manufacturing bases, developed infrastructure and established industrial clusters.
- **Savings over C&I category grid tariffs:** This parameter captures the potential cost savings available to C&I consumers from switching to renewable energy relative to prevailing grid tariffs. Maharashtra and Karnataka stand out because of their wide tariff differentials, making them attractive markets for open-access and behind-the-meter solutions.

Gujarat, Karnataka, and Maharashtra are expected to be the three leading states for C&I renewable energy market growth. Across India, renewable energy penetration in the C&I segment stood at approximately 9% in fiscal 2025 and is projected to increase to around 20% by fiscal 2030.

Table 7: State attractiveness Index

State	C&I Consumption	Regulatory environment	Resources	Industry	Savings for C&I	Level playing field	Overall
Karnataka	M/L	M	H	H	H	H	H
Gujarat	H	H	H	H	M	H	H
Maharashtra	H	H	H	H	H	H	H
Tamil Nadu	M	H	H	H	M	H	H
Madhya Pradesh	M/L	L	M/L	L	M	H	M/L
Rajasthan	M/L	L	M/L	L	H	H	M/L

H: High, M: Moderate, M/L: Moderate to low, L: Low

Source: Industry, CEA, Respective State Discoms and SERCs, Crisil Intelligence

Certain states are structurally more attractive than others, and suppliers that understand these differences and establish an early presence are better positioned to lead those markets.

C&I consumers increasingly seek more than low tariffs; they prioritise predictability, flexibility, and reliability in energy procurement. A key expectation is that contracted renewable energy tariffs remain below grid tariffs throughout the PPA tenure, thereby delivering sustained savings. Consumers also prefer developers with strong financial credentials, proven execution capabilities, and local market experience, as these factors provide greater confidence in timely project delivery and continuity of supply.

Minimising operational and compliance risks is equally important. Consumers expect adherence to central and state regulations, reliable plant performance, and minimal disruption to business operations.

Buyers are also favouring developers that offer a broad portfolio of technologies including solar, wind, and hybrid systems and procurement structures such as on-site, group-captive, third-party open-access, and virtual PPAs. This breadth enables companies to address their decarbonisation objectives more comprehensively.

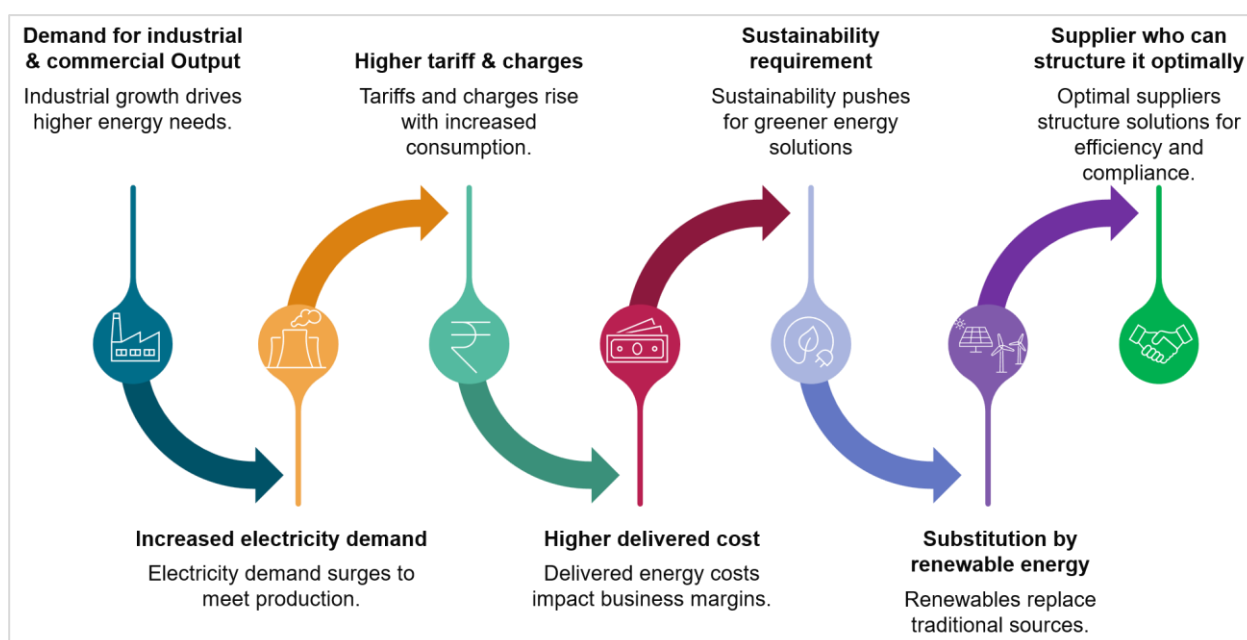
Contractual flexibility is another critical consideration, including adaptable tenures, exit provisions, and mechanisms to accommodate mergers and acquisitions or regulatory changes. Consumers also prefer early commencement of supply typically within six to nine months for open-access projects giving an

advantage to developers with shovel-ready capacity and the necessary land, connectivity, and regulatory approvals already in place.

4.2 Transition to Renewable power for C&I

Growth in C&I RE demand is closely linked to the expansion of industrial and commercial activity, which drives higher electricity consumption. As power demand increases, businesses face rising electricity expenditures due to the escalation of grid tariffs, open access charges, and other power procurement costs. This creates increasing pressure on delivered electricity costs, particularly for energy-intensive industries seeking to protect margins and improve cost competitiveness. In response, many C&I consumers are turning to RE solutions, such as open access and rooftop solar, to secure lower and more predictable power costs while advancing sustainability objectives. As RE procurement structures become more sophisticated, there is growing demand for integrated solution providers that can design, develop, finance, and manage customised renewable energy portfolios aligned with customers' operational and regulatory requirements.

Figure 30: RE demand creation for C&I consumers



RE is emerging as a key driver for industrial competitiveness, enabled by evolving OA and power-market reforms

Energy is a critical input to manufacturing competitiveness, influencing the cost structure, operational reliability and productivity of industrial enterprises. For energy intensive industries, the availability of reliable and competitively priced electricity can have a direct bearing on production economics, capacity utilisation and long-term investment decisions. As Indian manufacturing expands and businesses increasingly seek to improve cost efficiency and reduce carbon intensity, RE is emerging as an important component of the industrial energy mix, offering opportunities to manage electricity costs while supporting decarbonisation objectives.

India's evolving power market framework has progressively enabled greater participation of C&I consumers in renewable energy procurement. The expansion of open access increased regulatory focus on enabling competitive power procurement and reforms aimed at improving the ease and predictability of interstate

and intrastate renewable power transactions have reduced some of the structural barriers to C&I renewable adoption. The policy direction has increasingly shifted towards enabling consumers to access RE power through more flexible procurement and contracting mechanisms, thereby supporting the transition from conventional grid-supplied electricity to a more diversified and RE-intensive power portfolio. This evolving regulatory environment, together with growing corporate sustainability requirements and the increasing economic attractiveness of RE power is expected to support continued growth in C&I RE adoption.

Transition to RE power is shifting the challenge from simply procuring electricity to optimising the delivered cost, reliability and flexibility of electricity

For C&I consumers, renewable energy offers the potential to reduce delivered electricity costs while advancing sustainability goals. However, competitive headline renewable tariffs alone do not guarantee project viability or customer savings. Timely access to suitable land, grid connectivity and evacuation infrastructure along with effective management of regulatory approvals, supply chains, construction and commissioning determines whether contracted capacity becomes operational on schedule and performs as expected. Unlike conventional DISCOM supply, where the utility manages generation, transmission, distribution, scheduling and balancing, renewable-energy open-access procurement requires consumers or their solution providers to manage a considerably more complex power-procurement architecture.

Accordingly, delivered-power economics can differ materially from the quoted renewable-generation tariff. Actual savings depend on transmission and wheeling charges, open-access surcharges, banking provisions, scheduling and deviation costs, energy losses, residual DISCOM procurement, and the alignment of variable solar and wind generation with the consumer's hourly load. Reliability may also require hybridisation, storage or grid backup, while state-specific regulations and captive or group-captive compliance can materially affect project economics.

This increasing complexity creates a clear need for developers that can provide an integrated, lifecycle solution encompassing project development and sourcing, procurement, regulatory management, execution, energy optimisation, hybridisation and storage, portfolio balancing, and ongoing asset management rather than merely supplying renewable power.

C&I open access enables higher-value, flexible RE solutions through direct customer relationships and tailored power procurement

Supply to C&I consumers through open access segment has distinct advantages over the utility segment. C&I open access offers players direct access to creditworthy customers, potential higher realisations and greater flexibility in structuring long-term contracts compared with competitively bid utility-scale projects. C&I Consumers also get the required RE to suit demand profiles at competitive rates, reducing power purchase costs.

Some of the benefits are summarised below:

Figure 31: Advantages of C&I segment over the Utility segment

	C&I segment	Utility segment
 Electricity consumer	Electricity sale to C&I consumers under open access or group captive route with tariffs linked to industrial tariffs	Power supply to state distribution utilities / REIA structured on a fixed tariff with a term of 13 to 25 years
 Headroom for growth	Commercially driven and planned, not dependent on government auction targets	Growth is dependent on auction wins
 Technology	C&I can optimally utilize excess battery combined with WSH projects to provide firm supply and additional capacity	Capacity is fixed based on long term Power Purchase Agreements ("PPA")
 Pricing power	Dependent on discom's industrial tariff with the option to pass 50% to 100% of the variation depending on type of contract	Bid based fixed tariff
 Customer diversification	Contract with multiple and diverse set of offtakers	Contract with a single offtaker for each project
 PPA terms	C&I players have an option to not lock in tariffs or shorter lock in periods of 3 to 5 years	Bid based PPA's lock tariffs for long term (mostly for 25 years)

Source: Industry, Crisil Intelligence

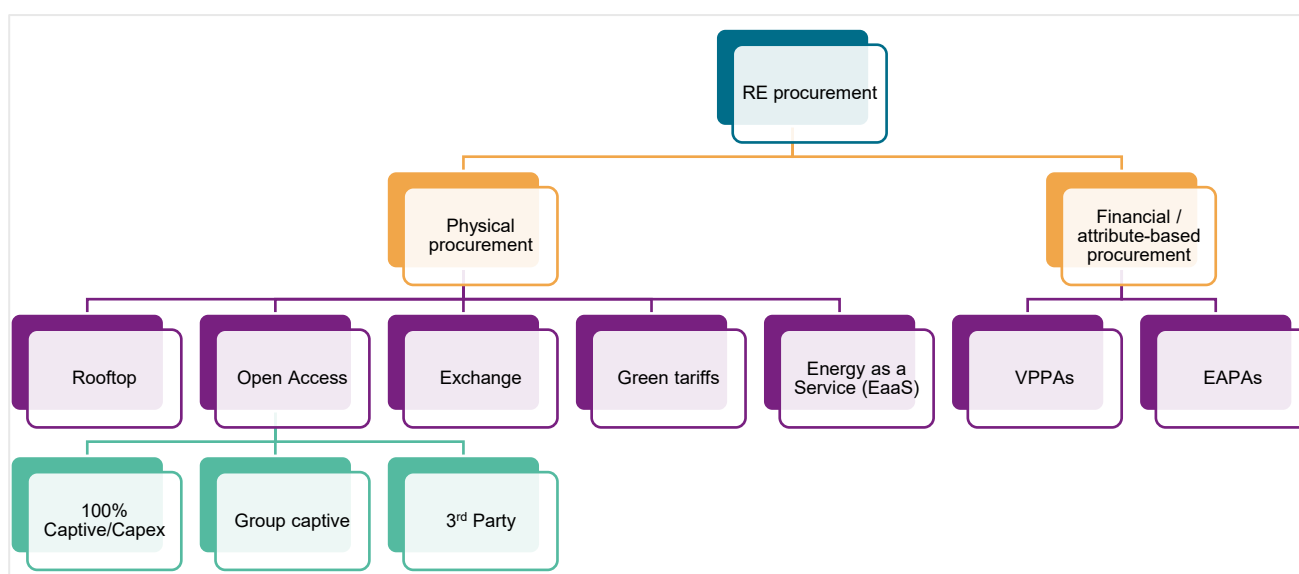
- **Reduction in electricity costs:** RE power, particularly solar and wind, has become cost competitive with conventional grid electricity. C&I consumers can potentially reduce their landed power cost by procuring RE through open access, captive/group-captive models or long-term PPAs.
- **Rising industrial electricity demand:** India's expanding manufacturing, infrastructure, data center, logistics and commercial sectors are increasing electricity consumption. C&I consumers therefore represent a significant and growing source of demand for RE power.
- **Decarbonisation and net-zero commitments:** Large corporates are adopting net-zero and science-based emissions reduction targets. Procuring RE power enables C&I consumers to reduce Scope 2 emissions associated with purchased electricity.
- **Regulatory support:** Policy initiatives such as RPOs and the GEOA have created a more supportive framework for C&I consumers seeking to procure RE electricity.
- **Energy security:** Captive and distributed renewable generation can reduce dependence on conventional grid power and provide consumers with greater control over their electricity sourcing.
- **Technological advancements:** The declining costs of solar modules, wind turbines, inverters and BESS are improving the economics of RE power and enabling more reliable renewable electricity supply.
- **Shift towards hybrid and FDRE:** Wind-solar hybrid projects and storage backed RE solutions can improve the generation profiles and reduce intermittency, making the RE power more suitable for C&I consumers with high reliability and RTC power requirements.

- **Increasing preference for predictable and clean power:** C&I consumers are moving beyond conventional electricity procurement towards clean, predictable and cost-competitive power, creating a structural opportunity for RE developers and power suppliers.

4.3 Types of OA models and procurement routes for C&I

Indian corporates and large consumers typically procure renewable energy through two broad routes: onsite procurement and offsite procurement. Rooftop or onsite solar/wind is a procurement where RE generation source is located near close proximity of the end use. Open Access is an offsite procurement mechanism where power is generated at a remote renewable energy project and supplied to the consumer through the transmission and distribution network. The choice of RE procurement route depends on load size, location, consumption profile, capital availability, risk appetite, and sustainability goals.

Figure 32: Various options available for RE procurement to C&I segment



Source: Crisil Intelligence

Rooftop or behind the meter or onsite solar is typically the first RE procurement route considered by corporates due to its low cost, scale and asset control. It is generally followed by the OA route to reduce power procurement costs and/or achieve sustainability goals.

4.3.1 Rooftop or onsite solar

Rooftop and onsite ground-mounted solar installations are a preferred RE option for C&I consumers with adequate space, offering 10-15% RE offset without the need for external land or transmission infrastructure. Adoption is supported by simple regulatory processes and proven implementation models across states, with metering structures such as net metering, gross metering or behind-the-meter systems.

C&I consumers typically choose between two models:

- **Capex model (dominant):** Consumers invest upfront and benefit from lower tariffs due to GST credits and accelerated depreciation, but bear performance and maintenance risk
- **Opex/RESCO model:** Popular with large consumers zero capex, risk transferred to the developer and assured savings, but limited by the need for creditworthy, large-scale off-takers

4.3.2 Open access

OA allows consumers to procure power directly from any power generator across the country. The route is suitable for consumers looking to procure a higher quantum of renewable power as there are no space constraints. Consumers may install renewable power projects themselves on a 100% captive basis, but procurement from IPPs under long-term 'group captive' PPAs is the most popular option.

Table 8: Key features of different OA business models

Particulars	100% Captive (Entire capex funding by consumer)	'Group' captive	Third-party sale
Key features	- Customer owns 100%, developer builds and operates - Consumer makes the entire upfront capital investment	- Customer owns minimum 26%, developer owning 74% or lower builds and operates - A third-party IPP may own the remaining equity and take all responsibility for developing and operating the power plant; consumers buy power from IPP as per a mutually agreed PPA	- Owned, built and operated 100% by developer - Consumers buy power from IPPs under a mutually agreed PPA
Benefits	- Streamlined project approval process - Competitive PPA price offered under the group captive model - Exemption from CSS and AS accounting for about Rs 1.30-2.50/kWh under the group captive model - Tax incentives available under 100% captive model - Very high EBITDA specifically in 100% captive / capex model.		- No upfront capital needed - No annual compliance requirements
Disadvantages	- Upfront capital investment required - Consumer(s) forego profits or dividends, which are adjusted in tariff - Annual compliance requirements to prove captive status		- Higher PPA price - Higher landed cost due to applicability of OA charges

Source: Crisil Intelligence

The MoP issued final green open rules in fiscal 2023, which are desirable for consumers, and most of the major states have already adopted these. The new rules allow small consumers with sanctioned loads over 100 kW to be eligible for OA with banking provisions subject to an annual cap of 30% of power consumed from the local discom. The rules propose that OA applications be approved within 15 days and that a uniform methodology for calculating OA charges be followed across all states. This is expected to propel capacity additions under this route over the medium to long term.

4.3.3 Point of connectivity including ISTS-connected projects

Open access projects can be connected at distribution level (11 / 33 KV) or at transmission level @ 66 KV or above on intrastate basis or at CTU or ISTS-connected projects on interstate basis. Each nature of connection carries its own advantages and disadvantages and is determined by the size of the project, nature and quantum of customer requirement. ISTS projects enable developers to support pan-India operations for corporate customers in multiple ways. Customers with multiple facilities across states can aggregate demand to purchase electricity under a combined PPA; customers with facilities in states where STU-connected projects are not viable or feasible can access power through CTU-connected projects; and customers that wish to contract for energy attributes through Environmental Attributes Purchase Agreements (EAPAs) without purchasing physical electricity. EAPAs are contracts that require developers to unbundle energy attributes and sell them to C&I consumers, often registered as I-RECs. The electricity generated from these projects is sold either on power exchanges in India or to other brown power PPA consumers.

There are two key categories of C&I customers that benefit from ISTS power:

1. Large manufacturing facilities in states without adequate renewable potential or high intrastate OA costs or policy landscape (e.g., Jharkhand, Odisha, UP, WB and Northeast)
2. Data center and technology companies are purchasing environmental attributes generated from green power in India to meet RE commitments for their operations in India and the broader Asia-Pacific region

The OA market has historically been limited to intra-state projects, where both the renewable project and the consumer are located within the same state. However, the market has been exploring inter-state projects given ISTS waiver. However, such ISTS waiver is on the way out with 25% reduction starting July 1, 2025, and a full phase-out by fiscal 2028 for OA projects, which will typically load Rs 0.50-0.80/kWh for a 25-year period for projects commissioned by June 2025 and more than Rs. 1.5 per unit on full phase out of ISTS waiver.

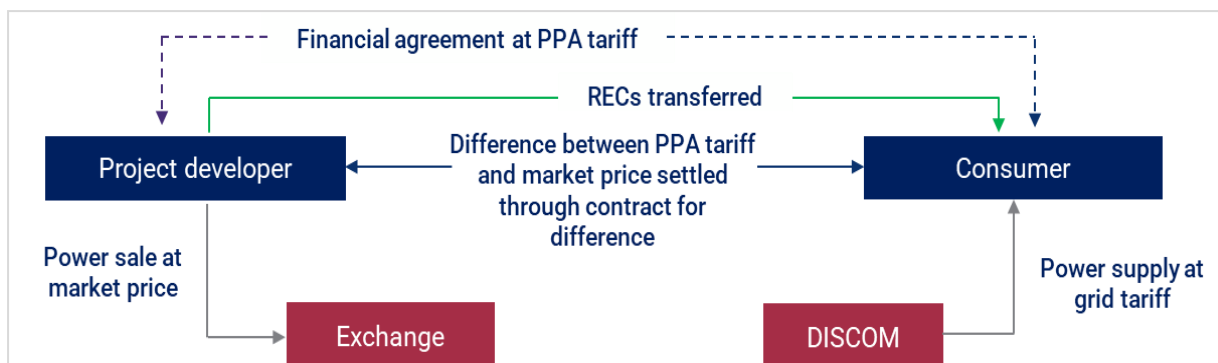
4.3.4 Virtual power purchase agreements

VPPAs are more flexible in their structure than physical PPAs as the corporate buyer and power producer do not have to be connected to the same electricity network. It is a contract for difference. In this mechanism, RE developers sell their power in wholesale market while transferring the green attributes of power to VPPA consumers. In return, consumers guarantee a fixed price (strike price) with positive and negative adjustments based on difference of open market price vs. strike price. The entire transaction is financial and would not require actual physical transfer of power from generator to consumer.

In October 2021, the Supreme Court passed the order wherein CERC was granted authority over all physical delivery-based forward contracts, while the regulation of financial derivatives was entrusted to SEBI. Consequently, any VPPA will be subject to joint regulation by both CERC and SEBI. In January 2025, SEBI stated that the regulatory authority for non-tradable and non-transferable VPPA contracts lies entirely with CERC, which led the MoP to instruct CERC to develop the new VPPA guidelines. As a result, CERC issued guidelines for VPPA agreements in December 2025, which provides clarity on regulatory approvals, implementation process and price settlements. It is a significant milestone for clean energy transitions for C&I consumers in India.

The investors, power producers, and consumers could now explore the benefits of such innovative power purchase mechanisms, leading to potential advancements in the adoption of RE sources, which is expected to further boost demand for corporate PPAs through the open access route.

Figure 33: VPPA transaction structure



Source: Crisil intelligence

VPPAs are emerging as a commercially alternate option specifically for data centers.

4.3.5 Power exchanges

The power exchange market in India, specifically the Green Term-Ahead Market (G-TAM) and Green Day-Ahead Market (G-DAM), provides consumers with opportunities to buy and sell renewable power on a short-term basis. G-TAM allows bilateral purchase of power for delivery duration up to one week, with options for same-day, next-day, next-week, or next 2-10 days contracts. In contrast, G-DAM enables trading of power through a closed online auction process for delivery on the next day. Green attributes are embedded with traded power to avoid double counting, ensuring the integrity of the market.

Indian Energy Exchange (IEX) achieved its highest-ever electricity traded volume of 141 BU in fiscal 2026, representing a 17% year-on-year increase. RTM posted growth of 41% YoY, while the green market grew by 23% YoY. IEX also recorded its highest-ever annual trade of RECs in fiscal 2026. A total of 187.20 lakh RECs were traded during the year, up 5% year-on-year.

During fiscal 2026, improved wind, hydro, and solar generation along with sustained supply from coal-based generation boosted supply liquidity on the exchange platform. This contributed to a significant decline in both DAM and RTM prices. For fiscal 2026, the Day-Ahead Market (DAM) Market Clearing Price (MCP) stood at Rs 3.86/unit, down 13.7% from fiscal 2025. In the Real-Time Market (RTM), the MCP was Rs 3.59/unit in fiscal 2026, declining 16% compared to fiscal 2025.

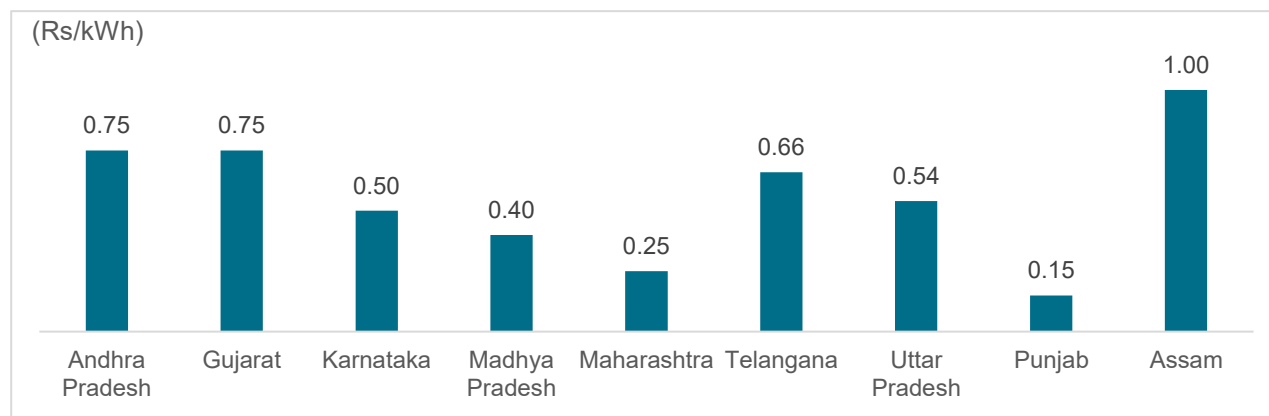
4.3.6 Green tariffs

A green tariff is a special power tariff determined by state regulators for consumers that want to purchase renewable power. It is a simple procurement route that does not require any changes in physical procurement, metering or infrastructure, making it attractive to consumers that are averse to relatively complex procurement options such as OA. All consumers are eligible to procure power from the local discom at green tariffs, but it entails an extra cost (over and above the respective consumer category grid tariff), which is a deterrent for many.

In line with the rules, the MoP directed SERCs to take necessary measures for the determination of green tariffs and clarified that the green tariff should not exceed the average power purchase cost of RE, including a surcharge equivalent to 20% of the average cost of supply, along with a margin of 25 paise. Majority of the states have notified green tariffs for their states which are over and above the normal tariff applicable to the consumer category. States like Andhra Pradesh and Gujarat have made the tariff applicable to all consumer categories. Tamil Nadu, Karnataka has determined green tariff only for willing HT consumers.

Despite their operational simplicity, the uptake of green tariffs has been poor due to the high cost, which is set at a premium of 5-10% over grid tariffs. Overall, the green tariff has remained a non-starter due to its high cost and lack of clarity over green attributes.

Figure 34: Prevailing green tariff premium in key states



Note: This is the premium which is to be paid over and above the prevailing tariffs for the category

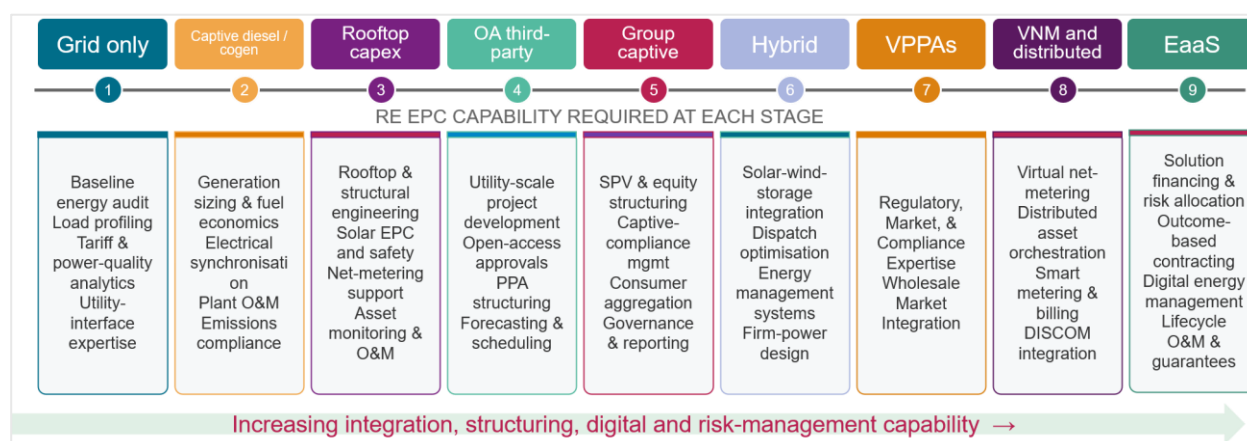
Source: Tariff Orders for FY2026-27 of respective SERCs, Crisil Intelligence

4.3.7 Energy-as-a-Service (EaaS)

Energy-as-a-Service (EaaS) is an emerging commercial model under which a customer procures an energy outcome or integrated energy solution rather than independently investing in and managing the underlying energy assets and infrastructure. Depending on the customer's load profile, regulatory environment and commercial requirements, the service provider may design, finance, develop, own, operate and optimise a combination of onsite or offsite renewable generation, grid power, energy storage and other energy-management solutions. The customer typically pays an agreed energy or service charge over the contract tenure, while the service provider assumes defined responsibilities relating to asset performance, operations and optimisation. Compared with a conventional renewable power PPA, an EaaS model can have a broader scope, with the objective of optimising the customer's overall delivered electricity cost, reliability and renewable-energy utilisation rather than merely supplying renewable electricity from a specified asset. Further an EaaS model blends the advantages of self-owned capex and group captive model and therefore making it financially more attractive without exposing the customer to additional risks.

For C&I consumers, EaaS can therefore provide an alternative to asset ownership or standalone power procurement, particularly where the customer seeks to reduce upfront capital deployment and outsource the complexity associated with technology selection, project development, regulatory compliance, operations and ongoing energy optimisation. The applicability and structure of an EaaS arrangement, however, would depend on the regulatory framework, customer load profile, financing structure and risk allocation agreed between the customer and service provider.

To summarise the evolution of C&I power procurement, it reflects a gradual shift from purchasing electricity as a commodity to managing energy as a strategic input. Historically, C&I consumers relied almost entirely on grid electricity, supplementing supply with captive generation for reliability. As RE costs declined and regulations evolved, procurement models expanded from self-owned assets to increasingly sophisticated structures involving third-party ownership, open access, hybrid solutions, and energy-as-a-service offerings. This evolution has increased the need for specialised partners capable of structuring, developing, financing, owning, operating, and optimising RE solutions throughout their lifecycle.

Figure 35: Procurement evolution timeline


Source: Industry, Crisil Intelligence

The journey from grid-only supply to Energy-as-a-Service (EaaS) reflects a shift from simple electricity procurement toward integrated, low-carbon, digitally managed, and performance-based energy solutions. Across the journey, the RE EPC player evolves from a technical contractor into an integrated energy-solutions provider. Its capabilities expand from audits and equipment installation to regulatory structuring, consumer aggregation, multi-technology optimisation, market-risk management, digital orchestration, financing, and long-term outcome guarantees.

Table 9: Summary of different procurement routes

Parameters	Rooftop	100% Captive	Group captive	3rd Party	Exchange	Green tariffs	VPPAs	EAPAs	EaaS
Procurement	Physical	Physical	Physical	Physical	Physical	Physical	Financial	Attributes	Energy outcome or service
Ownership	Consumer and/or developer	Consumer	Group of consumers and developer	Developer	Not Applicable	Not Applicable	Developer	Not Applicable	EaaS provider
Balance sheet treatment	On balance sheet	On balance sheet	Generally, off-balance sheet through SPV investment	off balance sheet	off balance sheet	off balance sheet	off balance sheet	off balance sheet	off balance sheet (depending on contract)
Onsite vs offsite	Onsite	Onsite or offsite	Offsite	Offsite	Offsite	Offsite	Offsite	Offsite	Onsite or offsite
Capex/Opex	CAPEX; RESCO/PPA: OPEX	Capex	Equity by consumer and LT PPA	Opex (LT PPA)	Opex	Opex	Opex	Opex	Opex
Contract form	Asset purchase / EPC contract	Asset purchase / EPC contract	PPA through SPV	Physical PPA	Spot or term market contract	Utility green tariff	Financial contract	Financial contract	Energy Service Agreement
Tenure	20–25 years	20–25 years	5-15 years	5-15 years	Short-term or spot	NA	10-20 years	10-20 years	5-15 years
Consumer Profile	Strong daytime demand	Large C&I with table, predictable demand	Multiple C&I users	Multiple C&I use	Consumers with variable demand, short-term requirements	Companies prioritising simplicity	Large corporates	Corporates seeking Scope 2 claims	Large C&I with table, predictable demand

Parameters	Rooftop	100% Captive	Group captive	3rd Party	Exchange	Green tariffs	VPPAs	EAPAs	EaaS
Cost competitiveness	High	Very high	Very high	Medium	Variable	Low	Uncertain	Cost-effective	High
Reliability & flexibility	Medium	High	High	High	High	Medium	High	High	High
Performance risk	Borne by customer/owner	Borne by customer/owner	Shared by SPV and consumers	Primarily borne by supplier	Customer bears market-price risk	Primarily borne by utility	Supplier/project performance and settlement risk	Supplier/project performance and settlement risk	Service provider
Regulatory & execution complexity	Medium	High	Very high	High	Medium	Very low	High	Medium	High
Typical suppliers	EPC contractors and solar OEMs	EPC contractors and solar/wind OEMs	RE developers and aggregators	RE developers	Power exchanges	Utilities	RE developers	RE developers	ESCOs, RE developers

Source: Industry, Crisil Intelligence

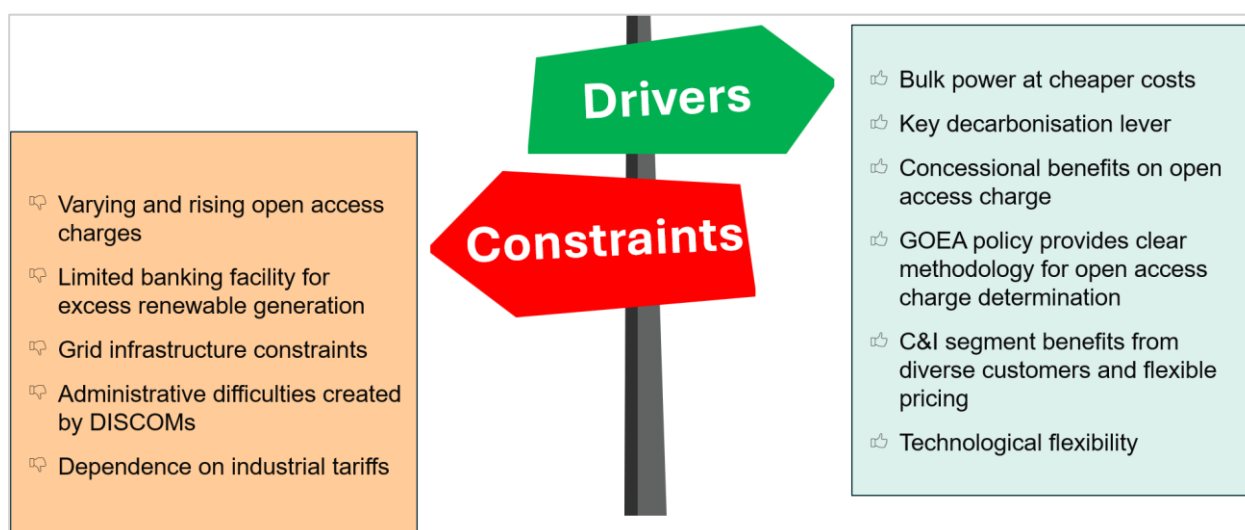
C&I RE procurement in India is evolving from simple, onsite rooftop solar installations toward more sophisticated, portfolio-based solutions. While rooftop solar remains attractive for reducing daytime grid consumption, its scale is constrained by roof availability and does not fully address round-the-clock demand.

Large consumers are therefore moving to offsite open-access models, including captive, group-captive and third-party PPAs, to access utility-scale renewables and achieve greater cost savings and decarbonisation. Procurement is also expanding beyond single-source solar to wind-solar hybrids, interstate supply and storage-backed solutions, which provide better alignment with industrial load profiles, reduce intermittency and support higher renewable penetration.

At the same time, C&I buyers increasingly use power exchanges, green tariffs and financial or attribute-based instruments such as VPPAs and EAPAs to address residual demand, dispersed sites and markets where physical open access is constrained. Consequently, procurement is shifting from a single-project decision to an integrated energy-portfolio strategy that balances cost, reliability, flexibility, regulatory risk and credible emissions reduction.

4.4 Drivers and constraints of C&I/open access market

Figure 36: Key drivers and constraints of OA Market



Source: Crisil Intelligence

Drivers:

- **Bulk power at cheaper costs:** One of the main advantages and driving forces behind open access is the significant reduction in energy costs. Open access solar power offers a more economical alternative to purchasing electricity from DISCOMs, making it an attractive option for industrial consumers. The cost of industrial tariffs from DISCOMs can be substantial, ranging from Rs 6-8 per kilowatt-hour (kWh), whereas open access solar power can be generated at a significantly lower cost of Rs 3-4 per kWh, excluding grid charges. This provides industrial consumers with the benefit of bulk energy consumption at affordable rates, making open access solar power a highly cost-effective solution.
- **Key decarbonisation lever:** For corporations, especially those dedicated to decarbonisation, reducing emissions is a top priority, and renewable energy has emerged as a readily available and effective solution. As a commercially and technologically mature option, renewable energy offers a straightforward way for companies to significantly decrease their Scope 1 and 2 emissions, making it an attractive lever for those seeking to minimise their environmental footprint. By adopting renewable energy, corporates can achieve substantial emissions reductions, bringing them closer to their decarbonisation goals.
- **Concessional benefits on open access charge:** Some states provide concessions on grid charges and captive consumers can avail the benefit of exemption from cross-subsidy surcharge and additional surcharge and third-party consumers can also avail benefit of additional surcharge waiver, making it attractive for industries to shift to open access solar. These benefits reduce the overall cost of electricity procurement compared to purchasing from DISCOMs.
- **GOEA policy provides clear methodology for open access charge determination:** The GOEA policy aims to streamline the calculation of charges for open access consumers, ensuring predictability and transparency. This helps industries and commercial establishments make informed financial decisions while transitioning to renewable energy.
- **C&I segment benefits from diverse customers and flexible pricing** - Unlike the utility-scale segment, where energy is sold to a single off-taker, open access projects can have multiple industrial and commercial customers. This diversifies revenue streams and reduces financial risks for developers. C&I players also have the option to choose between different pricing models, such

as short-term PPAs (3-5 years) or tariffs linked to industrial rates, rather than being locked into fixed long-term rates.

- **Technological flexibility:** Open access solar projects can incorporate battery storage or hybrid solutions (e.g., solar + wind) to enhance power reliability. Industries can optimise their power consumption by storing excess energy for later use, reducing dependence on the grid during peak hours. Waste heat recovery systems (WSH projects) can be integrated with solar for additional efficiency gains.

Constraints:

- **Varying and rising open access charges:** While open access solar is cost-effective, state regulations frequently change, leading to uncertainty in charges. Grid charges are revised annually which can reduce potential savings and impact project viability. Different OA charges for different states can be a constraint.
- **Limited banking facility for excess renewable generation:** In most states, there is a limitation on storing excess solar power generated during the day in the grid for later use, with restrictions on the duration for which it can be stored or total quantum of power that can be stored. This constraint compels consumers to either utilise excess power immediately or relinquish it, resulting in inefficiencies and potential losses. Furthermore, some states that previously offered banking facilities, which allowed excess energy to be stored and used later, have either modified, or withdrawn these provisions. This change has retroactive implications, affecting older projects that were established with the assumption of longer banking periods, thereby disrupting their operations and financial viability.
- **Grid infrastructure constraints:** Availability of transmission corridors is a major barrier, as open access projects need to use the existing grid to transport power. Congestion in T&D networks, particularly in states with high renewable penetration, can lead to curtailment of power generation.
- **Administrative difficulties created by DISCOMs:** State DISCOMs, fearing revenue loss from high-paying industrial customers, often create bureaucratic hurdles. These include delays in granting open access approvals, refusal to sign necessary agreements, and imposing restrictive policies.
- **Dependence on industrial tariffs:** Open access solar pricing is often linked to industrial electricity tariffs, which are volatile. If industrial tariffs fall due to government interventions, open access projects may become less competitive.

4.5 Policies and regulations driving growth

4.5.1 Captive/ group captive policies

As per electricity Rules, 2005 a power plant could qualify as a captive generating plant (CGP) if the captive user holds at least 26% ownership. A group captive mode is where a developer develops a power plant for collective usage of many consumers. A power project is considered 'captive' if the consuming entity or entities consume at least 51% of the power generated and owns at least 26% of the equity.

On March 13, 2026, the Ministry of Power, Government of India, notified amendments to Rule 3 (Amendment) of the Electricity Rules, 2005. These amendments significantly ease compliance and resolve structural ambiguities for CGPs and group captive renewable projects. They modernise captive regulations for C&I consumers.

The amendment upholds the "twin qualifying criteria" (26% ownership and 51% consumption) under Rule 3 of the Electricity Rules, 2005.

- Captive users now explicitly include end-users consuming electricity directly or through energy storage systems.
- A captive user, its subsidiaries, and its holding company (plus other subsidiaries) are treated as a single entity, mitigating risks if a group member under-consumes
- In case of a power plant set up by a registered co-operative society, the eligibility conditions shall be satisfied collectively by the members of the co-operative society
- Captive user(s) must ensure that the eligibility conditions are complied with during the financial year. If minimum captive consumption requirement is not met during any period, the entire electricity generated shall be treated as supply of electricity by a generating company and cross-subsidy surcharge and additional surcharge shall be levied on such consumption:
- Users can draw up to 100% of their proportional consumption share, and compliance can be assessed for specific units identified for captive use rather than the entire station.
- Where the ownership pattern of the power plant varies during the financial year, the proportionate consumption of each captive user shall be determined on the basis of the weighted average shareholding of such captive user during the financial year
- In case a power plant set up by an association of persons, any consumption by an individual captive user in excess of 100% of its proportionate consumption shall be treated as supply of electricity by a generating company and cross-subsidy surcharge and additional surcharge shall be levied on such excess consumption.
- Verification to be conducted by state-designated nodal agencies for intra-state projects and by the NLDC for inter-state projects.
- An appeal against the verification shall lie before a Grievance Redressal Committee constituted by the Appropriate Government.
- Cross-subsidy and additional surcharges are suspended while captive status is pending verification.

As far as benefits for captive users are concerned, as per section 42(2) of the Electricity Act, 2003, captive power plants, set up by end-users for their consumption, are exempted from payment of cross-subsidy surcharge and the additional surcharges, otherwise would be applicable for open access consumers. Thus, the central government's policy support has played a role in promoting the captive/group captive model, especially for C&I consumers.

4.5.2 Introduction of Green Energy Open Access rules

The Electricity (Promoting Renewable Energy through Green Energy Open Access) Rules, 2022 allow consumers to procure green power through open access for contract demand of 100 kW or above and provide no limit for captive consumers. Multiple consumers together with a total contract load of 100 kW are also eligible for open access, making access to OA easier for smaller consumers. As open access allows bulk procurement at low cost, open access has become one of the most popular routes among corporates.

Key provisions include:

- Cross-subsidy surcharge cap: cannot increase by more than 50% for 12 years from project commissioning.
- Additional surcharge: full waiver where fixed charges are being paid by the consumer.
- Banking: Banked quantum up to 30% of monthly consumption for DISCOM, permitted monthly payment of banking charges at 8% of banked energy; unutilised banked energy lapses at end of each cycle; generators receive RECs to the extent of lapsed energy.
- Discom procurement option: consumers may requisition green energy from their discom.
- Offshore wind: projects commissioned up to December 2032 will not attract additional surcharge for supply to open-access consumers.
- Deemed approval: if an open access application is not approved within 15 days, it is deemed approved.

These provisions improve long-term certainty on charges and reduce approval delays, supporting higher C&I uptake. However, outcomes remain dependent on discom cooperation, regulatory proactiveness, and the competitiveness of green tariffs.

Most of the large industrial and RE rich states have notified green energy open access regulations, largely in line with the original Rules.

Exemptions and provisions for third-party OA consumers

Consumers opting for open access to procure power from third-party suppliers must pay transmission and distribution utilities various charges to use their networks. These include the cross-subsidy surcharge, transmission and wheeling charges, losses, connectivity charges and load dispatch centre charges. The cross-subsidy surcharge, determined by state regulatory commissions, compensates distribution companies for revenue loss from high-paying commercial and industrial consumers shifting away from discom supply.

The National Tariff Policy (2016) ensures that this surcharge remains manageable to encourage competition. It caps the cross-subsidy surcharge at 20% of the ACoS for industrial consumers and exempts captive and group captive sales. Further relief comes under GEOA Rules, 2022, which limit any increase in the cross-subsidy surcharge to 50% for 12 years from the commissioning of the project and waive additional surcharges if fixed charges are paid. Since cross-subsidy and additional surcharges together account for roughly 50% of total grid charges, these measures significantly improve the viability of OA for consumers.

Exemptions and provisions for captive and group captive consumers

Captive and group captive routes remain the most attractive option as projects set up by end users for their own consumption are exempt from paying cross-subsidy and additional surcharges otherwise applicable to third-party consumers. Thus, the central government's policy support has played a role in promoting the captive/group captive model, especially for C&I consumers.

Banking and procedural clarity

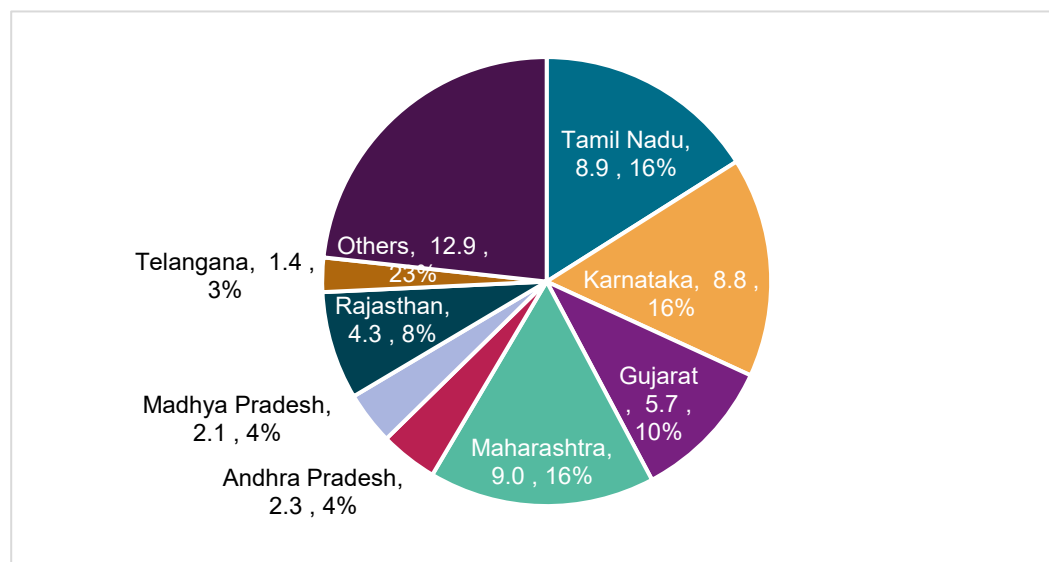
Banking is allowed on a monthly basis on payment of banking charges. The unutilised surplus banked energy lapses at the end of each banking cycle and the RE generating station gets RECs to the extent of

the lapsed banked energy. Consumers can also purchase green energy by placing a requisition with their discom.

The rules provide stakeholders long-term certainty on open access charges, which will help determine their returns from the project. Further, if an open access application is not approved within 15 days, it will be deemed approved to ensure timely execution of projects by minimising the risk of cost escalation. The demand for green energy open access from C&I consumers is likely to increase after implementation of these rules. However, discom cooperation, regulatory proactiveness in timely tariff orders and green energy tariffs are monitorable.

A significant portion of this capacity is concentrated in a few states such as Maharashtra, Tamil Nadu, Gujarat, and Karnataka, which together contribute over 55% of the total C&I RE open access capacity. These states are recognised for their high industrialisation levels and, consequently, account for a larger proportion of C&I power consumption compared to other states.

Figure 37: Share of RE open access capacity across states as of December 2025 (GW)



Source: Crisil Intelligence

The C&I consumers are exhibiting a growing preference for transitioning wind-solar hybrid and RTC energy sources. This inclination is primarily driven by the advantages of wind power supplying energy during peak hours and diversified output profile of different sources helping corporates in achieving higher renewable penetration. Furthermore, with the introduction of Electricity Amendment Rule 2023 wherein the ToD tariff regime has been specified, and a specific provision has been made to set the tariffs for solar hours at 20% lower than the normal tariffs. As a result, C&I consumers may be reluctant in procuring standalone solar power at higher tariffs, as they would realise comparatively less revenue during the solar hours due to the lower tariffs imposed for solar energy consumption. Moreover, a noticeable trend is emerging as more states are shifting away from annual banking mechanisms towards monthly banking arrangements. In light of this, consumers are increasingly opting for the adoption of hybrid solutions, which offer distributed energy generation capabilities throughout the day.

A fragmented and evolving regulatory framework increases the value of experienced energy partners capable of navigating state specific OA, captive and connectivity requirements

C&I renewable energy procurement in India is subject to a broad and evolving regulatory framework covering electricity and OA regulations, captive and group-captive structures, GEOA regulations, banking provisions, State Renewable Energy policies, CEA technical standards and guidelines, and ISTS/STU connectivity and transmission requirements. These regulations are administered across multiple central and state-level authorities and can differ materially across states, particularly with respect to open-access charges, banking, eligibility criteria, connectivity, scheduling and settlement mechanisms. The regulatory framework also continues to evolve, requiring consumers to continuously monitor amendments, tariff orders, technical requirements and state-specific policy changes. For C&I consumers, independently navigating this fragmented and frequently changing regulatory landscape can increase transaction complexity and execution risk. This creates a value proposition for experienced integrated energy solution providers that can monitor regulatory developments, assess their implications for individual customers, structure appropriate captive, group-captive or open-access arrangements, and coordinate connectivity, approvals and implementation. Such capabilities can enable C&I customers to identify and implement regulatory-compliant and commercially appropriate renewable energy procurement solutions while reducing the internal effort required to navigate multiple regulatory and institutional interfaces.

4.6 Energy storage can be a cost if not optimised properly

The economics of BESS in India is evolving rapidly. The MoP reported that the discovered cost of BESS through competitive bidding declined from Rs. 10.18/kWh to Rs. 2.1/kWh in recent bids. The government has also introduced VGF support for 43 GWh BESS, while ISTS charge waivers have further improved the economics of eligible storage projects. This does not mean that BESS is universally economic for C&I applications; rather, it strengthens the case for application-specific storage sizing based on the customer's load profile, renewable generation profile and value of avoided grid energy.

For energy-intensive C&I consumers, energy storage can bridge the mismatch between RE generation and the customer's consumption profile, but the economics of storage remain highly sensitive to utilisation, duration, battery cost and the value of the energy shifted. A more economical approach is to first optimise the RE portfolio through solar-wind hybridization, generation profile matching, demand-side optimisation and grid integration, and deploy BESS only for the residual requirement where the value of peak shifting, renewable firming, demand management or reliability justifies the incremental storage cost.

Wind and solar generation often show natural complementarity. Solar generation is strongest during daytime and summer months, while wind generation typically improves during evening/night hours and the monsoon season in several key renewable-rich states. By combining both resources in the right proportion, hybrid projects can deliver a smoother generation profile, improve plant load factors, and reduce intermittency.

Table 10: Illustration of landed RE price with different configurations

Particulars	Unit	Standalone Solar	Standalone Wind	Wind-Solar Hybrid	Solar+BESS	WSH + BESS
Total electricity requirement	Units	100	100	100	100	100
RE directly consumed	Units	40	55	85	40	85
Surplus RE available for BESS charging	Units	NA	NA	NA	5	15
RE discharged from BESS	Units	0	0	0	4.5	13.5
Total RE supplied	Units	40	55	85	44.5	98.5

Particulars	Unit	Standalone Solar	Standalone Wind	Wind-Solar Hybrid	Solar+BESS	WSH + BESS
Grid electricity purchased	Units	60	45	15	55.5	1.5
Illustrative LCOE	Rs/kWh	3.45	3.70	3.95	4.10	4.30
STU Charges	Rs/kWh	1.16	1.16	1.16	1.16	1.16
Wheeling Charges	Rs/kWh	0.76	0.76	0.76	0.76	0.76
STU Losses	Rs/kWh	0.11	0.12	0.13	0.13	0.14
Wheeling Losses	Rs/kWh	0.26	0.28	0.30	0.31	0.32
CSS & AS	Rs/kWh	0.00	0.00	0.00	0.00	0.00
Other Charges	Rs/kWh	0.10	0.10	0.10	0.10	0.10
Landed RE price	Rs/kWh	5.84	6.12	6.40	6.56	6.78
Grid electricity price	Rs/kWh	9.50	9.50	9.50	9.50	9.50
Cost of RE power	Rs.	233.68	336.54	543.64	292.01	668.18
Cost of grid electricity	Rs.	570.00	427.50	142.50	527.25	14.25
Total electricity cost	Rs.	803.68	764.04	686.14	819.26	682.43
Effective average cost per unit consumed	Rs/kWh	8.04	7.64	6.86	8.19	6.82

Note: The above calculations is only for illustrative purposes for an intrastate STU connected RE OA project located in Maharashtra. Illustrative LCOE is based on recent utility scale tenders. CSS and AS have been assumed at nil under captive/group captive structure. For Solar+BESS and WSH+BESS, additional RE capacity/oversizing & surplus generation during peak RE generation hours have been assumed to facilitate BESS charging; discharged energy has been considered after accounting for efficiency and losses.

Based on the above comparison, BESS can improve the dispatchability and utilisation of RE power, however, its economic attractiveness for C&I consumers is highly dependent on the consumer's load profile, location-specific electricity tariffs and the differential between peak and off-peak power costs. For consumers with predominantly daytime or relatively flat demand, the additional cost of storage may not be fully offset by the incremental RE utilisation. In contrast, for consumers with significant evening or peak-period demand, BESS can enable RE power to replace higher-cost peak electricity, thereby improving the overall economics of storage. The attractiveness of BESS is therefore expected to be highest where peak-period electricity tariffs, demand charges, backup power costs or reliability-related losses are sufficiently high to offset the levelised cost of storage, while hybrid configurations can further improve economics by providing a more complementary generation profile and reducing dependence on storage.

As a result, developers may be able to lower dependence on BESS, which involves high upfront CAPEX, replacement costs, and efficiency losses. That said, BESS still plays an important role where firm dispatch, peak-hour supply, grid balancing, or round-the-clock power obligations are required.

4.7 Emerging areas and sectors in C&I

- Companies in emerging industries are steadily increasing their RE procurement, driven by sustainability targets, reputational pressure, and long-term cost advantages.
- Large airports, metro networks, and IT campuses are increasingly building new RE capacities.
- India's data center industry is growing rapidly due to rising demand for cloud computing, data storage, and digital services, including GCC (Global Capability Centers).
- To meet increased demand, data center operators are likely to choose renewable power sources to reduce both costs and carbon footprint.

- Electricity costs represent about 30–40% of a data center's operating expenses.
- India is emerging as a key node on the global data center map, supported by rising internet penetration and growth in digital payments infrastructure.
- Data centers are expanding steadily, and India is generating about 20% of the world's digital data.
- Total data center capacity in India has grown from a few hundred megawatts in 2020 to roughly 1–1.5 GW by 2025.
- Capacity is expected to rise substantially further to 8–10 GW by fiscal 2030 (approximately 6–7x growth).
- Airports are particularly well-suited for renewables because they have large, continuous land footprints and high energy demand for lighting, HVAC systems, baggage handling, and terminal operations.
- Metro networks also drive RE adoption due to steady electricity consumption for traction and station power
- IT campuses often operating as power-intensive “digital factories” seek reliable, lower-carbon power to support data centers, offices, and 24/7 operations.

4.8 Emerging C&I RE based configuration/business models

C&I customers have diverse electricity consumption patterns, and the optimal RE configuration therefore varies based on the timing and continuity of demand, peak-load requirements, RE targets and the economics of grid power. Accordingly, as discussed in the previous sections, RE solutions can range from standalone solar or wind to combinations of solar, wind and BESS, with increasing system complexity deployed where required to improve temporal matching, manage peak demand, enhance reliability or increase RE penetration.

The table below presents the key RE models and configurations applicable to C&I customers, their key characteristics and the broad customer requirements or load profiles for which each configuration may be suitable. The suitability of each configuration is therefore determined on a case-by-case basis, considering the customer's hourly load profile, applicable tariffs and open-access charges, RE resource availability and the delivered cost of electricity.

Table 11: Key RE models and configuration applicable to C&I

RE model / configuration	C&I customer / load characteristics	Expected key customer requirement
Solar	Daytime or Solar-hour dominant manufacturing	Reduce daytime grid purchase cost; lowest possible delivered energy cost
Wind	Night-time dominant operation	RE generation during night / off-solar hours
Solar+BESS	Daytime intensive but with significant evening peak	Reduce peak period grid cost and manage demand charges
Solar+Wind	24x7 continuous load; seasonal /weather sensitive load; consumer with limited storage economics	High RE share with greater generation availability across operating hours; optimise RE supply against seasonal demand; improve utilisation of existing RE asset
Solar+Wind+BESS	24x7 load with high reliability / firm power requirement; consumer seeking maximum grid independence	Higher RE penetration with improved firmness and predictable supply; reduce grid reliance

Source: Crisil Intelligence

4.9 Cost optimisation using RE C&I business- case studies

The following case studies highlight the industry's ability to deliver cost-effective RE solutions towards the requirements of C&I customers. Through a combination of solutions, including RE generation, hybrid configurations and other customised energy models, the industry can enable customers to optimise their energy costs while improving the utilisation of RE and associated infrastructure.

Theme 1: Evolution of customer engagement from advisory to lifecycle asset management

As RE adoption matures, customer expectations are evolving beyond project execution to long-term strategic partnership. Businesses increasingly require support across the entire energy lifecycle, including procurement strategy, regulatory assessment, portfolio optimisation, and future expansion planning. This shift is transforming RE solution partners from project developers into trusted advisors capable of delivering sustained commercial and operational value over time.

Case study 1: Evolution of customer engagement from advisory to lifecycle asset management		
❑ Customer profile - Steel manufacturer ❑ Annual electricity consumption: 88-104 MU ❑ Objective: Transition from conventional power to sustainable energy while maintaining operational continuity	❑ Solution Approach ▪ A phased RE transition roadmap was developed over ~ 7 years, enabling the customer to gradually increase RE adoption while maintaining commercial competitiveness. ▪ Sourcing strategy evolved from Power Exchange to Group Captive structures and eventually Captive Hybrid solutions, allowing flexibility as market and regulatory conditions changed. ▪ Continuous optimisation of procurement models helped reduce transmission costs and banking dependence while improving RE penetration. The engagement evolved into a long-term energy management partnership rather than a one-time project implementation	❑ Business impact ▪ Transitioned from 100% conventional energy to 95% RE ▪ Achieved commercially competitive RE procurement ▪ Reduced transmission costs and banking dependence ▪ Established a scalable long-term energy platform
❑ Business challenge ▪ Achieve renewable transition without dependence on government incentives ▪ Maintain commercial viability throughout the transition ▪ Identify the optimal renewable procurement model amid evolving regulations and market conditions		❑ Key takeaway ▪ Customer engagement has evolved from project delivery to long-term advisory and lifecycle asset management, creating sustained value across the energy journey

Metric	Unit	Before	After
Annual Power Consumption (A)	MU	96	96
Renewable Capacity		Nil	Hybrid RE Portfolio (~30-35 MW equivalent)
BESS Requirement		NA	Not required
Renewable Energy Share		0%	~95%
Grid Power Consumption	MU	96	4.8
RE Power Consumption	MU	0	91.2
Grid tariff	Rs/kWh	8.0	8.0
Landed RE tariff	Rs/kWh	-	5.4
Effective Delivered Power Cost (B)	Rs/kWh	8.0	5.5

Metric	Unit	Before	After
Annual Electricity Spend (C=AxB)	Rs Cr	76.8	53.1
Annual Savings (Difference between electricity spend before and after)	Rs Cr	-	23.7

Theme 2: Anticipating regulatory changes to enhance project viability and development outcomes

The RE sector is heavily influenced by evolving policy and regulatory frameworks, making proactive engagement with regulators a critical success factor. Organisations that can anticipate regulatory developments, interpret emerging policies, and navigate approval processes effectively are often able to unlock new opportunities and enhance project economics. The following case study demonstrates how early action and regulatory expertise can create significant competitive advantages.

Case study 2: Anticipating regulatory changes to enhance project viability and development outcomes

❑ Customer profile - Leading chemical manufacturer ❑ Existing asset: 50 MW wind project ❑ Objective: Increase RE generation while maximising existing infrastructure utilisation	❑ Solution Approach • The solution focused on hybridising the customer's existing 50 MW wind project by integrating 70 MWp (50 MWp) solar capacity. • Existing pooling substations, transmission lines and evacuation infrastructure were utilised, eliminating the need for significant additional capital investment.	❑ Business impact • Annual transmission cost savings: ~₹7.62 crore • Achieved ~95% utilisation of available renewable generation • Eliminated need for new transmission infrastructure • Established benchmark for future hybridisation projects
❑ Business challenge • Expansion required significant new transmission investment • Project needed approval under a newly introduced state Hybrid Policy • Regulatory framework had not previously been tested at scale	• Extensive regulatory engagement was undertaken to secure approvals under a newly introduced Hybrid Policy and address implementation uncertainties. • The approach maximised renewable generation while significantly improving infrastructure utilisation	❑ Key takeaway • Early engagement with regulators and deep policy expertise can unlock pioneering projects and create strong first-mover advantages

Theme 3: Impact of regulatory changes on development assets and developer responses

Growing transmission constraints, changing OA regulations, banking restrictions, and evolving incentive structures are reshaping the RE development landscape. These changes require developers to continuously reassess asset strategies, technology choices, and geographic focus areas to maintain project viability. The case study highlights how proactive adaptation can help mitigate regulatory risks, avoid stranded assets, and preserve long-term returns.

Case study 3: Development asset strategy amid regulatory changes

❑ Industry context Renewable developers facing: • ISTS waiver sunset • Evolving open-access charges • Banking restrictions • Transmission congestion • Evacuation constraints in high-wind state	❑ Solution Approach • Developers are increasingly adopting next-generation WTG technology to expand into lower wind-density regions and reduce dependence on congested RE corridors. • Development strategies are shifting toward hybrid and co-located projects that optimise evacuation infrastructure and improve asset utilisation. • Portfolio structures are being designed with limited reliance on banking mechanisms to minimise exposure to regulatory changes. • Greater focus is also being placed on securing transmission connectivity early in the project lifecycle to safeguard project viability	❑ Business impact • Reduced congestion and regulatory risks • Improved utilisation of transmission infrastructure • Hybrid PLFs at ~ 40-45% • Enhanced resilience against tightening regulations
❑ Business challenge • Regulatory changes threaten project economics • Limited high-resource locations causing congestion • Increased risk of stranded assets and delayed projects		❑ Key takeaway • Proactive adaptation to policy and technology shifts is critical for preserving project viability and long-term returns

Theme 4: Increasing preference for hybrid renewable solutions among C&I consumers

C&I consumers are increasingly seeking RE solutions that provide a balance between cost efficiency, energy reliability, and infrastructure optimisation. Hybrid projects are emerging as an attractive option across a broader range of consumer segments. This case study illustrates how innovative project design can make hybrid solutions commercially viable even for medium-scale energy consumers including substantial reduction in transmission charges over the operating life cycle.

Case study 4: Hybrid renewable solution for medium-scale industrial consumers

❑ Customer profile • Leading two-wheeler manufacturer • Demand size approximately equivalent to one wind turbine • Sustainability-focused renewable procurement objective	❑ Solution Approach • A customised hybrid solution was designed to address the economics of RE procurement for a medium-scale consumer. • The project integrated a 2.7 MW wind turbine and a 1.5 MWp solar plant within the same land footprint, with solar installed beneath the wind turbine. • Shared evacuation infrastructure and a common metering arrangement reduced transmission and infrastructure costs while improving utilisation. • This innovative co-location model enabled commercially viable hybrid adoption at a smaller scale	❑ Business impact • Hybrid PLF: ~44% • Energy generation per evacuation capacity: ~175% of standalone solar • Transmission charges reduced for Hybrid: ₹0.84/unit, Solar: ₹1.49/unit, Wind: ₹1.09/unit • Long-term savings capable of offsetting >80% of wind turbine capital cost over project life
❑ Business challenge • Smaller load size made conventional hybrid projects uneconomical • Higher capital cost per MW • Underutilised transmission infrastructure • Higher evacuation charges impacting project economics		❑ Key takeaway • Hybrid solutions are commercially viable even for medium-sized consumers through efficient infrastructure sharing and co-location design

Metric	Unit	Before	After
Annual Consumption (A)	MU	10	10
Renewable Capacity		Nil	2.7 MW Wind + 1.5 MW Solar
Renewable Energy Share		0%	89%
Grid Electricity Consumption	MU	10	1.2
RE Power Consumption	MU	0	8.9
Grid tariff	Rs/kWh	8.5	8.5
Landed RE tariff	Rs/kWh	-	4.2
Effective Delivered Cost (B)	Rs/kWh	8.5	4.7
Annual Electricity Spend (C=AxB)	Rs Cr	8.5	4.7
Annual Savings (Difference between electricity spend before and after)	Rs Cr	-	3.8
Transmission charges	Rs/kWh	Standalone solar – 1.49	Co-located hybrid RE – 0.84
Savings in transmission charges based on co-located hybrid project	Rs/kWh	-	0.65

Theme 5: Storage optimisation and hybrid economics

While RE adoption continues to accelerate, achieving higher levels of renewable penetration requires solutions that address intermittency and demand variability. BESS are becoming an important enabler by improving energy reliability, increasing renewable utilisation, and enhancing overall project economics. The following case study demonstrates how integrating storage can unlock additional value and move organisations closer to RTC RE consumption.

Case study 5: Enhancing renewable penetration through Solar + Storage

Customer profile - Multi-location healthcare provider - Critical operations requiring uninterrupted power - Goal: Maximise renewable consumption and reduce electricity costs	Solution Approach - A distributed captive solar portfolio of approximately 9.2 MWp was deployed across 31+ healthcare facilities, supported by ongoing lifecycle energy management. - Consumption analysis identified significant dependence on grid power during non-solar hours despite substantial solar adoption. - To enhance renewable penetration, BESS integration was evaluated to store surplus daytime generation and supply power during evening demand periods. - The approach focused on improving both energy economics and operational reliability for critical healthcare facilities	Business impact - Renewable replacement projected to increase from ~56% to ~89% - Electricity cost reduced from ~₹15/unit to ~₹4.50/unit - Project payback ~1.9 to 3.5 years - Enhanced energy independence and reliability
Business challenge - Solar generation alone unable to fully replace grid power - Significant power consumption during non-solar hours - One region achieved only ~56% renewable replacement through solar alone		Key takeaway As storage costs continue to fall, BESS-enabled renewable solutions become a critical lever to improve renewable penetration and delivered energy economics

Theme 6: Customers evaluate delivered electricity cost, not just generation tariff

Energy procurement decisions are increasingly being driven by the total delivered cost of electricity rather than the headline generation tariff alone. Factors such as transmission charges, OA costs, banking requirements, surcharges, reliability, and alignment with consumption patterns all influence the true

economic value of a RE solution. This case study highlights the importance of adopting an integrated sourcing strategy that optimises overall delivered energy costs while maintaining supply reliability.

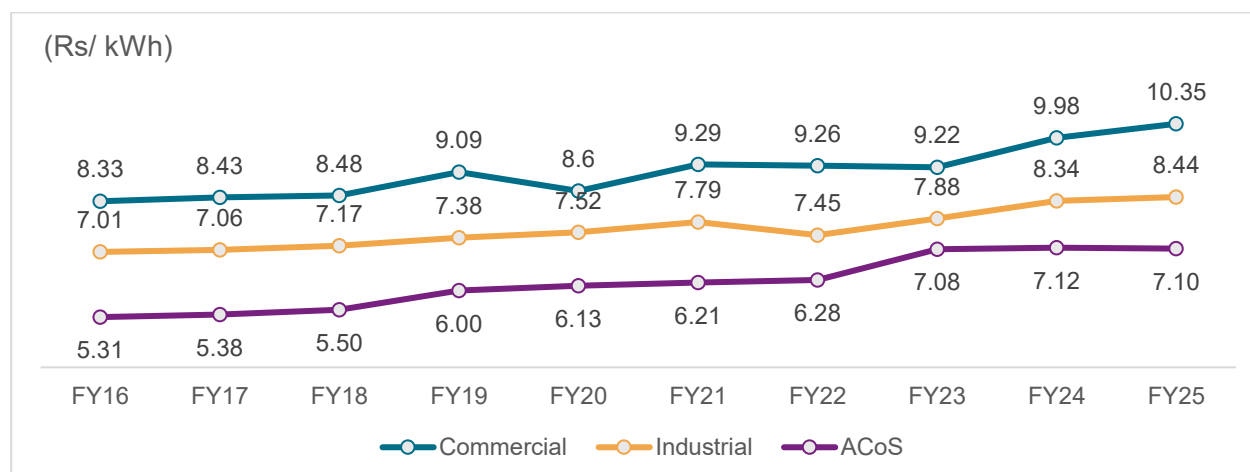
Case study 6: Optimising delivered electricity cost through hybrid open access

□ Customer profile - Energy-intensive manufacturing company - Annual consumption: ~80 MUs - Contract demand: 19.6 MVA	□ Solution Approach - A detailed assessment of the customer's load profile and delivered electricity costs was conducted to identify the optimal RE procurement mix. - A diversified Wind-Solar OA portfolio combining 18.9 MW wind and 8.8 MW solar capacity was designed to maximise RE penetration while ensuring supply reliability. - The structure was optimised to minimise dependence on banking mechanisms and reduce exposure to regulatory and surcharge-related risks. - The overall strategy focused on lowering the total delivered cost of electricity rather than simply achieving the lowest generation tariff.	□ Business impact - Achieved total RE replacement of ~80% - Monthly electricity bill reduced: from ₹7 crore to ₹2 crore - Annual savings in expenditure increased by ₹ 60 crore
□ Business challenge - Reduce electricity expenditure - Minimise dependence on grid power - Ensure uninterrupted supply for manufacturing operations - Evaluate energy procurement beyond headline generation tariffs		□ Key takeaway As storage costs continue to fall, BESS-enabled renewable solutions become a critical lever to improve renewable penetration and delivered energy economics

Metric	Unit	Before	After
Annual Consumption	MU	80	80
Renewable Capacity		3 MW Solar	18.9 MW Wind + 8.8 MW Solar
Renewable Energy Share		5%	80.7%
Grid Power Consumption	MU	76	15.4
RE Power Consumption	MU	4	64.6
Grid tariff	Rs/kWh	10.5	10.5
Landed RE tariff	Rs/kWh	4.0	4.7
Effective Delivered Cost	Rs/kWh	10.1	5.8
Annual Electricity Spend	Rs Cr	81.1	46.6
Annual Savings (Difference between electricity spend before and after)	Rs Cr	-	34.5

4.10 Utility C&I tariff and cost savings from RE power

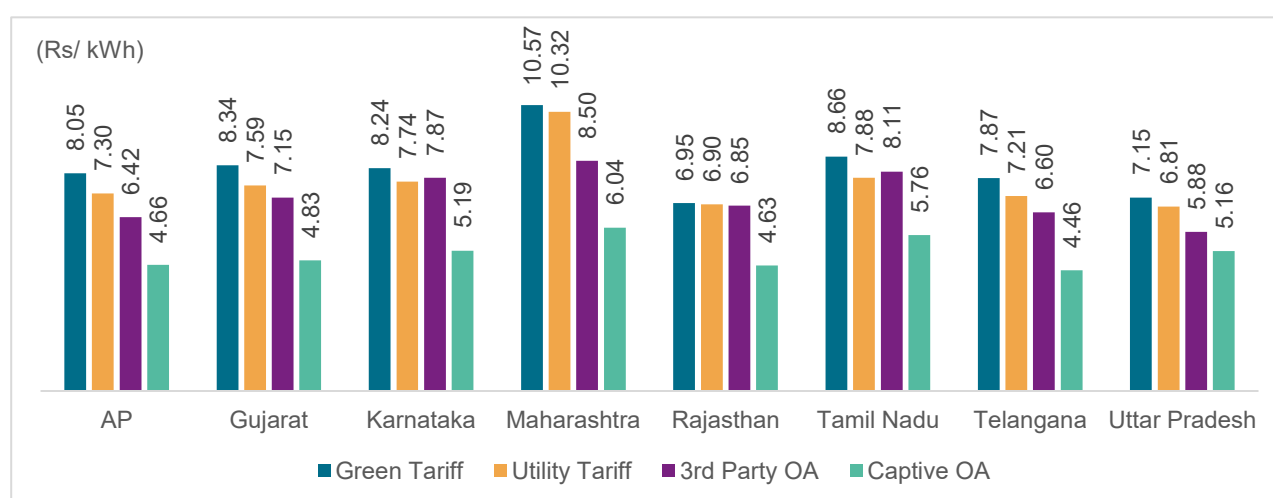
C&I grid tariffs logged a CAGR of 2.4% and 2.1%, respectively, between fiscals 2016 and 2025, led by the need to cross-subsidise agricultural and residential user tariffs. In fiscal 2025, the average cross-subsidisation level for Industrial consumers was ~119%. In the same period ACoS logged a CAGR of 3.3%.

Figure 38: C&I average billing rate at the national level (Rs/ kWh)


Source: NITI Aayog, Crisil Intelligence

Variable grid tariffs for HT industrial consumers showed an upward trend across states between fiscals 2022 and 2025. Although subject to slightly different tariff structures, HT commercial consumers followed a broadly similar trend. Despite some regional policy variations, this historic movement reflects increasing power procurement costs and persisting cross-subsidies.

The utility tariff for C&I consumers is significantly higher than other consumer categories and this trend is expected to continue in future. On the other hand, the landed RE tariff, which accounts for the levelised cost of RE projects and other open access charges to the consumers, has become increasingly competitive due to advancements in technology, higher CUF, lower cost of finance and supportive government policies and regulations for open access consumers. As demonstrated in the chart below, assuming a levelised WSH tariff of Rs. 3.70/kWh for key states, the difference becomes evident, offering businesses a compelling incentive to opt for sustainable and cost-effective RE solutions. In fact, it can be inferred from the chart below that the open access tariffs are still more competitive than variable utility tariffs even after removal/reduction of incentives for open access charges in majority of the states.

Figure 39: Utility vs open access landed tariff for Industrial consumer for FY2027


Note: Base tariff of Rs 3.70/kWh for third party and Rs. 3.50/kWh for CPP is considered for industrial consumer connected with Discom at 33 kV for long term intra state OA. The open access charges are as per Tariff Orders for FY2027. The landed tariff includes prevailing exemptions/concessions applicable for OA charges. No Additional surcharge considered for third party sale assuming that fixed charges being paid by the consumer.

Source: SERC Tariff Orders, Crisil Intelligence

Considering the various charges recovered from third-party OA consumers, discoms certainly benefit from providing open access versus continuing supply to C&I consumers. Discoms are not required to pay transmission costs for units not supplied. The concerned open access consumer/developer as the case may be pays the transmission charges to Transco for consumed units. Discoms recover CSS and ASC for loss of cross subsidy and stranded assets due to open access. Further, due to the backing down of higher variable cost sources, Discoms save more than cheaper marginal sources. Also, being connected to the grid, the open access consumers continue to pay the fixed charges. Thus, there is net gain to Discoms in the case of third party RE open access.

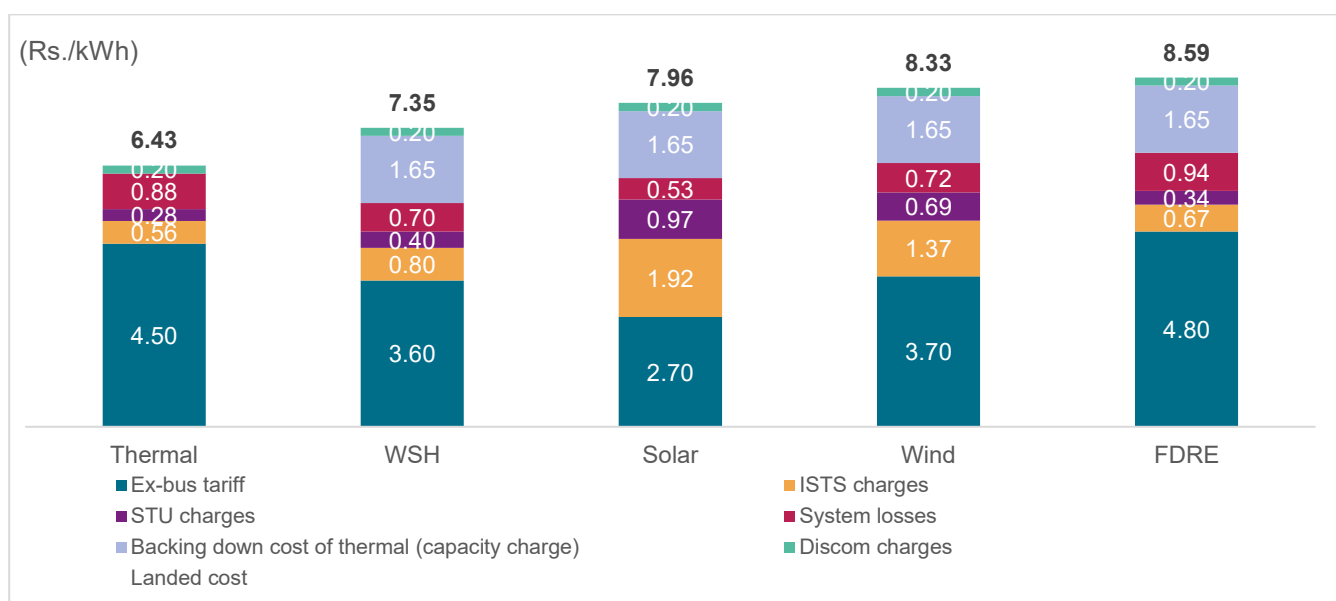
4.11 Outlook on cost of generation of multiple RE source vs cost of delivery

The tariffs for ISTS connected plain vanilla solar and wind projects were observed between Rs 2.5-2.7/kWh and Rs 3.0 – 3.5/kWh in 2023. Whereas the tariffs for WSH projects are in the range of Rs 3.3 – 3.6/kWh. Tariffs in Rajasthan, Maharashtra, Gujarat, Tamil Nadu and Karnataka are marginally lower than in other states due to availability of resources. The average CUF of solar projects can reach 25% or higher and for wind it could go up to 35% or higher in these regions.

While the ISTS connected ex-bus RE tariff for discoms appears attractive at Rs 2.5 – 3.5/kWh but the cost delivered to the discoms' periphery is significantly higher. This is because several additional costs are incurred during transmission and distribution of RE power. These include ISTS losses, ISTS drawal charges, T&D losses, STU charges, distribution service charges.

The delivered RE cost to discom can range from Rs. 8-8.5/kWh depending upon the ex-bus tariff and other costs which can vary from state to state. Similarly, the average levelised cost of a thermal plant is in the range of Rs 4-4.5/kWh which would also end up giving the landed cost of about Rs. 6.5 – 7/kWh. Hence, for discoms, the landed cost of RE power is at par with thermal despite lower ex-bus tariff and utilisation as compared to conventional source.

Figure 40: Delivered cost to Discom from various sources



Note: The ex-bus tariff of solar, wind, WSH, FDRE are as per the recent auctions; All calculations have been made assuming 1 MWac capacity. Assumed CUF of 25% for solar and 35% for wind. For WSH, 50% capacity of each solar and

wind has been assumed with a weighted average CUF of 30%, FDRE project CUF assumed as 72% and thermal PLF as 85% for CGS based plants. ISTS losses considered at 4.0%, State T&D losses considered at 15.5%, STU charges at Rs. 5,800/MW/day and ISTS charges of Rs. 0.35 Mn/MW/Month. Backing down cost of thermal (capacity charge) considered for RE projects to be paid by discoms for backed down capacity.

Source: Crisil Intelligence

4.12 Key challenges faced by C&I consumers

Contractual challenges:

- **Mismatch between PPA and loan tenure** – Typically the duration of long term PPA in C&I is of 10-15 years, while loan taken for the project might have a longer duration. The lenders may be hesitant to favourable loan terms if there is a significant tenor mismatch, potentially leading to higher interest rates. However, PPA tariff for supply of green power is generally at discount to Discom's tariff selling black power. Therefore, demand for green power should always be there.
- **Contract standardisation** – The government tenders usually have a standardised PPA which is a part of RfP document. However, in open access projects, PPAs are often customised based on mutually agreed terms between the parties after negotiation, leading to variety of terms and conditions. This can lead to inconsistencies in contractual terms and when any dispute arises, the lack of standardization can complicate the resolution process.
- **Contract enforcement** – It arises from the challenges associated with ensuring that all parties adhere to the terms of the PPA. However, inconsistent contractual terms, misinterpretation of any regulatory provision or ambiguity in policies and regulations can affect the enforcement of PPAs.

Operational challenges:

- **Design and structuring:** The success of a RE project depends not only on the availability of solar, wind or other resources, but also on how effectively the project is designed and structured. Appropriate choices regarding technology, configuration and financing determine the project's generation profile, reliability, cost competitiveness, bankability and long-term returns. Integrated technical, commercial and financial structuring is essential to convert a renewable resource into a reliable and investable infrastructure asset. Early optimisation helps achieve the right balance among generation, reliability, risk and cost. It also improves bankability, facilitates timely financial closure and enhances the project's ability to deliver sustainable returns throughout its operating life.
- **Performance risk** – RE sources are inherently variable. Unpredictable fluctuations in wind speed and solar irradiance due to extreme weather events lead to variations in power output which can affect the forecasted revenue for the period. Moreover, any unexpected breakdowns or faults can also lead to longer downtimes and reduced power generation. Advanced generation forecasting, remote monitoring and predictive analytics tools can help mitigate these risks.
- **Grid curtailment risk** – Integrating RE project into the grid in a RE rich state can pose challenges related to congestion, forcing grid operator to curtail power to avoid overloading the network. Moreover, intermittent supply of RE can also affect grid stability. Standalone solar is more prone to grid curtailment vis-à-vis wind or WSH since all solar projects peak at the same time which is not the case with wind.

Regulatory challenges:

- **Change in TOD tariff structure:** India has introduced a revised ToD tariff framework through the Electricity (Rights of Consumers) Amendment Rules, 2023. The change is intended to align retail tariffs with the increasing availability and lower marginal cost of solar power during daytime hours. The Rules provide that the SERCs may specify eight hours in a day as solar hours and the tariff during these

hours must be at least 20% lower than the normal tariff for the relevant consumer category. Lower retail tariffs during solar hours may reduce the savings available to C&I consumers.

- **Variable open access charges** – The open access charges are variable in nature; most of the state discoms are gradually increasing their open access charges which make it less attractive for the C&I consumer. Increase in open access charges can significantly increase the cost of RE power for a buyer. However, these charges being regulated cannot be increased on an ad hoc basis. The transmission and wheeling charges have remained consistent, albeit a few exceptions in most of the states. CSS is capped at 20% of the ACoS rate to industries and waived for group captive customers, full waiver of additional surcharge if fixed charges being paid by the consumer and not applicable for group captive customers. However, for projects offering fixed discount over discom tariff, any increase in CSS is advantageous. For a 20 paise per unit increase in CSS, an increase of Rs. 1.00 per unit in discom tariff would be required, which results in a clear advantage of 80%.
- **Addition of new open access charges** - There are various new charges other than the open access charges that have been introduced by many states on open access consumers. For example, in Gujarat, the hybrid policy of 2018 applied multiple wheeling charges of Re 0.05 per kWh on hybrid projects. In Madhya Pradesh, the Harit Urja Tax has been introduced in a new renewable energy policy at the rate of Re 0.10 per kWh. Some states have introduced grid support/parallel operation charges to captive power plants.

Captive project status - Captive projects are more attractive than 3rd party open access projects. Most C&I consumers are opting for captive open access and due to this Discoms are incurring revenue losses. Group captive customers need to abide by group captive guidelines as per the Electricity Act 2005 and compliance is tested at the end of the year. The Supreme Court issued a judgement on 09 Oct 2023 clarifying multiple aspects of the Electricity Rules 2005 relating to captive/group captive plants and captive users.

4.13 Key industrial states

India's industrial activity is concentrated across a set of economically large and manufacturing-intensive states, led by Maharashtra, Gujarat, Tamil Nadu and Karnataka. In terms of overall economic size, these industrial centres are also among India's largest state economies. For fiscal 2026 at constant prices, public estimates place Maharashtra's GSDP at roughly Rs 28.83 trillion, followed by Tamil Nadu at around Rs 19.2 trillion, Gujarat at ~ Rs 18.78 trillion and Karnataka at ~Rs 17.23 trillion.

Table : Key parameters of key industrial states for FY 2025

Parameters	Units	Maharashtra	Gujarat	Karnataka	Tamil Nadu	Rajasthan
GSDP at Constant prices (2011-12 series)	INR Tn	28.83	18.78	17.23	19.20	9.82
Industrial category electricity consumption from Utility (FY 23-24)	MUs	62,774	77,058	13,179	42,017	21,930
Industrial category electricity consumption- Captive (FY 23-24)	MUs	14,411	22,473	12,114	7,770	8,599
Total Industrial electricity consumption (Utility + Captive)	MUs	77,186	99,531	25,293	49,788	30,529
Tariff levels FY26-27						
Commercial	INR/kWh	13.49	5.32*	7.84	9.46	9.49
Industrial	INR/kWh	13.61	5.94*	8.26	10.22	8.06
Wind Potential	GW	173.9	180.8	169.3	95.1	284.3
Solar Potential	GW	486.7	243.2	223.3	204.8	828.8

Parameters	Units	Maharashtra	Gujarat	Karnataka	Tamil Nadu	Rajasthan
OA Charges						
CSS	INR/kWh	2.07	1.33	1.86	1.99	1.48
ASC	INR/kWh	NIL	0.76	NIL	0.10	0.50
RE Penetration (% Share in total installed capacity) March 2026	%	47%	64%	60%	57%	72%

Commercial: for 5kW, 500 Units per month including ED

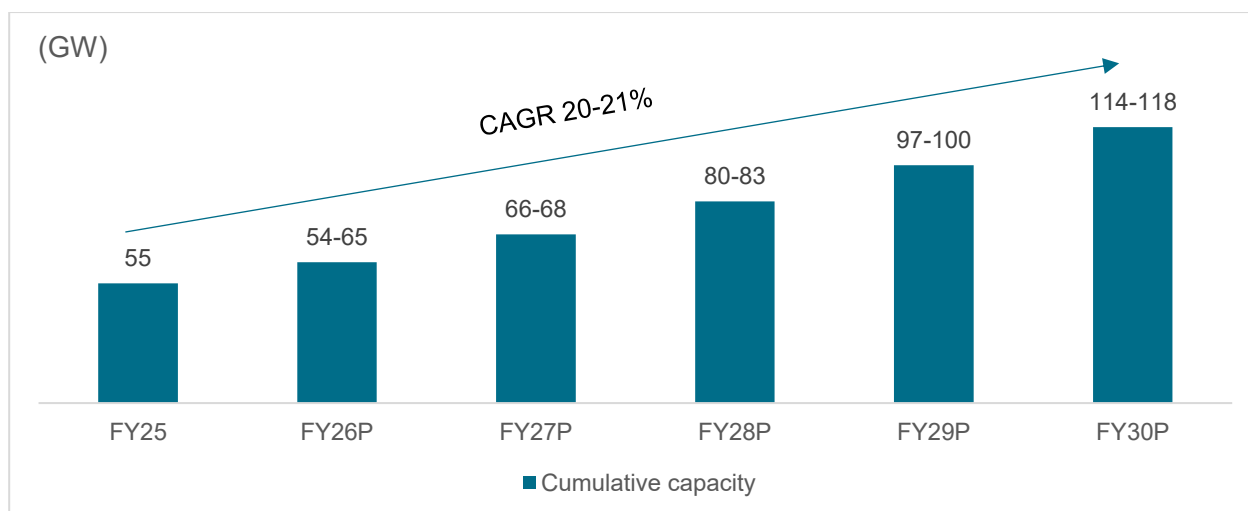
Large Industrial: 1,000 KW, 40% LF, 365,000 Units per month including ED *Excluding FAC,

Source: RBI, CEA, NIWE, NISE, REC

4.14 Outlook on RE based C&I capacity addition

Crisil Intelligence expects an installed capacity of about 114-118 GW of C&I driven RE capacity by fiscal 2030. About 68-73 GW is expected to be added between fiscal 2026-30 including the capacity required for green hydrogen production. The capacity addition of 12-13 GW of open access capacity is expected from wind projects to deliver RTC power. About 42-45 GW from solar projects (large scale projects), led by green energy open access rules 2022, sustainability initiatives/RE 100 targets of the corporate consumers, better tariff structures and policies of states such as Uttar Pradesh and Karnataka, which are more long term in nature. Further, 14-15 GW of rooftop solar projects (under the capex and opex mode) is expected to be commissioned by fiscal 2030, led by industrial and commercial consumers under net/gross metering schemes of various states.

Figure 41: Expected installed RE capacity outlook for C&I market



P: Projected; Source: Crisil Intelligence

5 Project and customer lifecycle in renewable energy

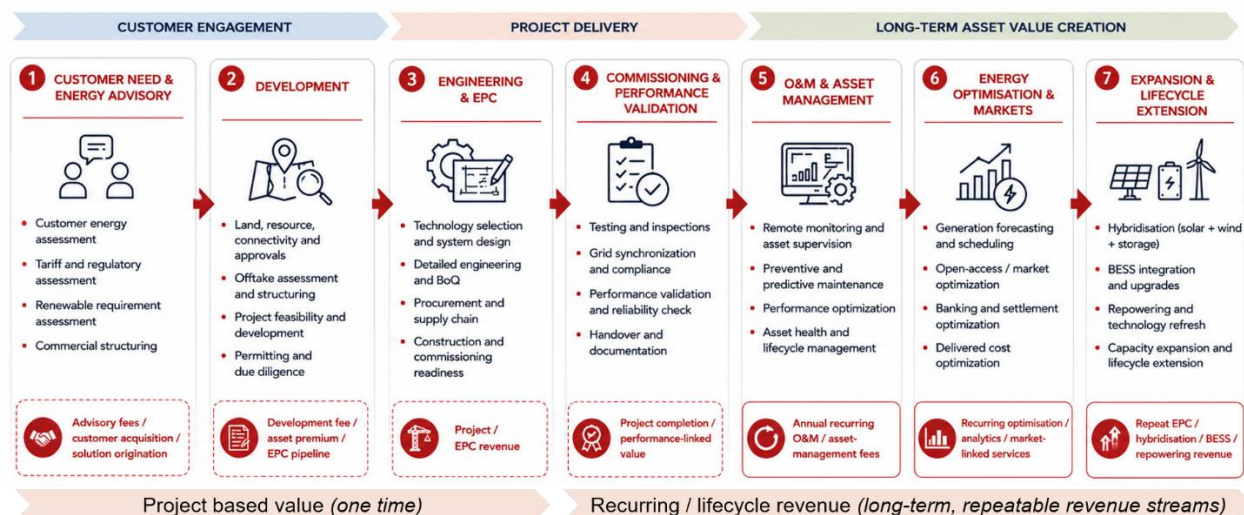
5.1 Project lifecycle

The renewable energy project lifecycle represents a comprehensive continuum that spans from initial advisory through long-term asset management, typically 25-30 years. This lifecycle is intrinsically intertwined with customer engagement, which commences well before project award and continues substantially beyond commissioning, creating multiple touchpoints for value creation and relationship deepening. For Integrated Energy Partner operating in this space, understanding this dual lifecycle is critical for competitive positioning, revenue optimisation, and establishing sustainable customer relationships that generate recurring revenue streams throughout the asset's operational life.

The RE project lifecycle comprises of a sequence of interlinked activities, commencing with the identification of the customer's energy requirements and project opportunity, and extending through project development, feasibility evaluation, technology selection, procurement, engineering, construction, commissioning, and long-term operations and maintenance. The lifecycle requires the participation of multiple stakeholders with specialised technical, commercial, financial and operational capabilities. Key stakeholders typically include IPPs or project developers, original equipment manufacturers (OEMs), project management consultants (PMCs), engineering, procurement and construction (EPC) contractors, asset managers and O&M service providers.

The RE project lifecycle can be divided into seven distinct yet interconnected phases that span the entire asset life, as illustrated in the detailed flowchart below.

Figure 42: Integrated RE project lifecycle and value pools



Source: Crisil Intelligence

The development stage represents one of the most complex, critical and value-intensive phases of the RE project lifecycle, requiring substantial effort across resource and site identification, land acquisition and aggregation, grid and transmission connectivity, statutory and regulatory approvals, project configuration, commercial structuring and bankability assessment. Successful execution of these activities can convert an early-stage opportunity into a de-risked, bankable and ready-to-build project, thereby creating significant development value and establishing the pipeline for subsequent execution.

5.2 Fragmented renewable value chain creates opportunity for Integrated Energy Solution partners

The RE value chain remains fragmented across equipment supply, project execution and power generation. This creates an opportunity for integrated energy partners that can bridge the gap between EPC and IPP models by supporting customers across the project lifecycle. Equipment manufacturers provide technology, EPC contractors execute projects, and IPPs provide long-term power; the underserved space is the independent, end-to-end energy partner that helps a customer assess, structure, optimise and manage RE power throughout its life cycle.

Table 12: RE value chain: market participants and the customer gap

Market participants	Conventional role	Key limitations /gap from customer perspective	How an integrated energy partner can address the gap
Equipment manufacturers (OEM)	Supply modules, wind-turbines, inverters, batteries and other RE equipment	Primarily technology/product focused; generally, do not take responsibility for customer specific energy economics, project development or lifecycle optimisation	Remain technology agnostic and select equipment based on lifecycle cost, generation performance and customer requirements, rather than being tied to a particular OEM
EPC contractors	Design, procure and construct RE projects and hand over the completed asset to the customer/developer	Focus is primarily on project delivery; customer support generally reduces significantly after commissioning (or expiry of EPC contract term)	Extend beyond EPC into pre-project advisory, energy assessment, project structuring, commissioning support, and post COD optimisation, creating an ongoing customer relationship
IPP/ RE developer	Develop, own and operate RE assets and sell electricity through long-term PPAs	Provides the customer with RE power, but the proposition can remain largely asset and PPA centric. The customer may still need to manage regulatory approvals, scheduling, portfolio optimisation	Act as an integrated energy partner, combining project development with regulatory management, RE portfolio optimisation, hybridisation and ongoing energy cost management
O&M provider	Responsible for the maintenance and upkeep of an RE asset following	Typically focused on asset availability and technical performance under O&M contract,	Integrate O&M with energy analytics and portfolio optimisation, thereby linking plant

Market participants	Conventional role	Key limitations /gap from customer perspective	How an integrated energy partner can address the gap
	commissioning, with the defined KPIs	with limited responsibility for the customer's overall energy economics	performance directly to the customer's delivered cost, reliability and long-term energy objectives
Conventional energy consultant/advisor	Provides feasibility studies, regulatory advice, energy audits or transaction support	Often advisory led and may not have direct responsibility for project execution, asset performance or long-term power delivery. This can create a gap between recommendation and implementation	Combine advisory, development, execution, and lifecycle management and allowing recommendations to be translated into operating RE assets and measurable customer savings
Customer	Procures electricity from the Discom / IPP	RE procurement introduces multiple variables – energy cost, reliability, OA charges, banking, scheduling, generation variability, regulatory compliance – that the customer may not have the capability or bandwidth to manage internally	Provide a single window RE solution that converts multiple technical, commercial and regulatory decisions into a simple delivered-energy proposition for the customer

Source: Crisil Intelligence

In the C&I segment specifically, the integrated project and customer lifecycle approach plays a particularly critical role for several compelling reasons. C&I customers typically lack in-house RE expertise and require comprehensive support throughout the asset's operational life, making them highly dependent on their EPC partner's capabilities beyond mere project execution. The financial stakes are substantial for C&I clients, as RE investments often represent significant capital and value commitments spanning for 20 – 30 years duration that directly impact their operational costs, sustainability credentials, and competitive positioning, necessitating assurance of long-term performance and support. C&I facilities face unique operational constraints including limited downtime tolerance, space limitations, and integration with existing electrical infrastructure, requiring EPC partners who understand these complexities and can provide responsive, ongoing support. The evolving regulatory landscape, and emerging sustainability reporting requirements create continuous advisory needs that extend well beyond project commissioning.

In today's market, C&I customers increasingly prioritise integrated energy partners spanning the entire value chain from initial advisory and feasibility studies through design, execution, commissioning, operations & maintenance, and long-term asset management support. This preference stems from the desire for single-point accountability, reduced coordination complexity, seamless knowledge transfer across project phases, and the assurance that the entity designing and building the system will be responsible for its long-term performance. Companies demonstrating comprehensive lifecycle capabilities

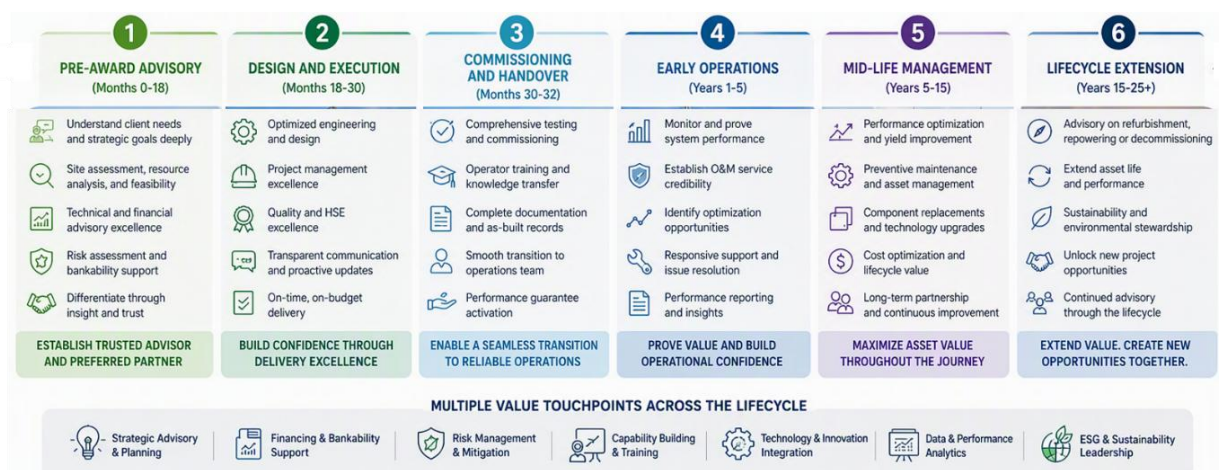
with proven track records in both project execution and multi-year O&M operations are gaining significant competitive advantage in customer selection processes. This shift toward lifecycle-oriented vendor selection represents a fundamental evolution in the C&I renewable energy market, where integrated energy partners are increasingly preferred over fragmented specialists, as customers seek to minimise risk, ensure accountability, and establish enduring partnerships that can adapt to their evolving energy needs over the asset's 25–30 year operational life.

5.3 Customer lifecycle

The customer lifecycle in the RE sector encompasses the complete journey from initial advisory and assessment of a customer's energy requirements through solution design, technology selection, procurement, project execution, commissioning, and long-term asset management. Unlike conventional equipment procurement transactions, RE projects involve enduring customer relationships that commence well before project award and continue substantially beyond commissioning, as the assets require continuous O&M, performance monitoring, optimisation, technology upgrades, mid-life refurbishment, and eventual lifecycle extension or repowering considerations.

The lifecycle generally begins with comprehensive pre-award advisory services assessing the customer's energy needs, which, particularly for C&I customers, may arise from rising electricity costs, decarbonisation commitments, RCO compliance requirements, energy security considerations, operational expansion plans, or the desire for greater predictability in long-term power costs and protection against grid tariff volatility. The EPC developer or IPP conducts detailed energy baseline assessments evaluating the customer's existing and projected electricity consumption profile, including historical consumption patterns, load profiling across different time periods, peak demand analysis, applicable tariff structures, existing power procurement arrangements, grid dependency, and future growth requirements, as the technical and financial viability of an RE solution depends critically on both the volume and timing of electricity consumption, load factor, and alignment between generation profile and consumption patterns. This initial advisory phase, typically spanning 12-18 months before project award, establishes the foundation for a trusted advisor relationship that evolves through design and execution (demonstrating project management excellence), commissioning and handover (ensuring smooth operational transition), early operations (proving system performance and O&M credibility), mid-life management (implementing enhancements and component replacements), and lifecycle extension phases (advisory on refurbishment or repowering), creating multiple value touchpoints and recurring revenue opportunities throughout the asset's operational life.

Figure 43: Customer lifecycle



Source: Crisil Intelligence

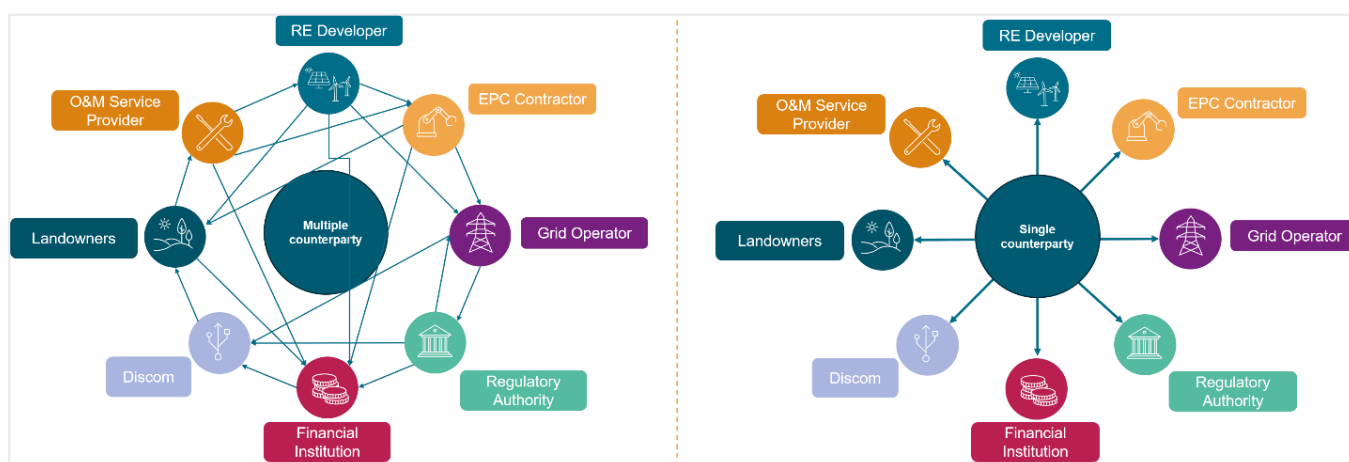
Different market participants cover different portions of this value chain, with integrated partners offering the broadest end-to-end coverage.

	Advisory	Development	Execution	Lifecycle services	Optimisation
Consultant	High	Medium	Low	Low	Medium
IPP	Medium	High	Medium	High	High
EPC contractor	Medium	Medium	High	Medium	Medium
OEM	Low-Medium	Low	High	High	High
O&M provider	Low	Low	Low-Medium	High	High
Integrated partner	High	High	High	High	High

The RE sector is witnessing the emergence of integrated energy partners who represent a paradigm shift from transactional project execution to comprehensive lifecycle partnership. The integrated partners are combining development, construction, financing, asset management and operations under a single platform. The integrated players strategically position themselves between traditional EPC contractors and pure-play IPPs thereby addressing the critical market gap for end-to-end support that customers increasingly demand. By serving as orchestrators across the entire value chain, integrated energy partners coordinate seamlessly with equipment manufacturers, specialised contractors, financial institutions, regulatory authorities, and DISCOMs while maintaining unwavering focus on customer outcomes throughout the asset life. This orchestration capability delivers single-point accountability, eliminates multi-vendor coordination complexity, ensures technical knowledge continuity, and provides customers with a stable long-term partner who evolves with their changing energy needs, regulatory landscape, and business growth trajectory.

For customers managing energy as a non-core function, particularly in the C&I segment, integrated energy partners represent the right solution model, transforming renewable energy adoption from a capital investment decision into a strategic partnership that delivers sustained value, performance assurance, and operational simplicity across decades of asset operation, as comprehensively illustrated through the project and customer lifecycle frameworks presented above.

Figure 44: A single Integrated energy partner dealing with multiple stakeholders



Source: Crisil Intelligence

When no single provider spans the full value chain, the customer must coordinate multiple parties and bear the resulting integration risk creating a clear need for an end-to-end energy partner.

For purposes of assessing the customer lifecycle, the revenue opportunity can broadly be divided into two categories:

Transactional / project-based revenue includes development, EPC, hybridisation, BESS deployment, augmentation, repowering and refurbishment. These activities typically generate larger but intermittent revenue opportunities linked to individual projects or capital investments.

Recurring / annuity-like revenue includes O&M, asset management, remote monitoring, performance optimisation and certain energy-management, scheduling or market-linked services. These activities can generate revenue over the operating life of the asset and provide greater continuity of customer engagement.

The distinction is important because an integrated energy partner can potentially combine higher-value project transactions with lower-value but recurring lifecycle services, thereby broadening the revenue opportunity associated with each customer.

5.4 Traditional EPC vs advisory led integrated model

Integrated energy models provide customers with a single point of accountability across project lifecycle

Traditional EPC models are generally execution-led, wherein the customer determines key project parameters such as technology, capacity, location and commercial structure before inviting EPC contractors to compete primarily on price, execution capability and delivery timelines.

In contrast, an advisory-led integrated model is requirement-led, wherein the Integrated partner first evaluates the customer's load profile, grid tariffs, energy requirements, expansion plans, RE targets, regulatory environment and available capital before determining the optimal combination of RE technologies and commercial structures. The resulting solution may include solar, wind, BESS, hybrid configurations, OA/captive procurement or energy-as-a-service models, followed by integrated project development, EPC, commissioning, O&M and ongoing energy optimisation. This model can provide

customers with a single point of accountability across the project lifecycle, reducing the need to coordinate multiple counterparties and shifting the focus from procuring a renewable energy project to optimising the customer's overall delivered energy cost, reliability and decarbonisation objectives.

The table below presents the key differences between traditional EPC model and advisory-led integrated model across different parameters and scope of services.

Table 13: Key differences- traditional EPC and Advisory-led integrated model

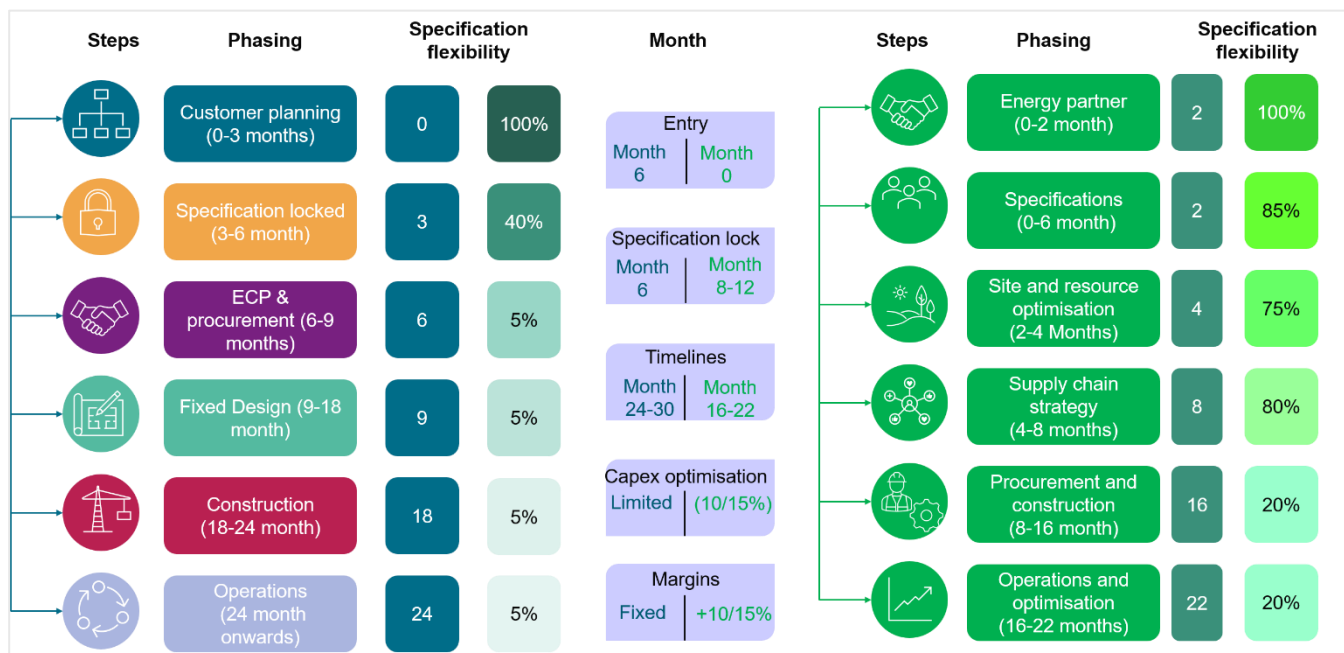
Particulars	Traditional EPC model	Advisory-led integrated model
Project definition	Customer typically determines technology, capacity, location, configuration and commercial structure before tendering	Assesses the customer's requirements and determines the optimal technology, capacity, location, configuration and commercial structure
Advisory & optimisation	Limited pre-tender advisory: EPC contractor primarily focuses on execution against defined specifications	End-to-end energy advisory and optimisation covering RE sourcing, open access, captive/third-party structures, storage, hybridisation and energy-as-a-service options.
Solution configuration	Solution is largely predefined by the customer; EPC bids primarily on price, execution capability and delivery timelines.	Solution is designed around the customer's economics and operating profile, including solar, wind, BESS, hybrid configurations, OA/captive and other suitable models
Project development	Customer may separately appoint consultants/developers for land, connectivity, approvals, offtake and other development activities	Can coordinate or integrate resource assessment, land, connectivity, approvals, project development, offtake structuring and bankability into a single solution
Execution responsibility	EPC contractor is primarily accountable for engineering, procurement, construction and commissioning	Assumes broader accountability across advisory → development → engineering/EPC → commissioning → O&M → optimisation and lifecycle management
Customer interface	Multiple counterparties may be involved across consultancy, development, land, OEM, EPC, O&M	Single-point accountability through an integrated energy partner, reducing coordination complexity for the customer
Post-commissioning engagement	Engagement may reduce substantially after project completion and handover, with O&M generally contracted separately	Longer-term engagement through O&M, performance monitoring, energy optimisation, market participation, asset management and subsequent expansion/hybridisation/BESS additions
Primary value proposition	Project execution and cost competitiveness	Delivered energy economics, reliability, decarbonisation and lifecycle optimisation

Source: Crisil Intelligence

In a traditional EPC model, the customer typically defines the project requirement, technology, and specifications before appointing an EPC contractor. The EPC enters after the solution has largely been determined and focuses on delivering the specified asset.

In an advisory-led model, an energy partner is involved earlier before the technology, capacity, configuration, and commercial structure are fixed. The partner helps translate the customer's energy objectives into an optimised solution and may remain involved through development, execution, operations, and performance optimisation.

Figure 45: Renewable energy project delivery: Two paths to value



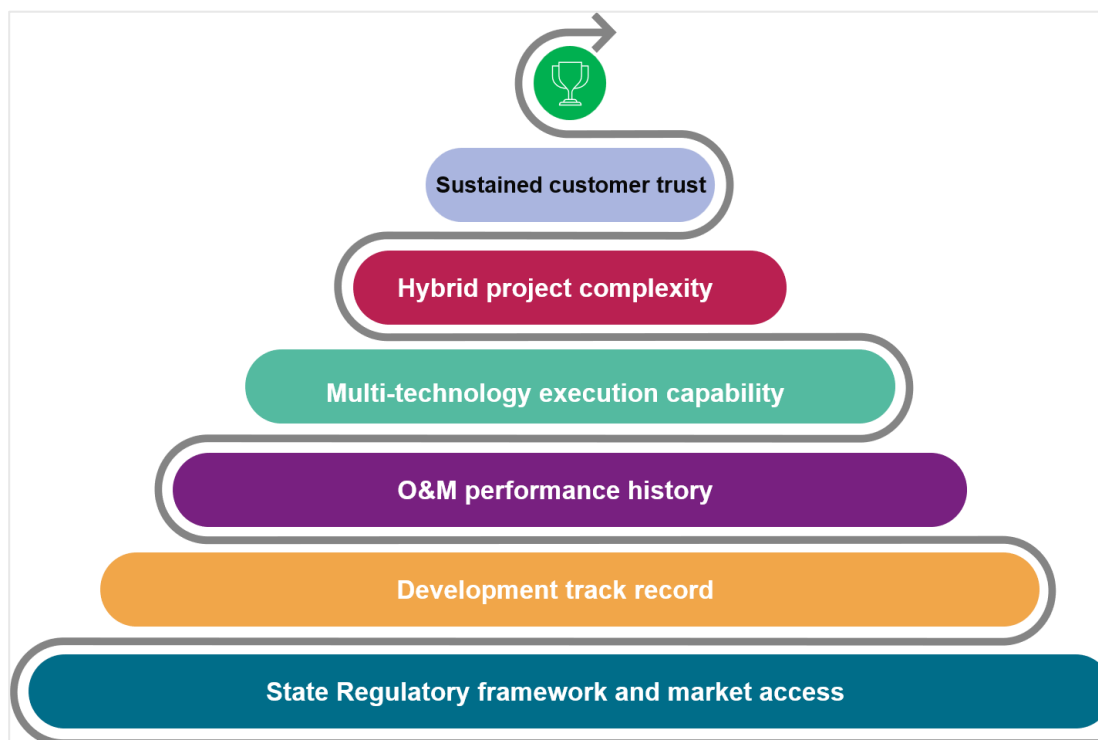
Source: Crisil Intelligence

The absence of an early-stage energy partner may result in a solution that is technically deliverable but not fully optimised for the customer's needs. Potential consequences include:

- specification lock-in before the best technology or configuration has been assessed.
- additional engineering, procurement, or construction costs arising from late changes.
- delayed commissioning due to fragmented responsibility for land, approvals, grid connectivity, or equipment interfaces.
- lower utilisation or energy yield because the asset does not align with the customer's demand profile.
- reduced flexibility to integrate storage, hybrid generation, or future expansion.
- separate contracts and accountability gaps across development, EPC, OEM, and O&M providers; and
- higher risk of stranded or underutilised assets if demand, tariffs, regulations, or operating patterns change.
- Being tied to a sub-optimal solution for the operating life of the asset

The barriers to building a credible RE platform increase from regulatory eligibility at the base to sustained customer trust at the apex.

Figure 46: Barriers for credible RE integrated platform



Source: Crisil Intelligence

- **State regulatory qualification:** Compliance with state-specific regulations, licensing requirements, grid-connection procedures, open-access frameworks, and other eligibility criteria is a fundamental market-entry requirement. Without the necessary approvals and qualifications, participation may be limited, irrespective of technical capabilities.
- **Development track record:** A proven track record in site identification, land acquisition, permitting, grid connectivity, offtake contracting, financing, and project development demonstrates the ability to convert projects from concept to bankability.
- **O&M performance history:** Demonstrated performance in plant availability, response times, safety, maintenance, and operational reliability is critical to building confidence in long-term asset performance and energy yield.
- **Multi-technology execution:** Delivering solar, wind, storage, and associated balance-of-system components across varied project configurations requires broad technical expertise, reliable supply-chain access, strong project management capabilities, and proven commissioning experience.
- **Hybrid project complexity:** Hybrid and firm RE projects involve additional execution challenges, including technology integration, generation forecasting, dispatch optimisation, storage management, grid compliance, and coordination among multiple equipment and service providers.
- **Long-term customer trust:** Sustaining customer confidence throughout the project lifecycle is the most significant barrier. This is built through consistent project delivery, transparent performance reporting, reliable service, predictable commercial outcomes, and timely issue resolution.

Overall, regulatory qualification represents the fundamental barrier to market entry, while long-term customer trust serves as the strongest differentiator and the most difficult capability to replicate. The

inherent barriers across the renewable EPC value chain have led to a fragmented market dominated by specialised providers rather than end-to-end integrators.

5.5 Emerging Industry trends & key customer requirements

C&I customers are increasingly moving from a generation-centric approach to an integrated energy management model, seeking to optimise the cost, reliability, flexibility, and sustainability of their electricity supply. This is driving demand for solutions that combine RE generation, hybridisation, storage, digital energy management, and flexible commercial structures rather than standalone RE projects. For an integrated energy partners, the opportunity lies in designing and continuously optimising the combination of generation, storage, grid power and commercial arrangements around each customer's specific load profile and energy objectives.

Table 14: Key customer requirements

Emerging industry trends	Key customer requirement	Implication for an integrated energy partner
Energy management over power generation	Lower delivered energy costs while maintaining reliability, regulatory compliance and predictable long-term power economics rather than simply adding RE capacity.	Provide end-to-end energy assessment, solution design, technology selection, project execution and ongoing energy analytics to optimise the customer's delivered cost and reliability over the asset lifecycle
Hybridisation of RE assets	Looking to improve RE utilisation and supply profiles without proportionately increasing storage or replacing existing assets.	Provide hybridisation, retrofit and repowering solutions that evaluate the optimal combination of solar, wind, and existing assets based on the customer's load profile and economics.
Storage optimisation & regulatory evolution	Greater flexibility to respond to changing banking, open-access and storage regulations while avoiding unnecessary upfront investment in BESS.	Combine regulatory advisory with hybrid-led system design and modular BESS integration, deploying storage only where its incremental value exceeds its cost
Digital energy management	Continuous visibility and optimisation of energy generation, consumption and power procurement, including forecasting, scheduling, deviations, performance monitoring and identification of cost-saving opportunities	Deploy digital monitoring and analytics across the asset lifecycle for performance optimisation, predictive maintenance and continuous improvement after commissioning.
Flexible commercial models	Ownership and financing structures aligned with their capital availability, risk appetite and energy objectives, rather than a single standardised procurement model.	Offer multiple commercial structures and tailor the ownership, financing and risk allocation to the customer's requirements, allowing customers to adopt renewable energy without necessarily taking full project-development or asset-ownership risk
Expansion beyond traditional C&I	Renewable adoption is expanding beyond large industrial consumers to smaller commercial customers and	Develop modular renewable, energy-service and distributed-energy offerings that can serve smaller customers while

Emerging industry trends	Key customer requirement	Implication for an integrated energy partner
	emerging consumer segments, creating demand for standardised, scalable and lower-complexity solutions	retaining C&I as the core market. This expands the addressable market without requiring every customer to adopt a large, bespoke renewable project

Source: Crisil Intelligence

The evolution of C&I RE procurement is creating customer requirements that extend beyond individual project execution, with increasing demand for development readiness, portfolio-level asset management, distributed energy solutions and ongoing energy optimisation. These trends create opportunities for integrated energy partners to participate across the customer lifecycle and address requirements spanning project development, execution, operations and long-term energy management.

Table 15: Customer requirements

Emerging industry trends	Key customer requirement	Implication for an integrated energy player
Development-ready RE projects	Reduce project lead time and execution risk through access to suitable land, resource, connectivity, approvals and bankable project structures	End-to-end project development covering site and resource assessment, land aggregation, grid connectivity, approvals, project configuration and offtake structuring to deliver development-ready projects.
Independent O&M and asset management	Improve asset performance, reduce downtime and obtain consistent performance across assets, including multi-brand portfolios	Integrated O&M and asset management with remote monitoring, predictive maintenance, performance analytics and lifecycle optimisation across renewable assets
Multi-site C&I energy procurement	Optimise RE procurement across multiple facilities and electricity connections while reducing the complexity of managing individual projects	Portfolio-level energy solutions through customer aggregation, energy assessment, digital monitoring and standardised commercial structures across multiple sites
Distributed energy and VNM / portfolio-based procurement	Access RE across multiple locations where individual rooftop or project-level deployment may be constrained by site availability, load profile or regulations	Integrated distributed-energy solutions combining customer aggregation, regulatory structuring, renewable allocation, digital monitoring and settlement support, where permitted under applicable regulations
Power markets and energy portfolio optimisation	Manage residual grid power, deviations, changing tariffs, renewable intermittency and evolving electricity-market requirements	Energy optimisation and market-linked services covering forecasting, energy balancing and scheduling, procurement optimisation, portfolio management and integration of renewable generation with storage and grid power

Source: Crisil Intelligence

Land-and-expand customer model

An integrated energy partner can potentially follow a “Land-and-Expand” approach, wherein the initial customer engagement provides an entry point for progressively expanding the scope of services. The relationship may begin with advisory and energy assessment, followed by development and execution of the customer's initial renewable energy project. As the customer's energy requirements evolve, the integrated energy partner can subsequently add wind, solar, BESS or hybrid capacity, undertake O&M and asset management, optimise energy procurement and support additional facilities or sites.

This approach can potentially create a substantially different customer economics model from a purely transactional EPC business. Rather than resetting the customer relationship for every new project, the RE solution partner can leverage its understanding of the customer's load profile, tariffs, RE requirements, operating assets, and expansion plans to identify subsequent opportunities. It therefore represents a potential progression from an initial customer acquisition to a broader, longer-term energy partnership.

To summarise: An integrated lifecycle offering can transform a one-time C&I RE project into a long-term customer relationship with multiple revenue opportunities

As C&I renewable energy procurement evolves from standalone capacity addition towards integrated energy management, customer value is increasingly extending beyond the initial EPC transaction. Customers may require a combination of development support, technology and commercial structuring, project execution, O&M, asset management, energy optimisation, and subsequent capacity expansion over the life of their RE portfolio. An integrated energy partner capable of maintaining a single point of accountability across these stages can potentially convert an initial project engagement into a longer-term customer relationship, with revenue opportunities spanning both transactional activities such as development, EPC, hybridisation, BESS and repowering, and recurring activities such as O&M, asset management and energy optimisation. This lifecycle-oriented model can provide opportunities to deepen customer relationships, increase repeat business, and diversify revenue beyond one-time EPC execution.

6 RE Project development, EPC and lifecycle services in India

6.1 Renewable project development

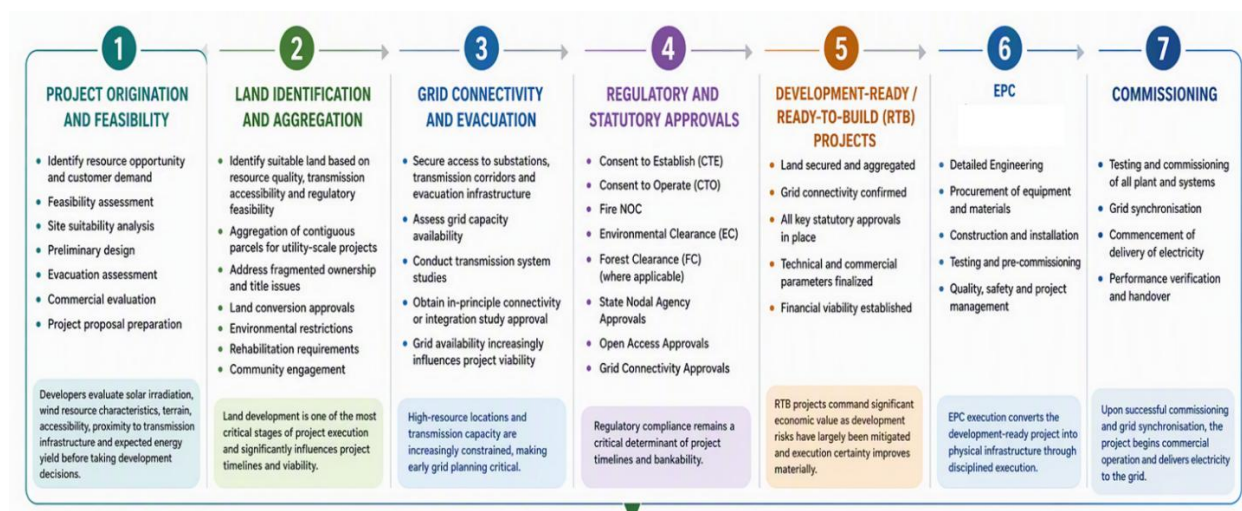
Renewable project development represents the foundation of the RE value chain. The development stage determines project viability, execution certainty and long-term investment attractiveness. Successful project development requires integration of technical, commercial, regulatory and infrastructure considerations before construction can commence.

The development of RE projects in India requires a detailed and extensive process, from conceptualisation to bringing it to Ready to Build (RTB) status. The processes can be broadly divided into the following stages:

- Origination
- Resource/site identification
- Land
- Connectivity
- Approvals
- Project configuration
- Offtake/commercial structure and Bankability
- Ready to Build (RTB)

As high-quality renewable resource sites, land and grid evacuation capacity become increasingly constrained, project development activities are emerging as a distinct source of economic value within the RE value chain. Prime wind and solar resource zones are concentrated in a limited number of states, primarily Rajasthan, Gujarat, Tamil Nadu, Karnataka, Maharashtra and Telangana, where the availability of contiguous land parcels has reduced, land aggregation has become more complex, and connectivity at key pooling substations is often subscribed in advance of commissioning timelines. As a result, the ability to aggregate land with clear title, secure firm grid connectivity and obtain the requisite statutory and regulatory approvals has become a differentiated capability rather than a routine procedural step.

Figure 47: Development of RE Project



Source: Crisil Intelligence

An early-stage project carries high development risk and low certainty of reaching construction, whereas a RTB project reflects substantially lower execution risk and correspondingly higher economic value, as the principal pre-construction uncertainties around land, connectivity and approvals have been resolved.

6.1.1 Project origination

Project origination is the initial phase of the RE project lifecycle, in which a developer identifies a commercially viable opportunity and converts it into a defined project concept capable of being progressed to development. Origination combines resource identification, demand assessment, technical screening, and commercial evaluation, and is a principal determinant of the eventual competitiveness of a project.

Resource identification and assessment: For solar projects, developers evaluate global horizontal irradiance, diffuse and direct normal irradiance, ambient temperature, and soiling profiles, together with satellite-derived and ground-measured data. For wind projects, assessment involves wind speed and wind power density at hub height, wind shear and turbulence intensity, directional distribution, and seasonality, generally supported by mast- or LiDAR-based measurement campaigns and micro-siting studies. Wind resource assessment is typically more data-intensive and requires longer measurement periods than solar and correspondingly involves higher upfront development expenditure.

Site screening and technical evaluation: Preliminary screening considers terrain and slope, soil and geotechnical conditions, flood and drainage characteristics, accessibility for heavy and over-dimensional cargo (particularly relevant for wind turbine components), water availability for construction and module cleaning, and proximity to habitation, forest areas, protected areas, defence installations and eco-sensitive zones. For WSH and FDRE projects, developers additionally assess the complementarity of wind and solar resource profiles at the same location and the feasibility of co-locating energy storage.

Evacuation and connectivity screening: Proximity to existing or planned transmission infrastructure is assessed at the origination stage, including the availability of substation capacity, the voltage level of interconnection, the length and route of the proposed dedicated transmission line, right-of-way considerations, and the anticipated timeline for upstream network availability. Given the increasing scarcity of connectivity at high-resource locations, evacuation screening has become a determining factor in site selection rather than a subsequent consideration.

Commercial evaluation and project proposal. Screening culminates in preliminary design and energy yield estimation, capital cost and operating cost estimation, assessment of applicable transmission charges, losses and other statutory levies, financing assumptions and tariff or return analysis. Sensitivity analysis is generally undertaken on resource variability, curtailment, capital cost, interest rates, module or turbine pricing and commissioning timelines. Projects that meet the developer's internal return and risk thresholds are approved for development expenditure, following which land, connectivity and approval activities commence.

Origination is characterised by the highest degree of uncertainty and the lowest capital commitment in the lifecycle. A significant proportion of originated opportunities do not progress to development, and the capability to identify and prioritise viable opportunities efficiently is an important differentiator among developers.

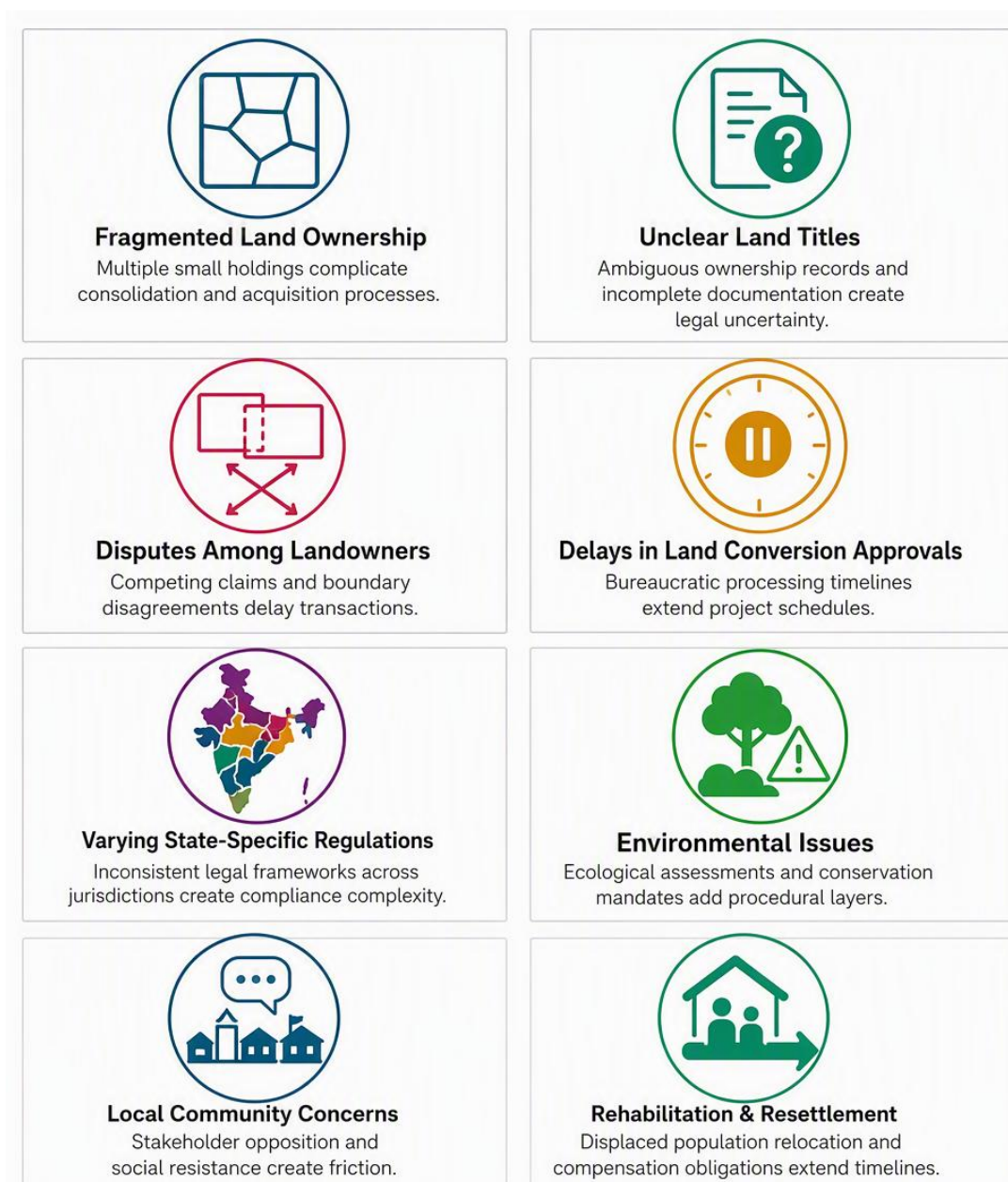
6.1.2 Land and site development

Land aggregation and site development represent one of the most critical, time-consuming, and execution-intensive stages of RE project development and are frequently cited among the principal causes of project delay.

Modes of land acquisition: Land is typically secured through one or more of the following routes: (i) allotment or lease of government or revenue land through state governments, state nodal agencies or RE development agencies; (ii) allotment within notified solar parks, wind parks or RE parks developed on a plug-and-play basis, wherein land, internal infrastructure and evacuation may be provided by the park developer; (iii) direct purchase from private landowners; and (iv) long-term lease of private land, including through registered lease deeds and, in certain states, through agreements with farmer collectives. Developers frequently adopt a combination of these routes, and the mode selected has implications for cost, timeline, title certainty, community relations, and financing.

Land identification and aggregation remain among the most significant challenges for RE developers, particularly for large scale utility projects that require contiguous land parcels. Fragmented land ownership, unclear land titles, disputes among landowners, delays in land conversion approvals, and varying state-specific regulations often prolong the acquisition process. In addition, environmental sensitivities, forest land restrictions, local community concerns, and rehabilitation & resettlement requirements further delay project implementation. Developers therefore adopt a structured land aggregation strategy involving detailed due diligence, stakeholder engagement, and geospatial assessment to minimise project risks, optimise site selection, and ensure timely project execution.

Figure 48: Key Issues in land acquisition in India



Source: Industry, Crisil Intelligence

Following aggregation, site development typically includes topographical and geotechnical survey, land levelling and grading where required, boundary demarcation and fencing, development of approach and internal roads, water and construction power arrangements, and establishment of construction facilities. For wind projects, site development additionally involves road widening, curve improvement and bridge or culvert strengthening to permit movement of blades, nacelles and towers, and preparation of crane hardstands.

Because land aggregation is sequential, negotiation-intensive, and dependent on local factors that cannot be fully controlled by the developer, early and secure land control is a significant contributor to project value and a meaningful barrier to entry.

6.1.3 Connectivity and regulatory approvals

Connectivity to the electricity grid and timely acquisition of the requisite regulatory approvals are critical components of RE project development, as delays in these areas can directly affect project timelines, commercial commitments and the ability to achieve scheduled commissioning.

The complexity varies depending on whether a project is connected to the ISTS or STU/intra-state network, and increases further for projects involving multiple states, open access, captive structures or hybrid configurations. Project developers may need to coordinate with multiple stakeholders, CTU, STUs, SERC, LDCs, Discoms, nodal agencies and other state or local authorities, depending on the project structure and location. The process may involve assessment and grant of connectivity, transmission-system planning and augmentation, OA approvals, evacuation arrangements, metering and protection approvals, scheduling and grid-code compliance, as well as other land, environmental and statutory permissions.

Securing connectivity and key regulatory approvals sufficiently in advance can therefore materially reduce execution and commissioning risk and provide greater certainty over the project's evacuation capacity, development schedule and commercial commitments. An integrated project developer can address this complexity through early-stage site and grid assessment, identification of the optimal connectivity point and transmission configuration, preparation and submission of applications, engagement with relevant authorities and utilities, and continuous monitoring of approval and augmentation milestones. This capability can enable customers to access development ready RE projects with greater visibility on grid connectivity, regulatory readiness and project timelines, while reducing the need for customers to independently navigate multiple regulatory and technical interfaces.

6.1.4 Ready-to-build projects and development value

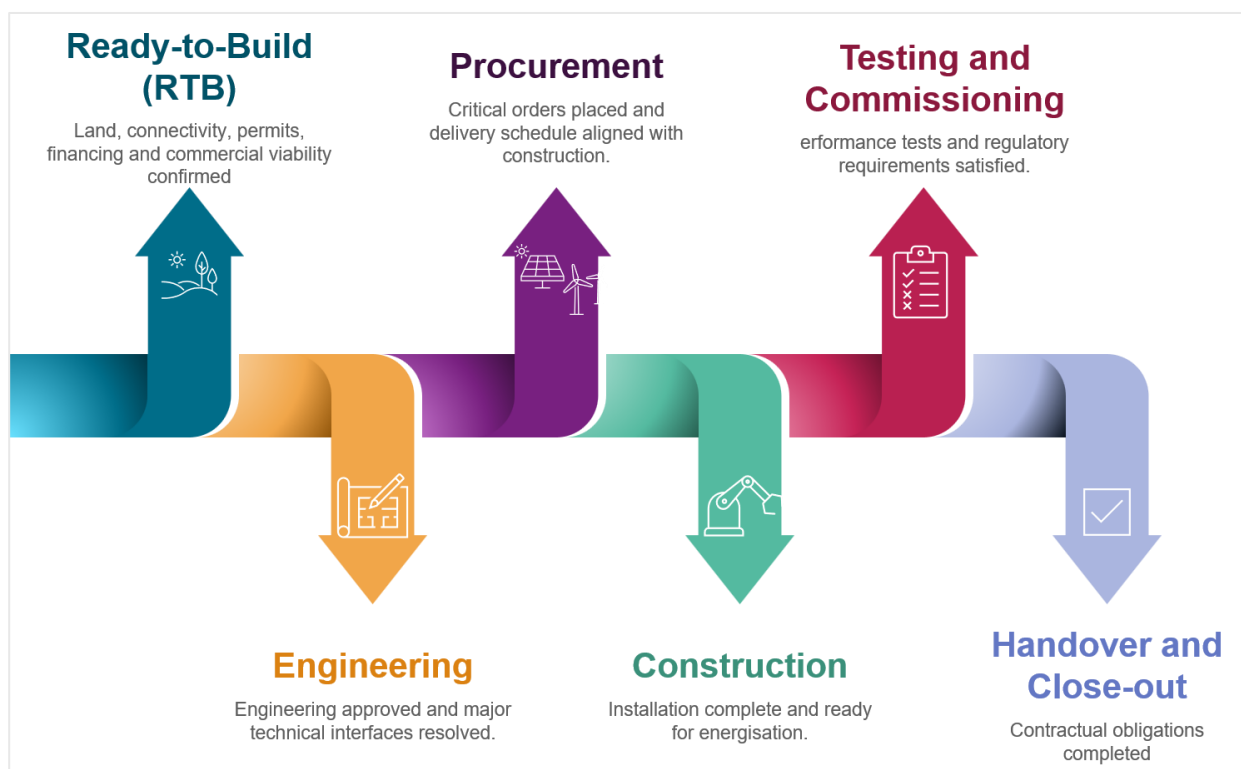
A project is generally regarded as RTB when the material pre-construction requirements have been secured such that construction can commence without further significant development risk. An RTB project typically comprises:

- land secured through ownership, registered long-term lease or government allotment, with satisfactory title diligence, adequate contiguity or micro-site coverage, and requisite land use conversion.
- bankable resource assessment and energy yield estimation, and a substantially frozen technical configuration, including capacity, technology, layout and, where relevant, storage sizing.
- firm grid connectivity or network access at a defined substation and voltage level, with a defined interconnection scheme and identified route and rights for the dedicated transmission line.
- material statutory and regulatory approvals obtained or clearly identified as available on application in the ordinary course, with adequate residual validity.
- identified or contracted offtake, or an established route to market, together with clarity on applicable charges, losses, and open access framework; and
- defined project cost, indicative construction schedule and, in many cases, advanced or concluded discussions on principal equipment supply and EPC arrangements and a defined financing plan.

6.2 Renewable EPC market

The renewable energy EPC lifecycle begins with an RTB project and progresses through engineering, procurement, construction, testing and commissioning, followed by handover to operations. Effective stage-gate governance and interface management are essential to deliver the project safely, on time, within budget and at the contracted performance level.

Figure 49: Renewable energy EPC project stages

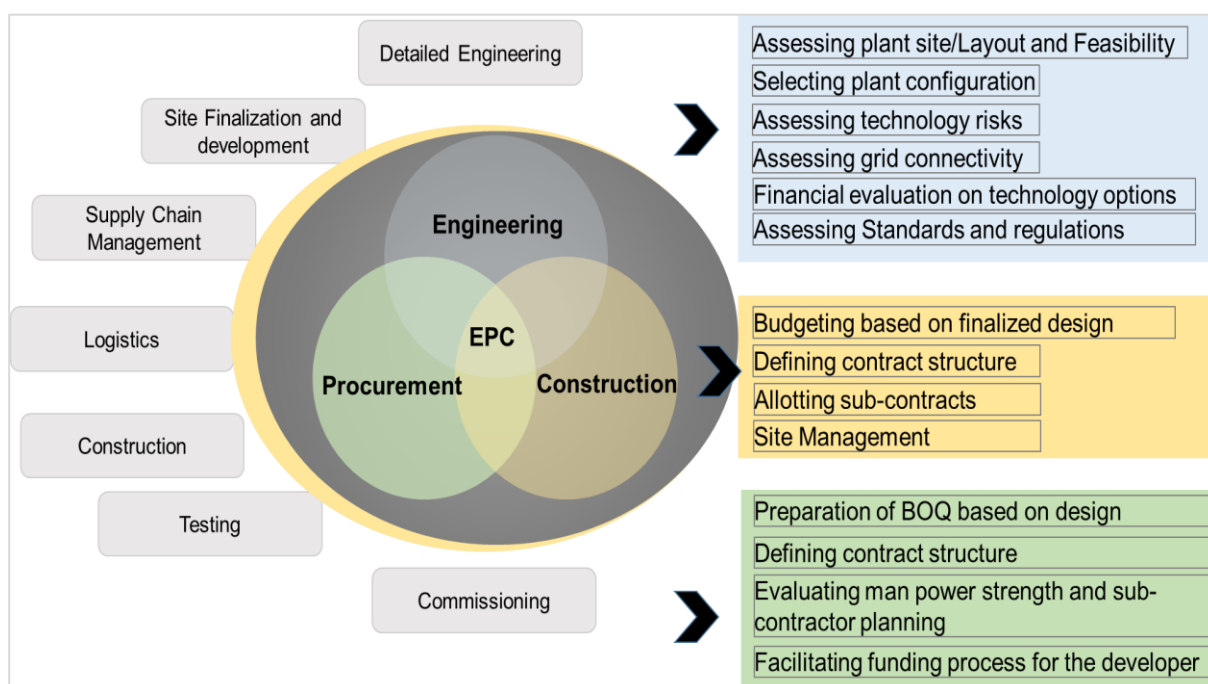


Source: Crisil Intelligence

6.2.1 Scope and execution models

Nations, majorly developing ones, have been investing heavily on large infrastructure projects through public as well as private investments. To ensure efficient and timely construction, it is imperative to have an effective model which ensures timely project execution, minimise construction delays and improve transparency. The EPC model is primarily used in construction of RE plants.

Figure 50: Checklist of an EPC model



Source: Crisil Intelligence

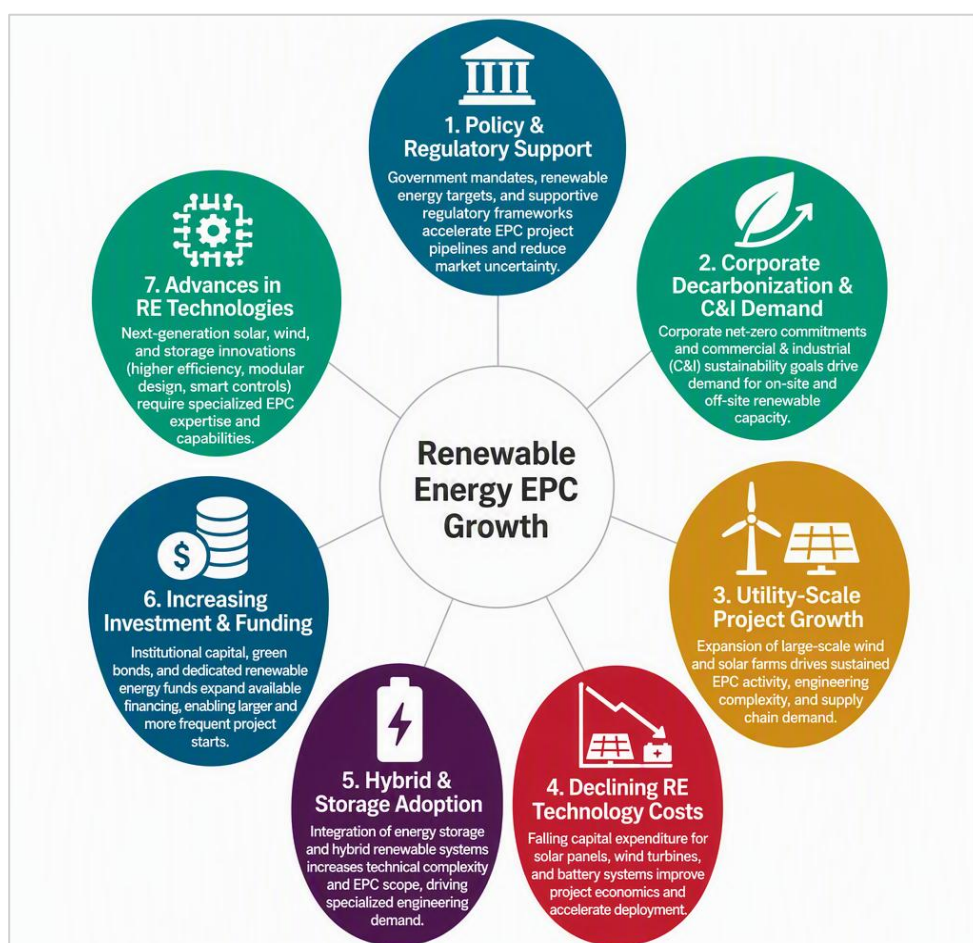
Under turnkey project structure, the contractor holds full responsibility of design and execution of the works, including EPC. Therefore, the contractor makes the facility ready to be used at the turn of a key. The project must be delivered at a pre-determined time and pre-determined cost, and the contractor must adhere to project specifications. In case of deviations, the contractor is liable to pay monetary compensation.

In case of balance of plant (BoP) structure, the entire project is broken into multiple packages with a major chunk contracted through EPC route and the rest through BoP.

6.2.2 Key drivers for RE EPC Services

Growing demand for clean energy from industries, commercial establishments, and residential sectors is driving the growth of the RE EPC market. Declining costs of RE technologies, such as solar and wind, are making them more competitive with fossil fuels, and driving demand for RE EPC services. Increasing investment and funding from government and private sources are supporting the growth of the RE EPC market. Advances in RE technologies, such as bifacial solar panels and larger wind turbines, are improving efficiency and reducing costs, and driving demand for RE EPC services.

Figure 51: Key drivers for RE EPC services



Source: Industry, Crisil Intelligence

However, RE EPC services also face certain challenges. Intense competition in the RE EPC market is leading to pricing pressure, making it challenging for EPC players to maintain margins. Land acquisition and permitting issues can delay project timelines and increase costs, making it challenging for EPC players to deliver projects on time and within budget. Shortage of skilled labor can hinder project execution and quality. Ensuring grid integration and connectivity for RE projects can be challenging, particularly in areas with limited grid infrastructure.

Overall, the RE EPC market in India is expected to grow significantly in the coming years, driven by government policies, declining costs, and increasing demand for clean energy.

6.2.3 EPC and integrated energy partner selection criteria

RE projects are characterised by fixed commissioning deadlines that leave limited headroom for cost overruns, and long asset lives during which generation performance directly determines returns. The selection of an EPC and solutions partner is therefore a significant decision for project owners, and is typically evaluated across the following parameters:

Figure 52: Partner selection criteria

	Execution track record	Demonstrated experience in commissioning projects of comparable scale, technology and terrain, including a history of meeting scheduled commercial operation dates. Cumulative megawatts executed, repeat business from established clients and performance on projects in challenging geographies are commonly used indicators of execution reliability.
	Financial capability	Adequate net worth, liquidity and access to working capital and non-fund-based limits, as EPC contracts require the contractor to fund procurement cycles and furnish performance and advance bank guarantees. Balance sheet strength also determines the credibility of the contractor's warranty and liquidated damages obligations, and is closely examined by lenders during appraisal.
	Technology and OEM relationships	Established relationships with module, inverter, turbine, tracker, transformer and battery manufacturers, which support access to preferred technology, priority allocation during periods of supply tightness, competitive pricing and enforceable warranty and after-sales support. The ability to remain technology-agnostic while advising on appropriate technology selection is an added consideration.
	Procurement capability	Ability to manage a multi-vendor, partly import-dependent supply chain, including vendor qualification, order sequencing, logistics and customs handling, inventory planning and management of price and currency volatility. Procurement capability also encompasses compliance with domestic content and approved model and manufacturer requirements where applicable under specified schemes.
	Regulatory and grid understanding	Familiarity with central and state regulatory frameworks, connectivity and open access procedures, grid code and metering requirements, and approval processes across states. This supports timely clearances, compliant design and smooth grid synchronisation and charging.
	Project management systems	Structured planning, scheduling and progress monitoring systems, digital project controls, quality assurance and quality control protocols, subcontractor and manpower management, and transparent reporting to owners and lenders' independent engineers. Robust systems are particularly important where multiple projects are executed concurrently across locations.
	Multi-technology capability	Capability across solar, wind, hybrid, energy storage and associated transmission and substation works, enabling integrated delivery of increasingly complex configurations such as firm and dispatchable renewable supply. Single-vendor capability across technologies reduces interface risk and coordination effort for the owner.
	Safety record	Demonstrated health, safety and environment performance, including safety management systems, training, incident reporting and lost-time injury metrics. Safety performance is a material consideration for institutional owners and lenders, and is increasingly linked to environmental, social and governance reporting requirements.
	Performance guarantees	Willingness and ability to provide contractual assurance on capacity, performance ratio or plant load factor, availability and, where relevant, degradation, supported by defined testing protocols, liquidated damages for delay and shortfall, defect liability obligations and appropriately sized security. The credibility of such guarantees depends on the contractor's financial standing.
	Operations and maintenance capability	In-house O&M capability, including preventive and predictive maintenance, remote monitoring and analytics, module cleaning and vegetation management, spares management and response times. Continuity between construction and O&M improves accountability for long-term generation performance and preserves design intent over the asset life.
	Single-point accountability	Preference for an integrated turnkey scope covering design, engineering, supply, construction, testing, commissioning and, in many cases, subsequent O&M, so that interface and coordination risk rests with the contractor rather than the owner. Single-point responsibility simplifies contract administration, clarifies recourse in the event of underperformance and is generally viewed favourably by lenders.

Source: Crisil Intelligence

6.2.4 Integrated vs fragmented execution

In C&I renewable energy deployments, project execution efficiency serves as a primary determinant of long-term financial viability. As C&I consumers increasingly transition from standard PPA to complex multi-technology setups, the delivery mechanism dramatically dictates asset lifecycle performance.

The industry is undergoing a structural transition from the traditional Fragmented Project Delivery paradigm to an evolved Integrated Execution Model. Choosing the right project framework directly impacts timeline, costs, and lifecycle performance. The table below delineates the structural, operational and commercial distinctions between these two models.

Table 16: Delivery models for project execution

Particulars	Fragmented Project Delivery	Integrated Execution Model
Contractual structure and risk	Multiple sub-contractors split across discrete packages (e.g., module/WTG procurement, civil works, electrical and BoP) resulting in fragmented warranties and split legal recourse	Single-point accountability under a turnkey framework. Minimises execution leaks and provides comprehensive recourse for project milestones
Interface management	High project management overhead for the customer to visualise and manage diverse technical, contractual, and coordination interfaces. Requires in-house engineering and regulatory teams.	The integrated energy partner fully manages underlying technical and operational complexity, presenting a simplified interface to corporate clients who can focus on core competencies
Technology integration	Technology choices locked prematurely at the tender stage. EPC players do not focus on suggesting the technology configuration, acting strictly as build-to-print contractors	Technology agnostic selection. The integrated partner understands the load profile first and then suggests the best mix of RE technology for maximising savings and returns
Engineering & design optimisations	Disjointed execution where plant engineering is decoupled from direct procurement, often leading to site-level alterations, component mismatches, and sub-optimal land/rooftop utilisation	Utilises an architecture data-driven and integrated engineering core. Procurement realities, local geographic constraints, and simulation data directly drive real-time design optimisations
Commercial structures & cost certainty	Vulnerable to multiple cost layers, stacked margins, and frequent change orders. Delays in one package trigger financial penalties and cascading delays across other sub-vendor contracts	Characterised by fixed-price turnkey contracts or unified tariff structures under long-term corporate PPAs (Capex or Opex/RESCO models). Provides absolute capital cost certainty to institutional investors and C&I consumers
Regulatory, DISCOM, & Open Access Approvals	Segmented management of statutory permissions. Separate vendors coordinate net-metering, state-level OA approvals, CEIG clearances, and bay allocations. High risk of regulatory roadblocks	Centralised regulatory compliance management. Handled by dedicated liaisoning team navigating inter and intra-state related approvals and grid connectivity related aspects seamlessly

Particulars	Fragmented Project Delivery	Integrated Execution Model
Lifecycle commitment	Limited post-COD involvement from EPC vendors once the full commencement of the project along with performance test is achieved and the liability period is over (max up to 2-3 years)	O&M and optimisation continue seamlessly across the asset's active life. Overall delivered-energy optimisation focused on increasing CUF and maximising commercial returns

Source: Crisil Intelligence

The integrated execution model serves as a vital risk-mitigation vehicle against volatile open access policies, banking regulations, varying charges and strict DSM regulations. By anchoring the entire value chain – from initial feasibility screening and structural vetting to commercial commissioning and digital asset management – integrated energy partners consistently ensure optimum cost of capital, compressed commissioning timelines and predictable tariff benefits over the standard 25-year asset lifespan. Project execution is essential but not sufficient, as the customer's needs extend well beyond commissioning to ongoing operations, optimisation, regulatory support, and lifecycle performance management.

Because the optimal renewable energy configuration depends on the customer's load profile, suppliers must remain technology- and configuration-agnostic. Most providers, however, are constrained by their existing offerings, creating an opportunity for an advisory-led partner to solve the customer's broader energy challenge rather than sell a predetermined solution.

6.2.5 Solar, wind, hybrid and BESS execution capabilities

EPC execution requirements vary significantly across standalone and hybrid RE projects, with hybrid configurations requiring a broader set of engineering, integration and commissioning capabilities. Standalone solar EPC generally involves relatively standardised and technology specific execution, while wind projects require greater site, resource and technology integration expertise.

The complexity of hybrid projects increases as it requires optimisation of the wind-solar configuration and shared evacuation infrastructure, coordinated execution across multiple technologies and OEMs, integration of SCADA/EMS and plant-level controls, and development of operating logic for BESS charging, discharging and renewable-energy dispatch. The contractor must also undertake integrated scheduling, grid-compliance testing and performance validation across technologies, while ensuring that the combined plant achieves its contracted generation and availability requirements.

Accordingly, multi-technology EPC capability represents a higher execution threshold than standalone renewable EPC capability. The ability to consistently execute such projects therefore represents an additional barrier to entry for EPC players whose capabilities are primarily limited to standalone renewable projects.

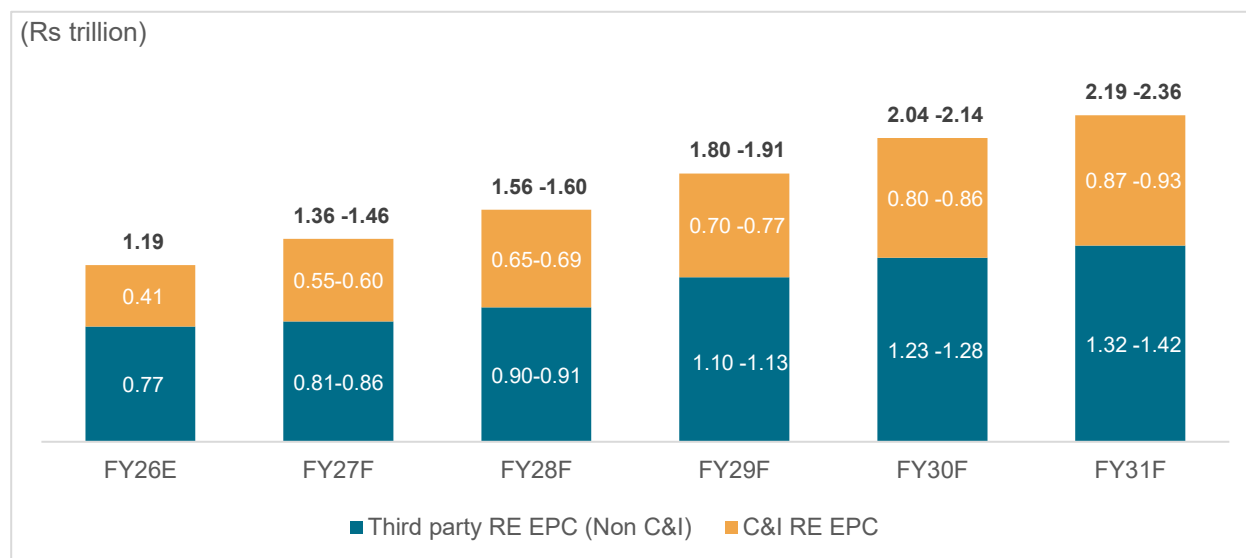
6.2.6 EPC market size FY27–31

The Indian RE EPC market is expected to witness sustained growth over the next five years, supported by continued capacity additions across utility scale, open access, rooftop and distributed RE projects. The total solar and wind capacity addition during fiscal 2027-2031 is estimated to be in the range of 330 – 380 GWp. Of the overall capacity additions, approximately 180 – 210 GWp is expected to be executed through third-party EPC providers. Within the third-party EPC market, C&I linked projects are estimated to account for approximately 40%, driven by increasing adoption of RE by C&I consumers, while the remaining 60%

is expected to comprise non-C&I EPC opportunities, primarily across utility-scale and other non-C&I RE projects.

A solar EPC cost typically ranges from Rs. 40,000 -45,000/kWp and a wind EPC ranges from Rs. 80,000 -90,000/kW. The total cumulative EPC cost is estimated at ~Rs 9.0 – Rs 10 trillion over fiscal 2027-2031 with annual average of ~Rs 1.8 – 1.9 trillion and representing an estimated CAGR of 12-13%.

Figure 53: India- annual RE EPC demand forecast during Fiscal 2027 to 2031



E: Estimated; F: Forecast; Source: Industry, Crisil Intelligence

6.3 O&M services for RE projects

Operations and maintenance cost form a critical component in the Levelised Cost of Electricity (LCOE) of solar, wind as well as hybrid energy projects. Therefore, renewable power producers adopt various strategies to lower this cost and improve the reliability of the services and to make the LCOE competitive with other producers and technologies. Typically, the O&M cost accounts for 7-10% of the LCOE for solar projects and 13-16% for wind projects involving plant operations and monitoring, preventive, corrective and breakdown maintenance, routine consumables and maintenance materials and associated site administration, reporting and performance monitoring activities. However, the actual scope of services varies by contractual agreements. These costs generally rise with the years of operation, however, with more experience various approaches have been formulated that generators follow to keep a check on the O&M costs.

The industry started with reactive maintenance, mostly resulting in downtime and major service or overhaul. The major reason behind this was late detection of the fault or part failure. The cost of service due to this late detection increases compared to early detection and repair. Moreover, with an increase in the life of the plant's generation the cost of major repairs escalates. Currently, the O&M service providers offer solutions to minimise the major repairs and replacements while moving towards predictive maintenance resulting into decrease in cost of maintenance.

6.3.1 O&M models: OEM / in-house / independent

There are three channels to obtain the operation and maintenance for RE projects:

Original Equipment manufacturers: The role of OEMs in O&M differs materially between solar and wind projects. For solar projects, OEM-provided O&M is generally concentrated on critical equipment such as solar inverters, with the initial equipment supply typically accompanied by a 5-year warranty/service period, after which the IPP may procure a separate AMC from the OEM. For wind projects, O&M of WTG is more closely integrated with the WTG supply agreement, with OEMs typically providing an initial free-O&M period of up to two years, followed by paid O&M arrangements of up to 5-25 years that may be extended depending on the contractual terms.

Renewable Energy Developers: Large renewable energy developers have also started to maintain their own capacity instead of traditionally OEM based O&M contracts. Currently, there is limited capacity from some of the key renewable energy developers being maintained in house. The major reason for this is to increase control over the operations of the generation and further reduce the dependence on the OEMs. This also reduces the risk of extended periods of shutdowns resulting in no drop in the availability of the plant. Moreover, due to financial hurdles for OEMs, IPPs have opted to carry out O&M activities in-house to avoid dependence on any external entity.

Third-party or Independent Service Providers: O&M services are also offered by third-party service providers. This is usually taken at the time of warranty expiration of the Annual Maintenance Contracts (AMC) with the OEM or when the OEMs are not preferred with the equipment acquisition due to financial constraints.

EPC Players offering O&M: An EPC player providing O&M services delivers end-to-end lifecycle support from initial design and plant construction to long-term performance monitoring, preventive upkeep, and prompt technical troubleshooting.

Figure 54: Value chain for renewable energy



Source: Industry, Crisil Intelligence

6.4 O&M and asset-performance-management market

As the Indian RE fleet has scaled, aged and diversified across technologies and OEMs, and as ownership has shifted towards institutional and financial investors with defined return and reporting obligations, O&M has evolved into a discipline that directly influences generation, revenue realisation, contractual compliance and asset valuation. In parallel, the emergence of the C&I open access segments where every

unit of shortfall translates into a lost contracted sale or an unhedged market exposure has increased the economic sensitivity of asset performance.

The transition from obligated O&M towards professional independent providers is being driven by a set of mutually reinforcing structural factors rather than by any single cyclical development.

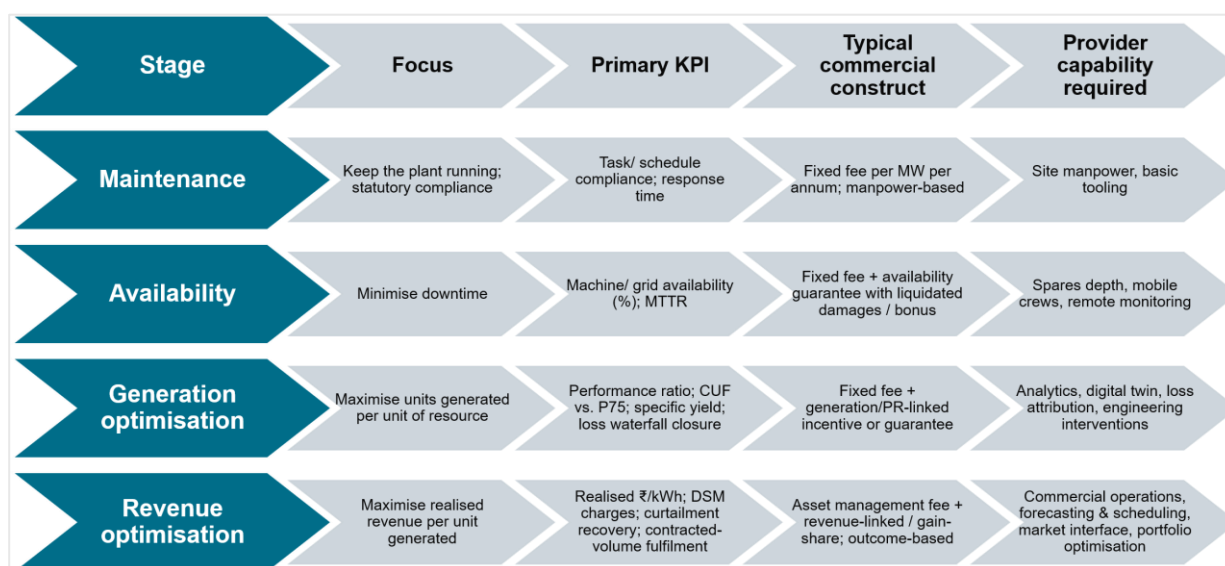
6.4.1 Broadening of scope: O&M with asset management and performance optimisation

The service scope contracted extends well beyond preventive and corrective maintenance and comprises three broad interlocking layers:

Layer 1 — Technical O&M (the maintenance foundation): Preventive and scheduled maintenance; corrective and breakdown maintenance; predictive and condition-based maintenance using vibration, thermal, oil-quality and electrical analytics; solar module cleaning (manual, semi-automated and robotic, including waterless systems); vegetation, drainage, erosion and site management; wind-turbine blade inspection and repair; inverter, transformer, switchgear and HV apparatus maintenance; BoP, pooling substation and transmission-line maintenance; spares management, inventory optimisation and reverse logistics; statutory and HSE compliance; and OEM warranty administration and claim enforcement on behalf of the owner.

Layer 2 — Asset management (the digital and administrative layer): 24x7 remote monitoring and centralised operations control; asset digitisation, tagging and creation of a single digital asset register and document repository; digital twin and performance-model construction; automated loss accounting and waterfall decomposition of generation shortfall into technical, environmental, grid, curtailment and contractual causes; energy accounting, invoicing support, open access scheduling and deviation management; contract, land, statutory and lender-covenant compliance management; insurance administration and claims support; and standardised, audit-ready owner, lender and rating-agency reporting.

Layer 3 — Performance optimisation (the commercial layer): Data-driven yield enhancement including inverter and tracker set-point optimisation, string-level fault elimination, cleaning-cycle optimisation against soiling and irradiance data, reactive power and clipping management; wind-farm-level wake, yaw-misalignment and power-curve optimisation and re-calibration; targeted retrofit, uprating, repowering and hybridisation studies and execution; BESS dispatch and augmentation planning; portfolio-level benchmarking and identification of underperforming assets; and, at the frontier, revenue optimisation across contracted, exchange and DSM interfaces.

Figure 55: O&M value ladder


Source: Crisil Intelligence

First two stages are increasingly commoditised, manpower-intensive and price-competitive, whereas last two stage services require accumulated fleet data, proprietary analytics, engineering depth and the ability to underwrite performance attributes that are difficult and time-consuming to replicate and that support higher realisations, longer contract tenures, higher renewal rates and greater revenue visibility.

6.4.2 O&M and asset-performance-management market drivers

Rapid growth in the installed RE base: India's installed solar and wind capacity cumulatively has increased from approximately 34 GW as of March 2016 to approximately 219 GW as of June 2026, comprising solar of 162 GW and wind of 57 GW. With over 170 GW of capacity under construction or awarded and pending commissioning, the serviceable asset base is expected to expand to over 500 GW by fiscal 2030.

The fleet is ageing and maintenance intensity rises with vintage: An estimated 60% of India's operating wind capacity and approximately 18% of its operating solar capacity is more than 7-8 years old. Ageing assets exhibit higher failure rates in critical sub-systems (gearboxes, generators, blades, pitch and yaw systems in wind; inverters, transformers, trackers, cabling and combiner boxes in solar), progressive module degradation and soiling-related losses, and increasing incidence of BoP and HV equipment issues. Maintenance intensity, spares consumption and the technical sophistication required therefore increase over the asset life, precisely at the point at which OEM cover typically lapses.

Expiry of OEM warranties and long-term service agreements: Comprehensive OEM warranty and long-term service agreement periods in India typically extend to 2–5 years for solar and 5–10 years for wind. At the point of expiry, the asset owner is required to make an explicit make-or-buy decision, and OEM renewal quotations frequently repriced upwards and restricted to the OEM's own scope create a natural entry point for independent providers offering broader scope at competitive commercial terms.

Portfolios are increasingly multi-OEM and multi-technology: Asset owners have procured across successive tender cycles from a wide range of WTG OEMs, inverter manufacturers, module suppliers and tracker vendors, and are increasingly adding BESS to existing solar and wind sites. Managing such a portfolio through separate single-brand OEM contracts results in fragmented accountability, non-

comparable performance reporting, duplicated site manpower, uneconomic spares inventories and disputed root-cause attribution. This fragmentation creates a structural preference for a single, brand-agnostic service counterparty.

The secondary market for operating assets is deepening: Transaction activity in operating RE assets in India has increased and acquirers of operating assets characteristically (i) lack legacy site organisations and therefore outsource O&M from day one; (ii) require independent, bankable technical due diligence, performance benchmarking and remedial capital expenditure plans; and (iii) mandate standardised reporting across heterogeneous acquired assets. Secondary market churn therefore converts captive O&M scope into contestable third-party scope.

Institutional, private equity, InvIT and sovereign-fund ownership has raised governance standards: A significant share of India's operating RE capacity is now held by financial and institutional investors, including infrastructure funds, pension and sovereign wealth funds, and listed InvITs. Such owners typically require independent verification of performance, contractual KPI regimes with quantified liquidated damages, auditable energy accounting, standardised HSE systems, ESG and sustainability-linked disclosures, and lender- and rating-agency-ready reporting. These requirements favour professional service providers with institutional-grade systems, documentation and governance over both OEM service arms (whose reporting is oriented to their own equipment) and sub-scale local contractors.

Owners are increasingly intolerant of downtime and are contracting for it: With tariffs having compressed across successive competitive bidding cycles and, in the C&I open access segments, with realisations linked to bilateral contracts, exchange prices and avoided grid tariffs, the revenue value of avoided downtime has risen relative to the cost of the O&M contract itself. Reduction in mean time to repair, faster fault triage, pre-positioned critical spares and effective vegetation, soiling and thermal management have therefore become primary evaluation criteria in O&M procurement, alongside price.

Accountability is shifting from availability to yield: Legacy O&M contracts guaranteed machine or plant availability. Owners are consequently migrating towards generation-based, performance-ratio-base and, increasingly, revenue-linked constructs. This shift structurally favours partners with analytics capability and the willingness and balance sheet to underwrite generation outcomes, and disadvantages providers offering manpower-based services alone.

Digitalisation has reduced dependence on OEMs: The proliferation of SCADA and plant-controller data access, IoT-enabled sensing and condition monitoring, drone- and robot-based thermography and module inspection, string- and module-level monitoring, digital twins, and machine-learning-based anomaly detection has materially narrowed the historical information asymmetry between OEMs and third-party service providers. Independent providers operating centralised remote operations centres are now able to detect, diagnose and, in many cases, remotely rectify faults across multiple OEM platforms and geographies, and to benchmark performance across a fleet a capability that single-brand OEM service organisations are structurally unable to replicate across a mixed portfolio.

However, the O&M services in India also face certain challenges. Most of the RE assets, such as wind farms and solar plants/parks, are located in remote areas, making it challenging to access and maintain them. The shortage of skilled labor in the RE O&M sector, particularly in areas such as turbine maintenance can hinder the growth of the industry. The lack of standardised data management practices and limited use of digital technologies can make it difficult to optimise RE O&M services and predict maintenance needs. RE sources, such as solar and wind, are subject to seasonal variations, which can impact performance and require specialised O&M strategies. The integration of RE sources into the grid can be challenging, requiring specialised expertise and equipment to ensure smooth and efficient operation.

Overall, the RE O&M services market in India is expected to grow significantly in the coming years, driven by government initiatives, technological advancements, and increasing demand for efficient and effective maintenance services.

6.4.3 Growth of independent multi-brand O&M

Independent providers with genuine multi-brand capability are able to offer:

- **Multi-OEM wind:** servicing of multiple WTG platforms and ratings, including legacy and orphaned fleets, with cross-platform blade, gearbox, generator, converter and control-system competence, and access to reverse-engineered, refurbished and alternative-source spares.
- **Multi-vendor solar:** servicing across central, string and hybrid inverter brands, multiple module technologies and generations, and multiple tracker and mounting-structure vendors.
- **Cross-technology and hybrid assets:** integrated solar, wind, WSH and BESS servicing at a single site or pooling point, including co-located hybrid dispatch optimisation and BESS state-of-health management, capacity augmentation planning, and cycle-life management.
- **BOP, HV and evacuation infrastructure:** inverter and pooling stations, transformers, switchyards, HV/EHV switchgear, protection and metering, internal cabling, and transmission lines up to the interconnection point scope that is typically outside OEM cover but is a material contributor to unavailability and loss.
- **Single-window accountability:** one counterparty, one KPI regime, one reporting standard, one HSE framework and one escalation matrix across a heterogeneous portfolio, eliminating inter-vendor attribution disputes.

Multi-brand capability generates advantages:

- a) **Contract addressability:** A brand-agnostic provider can bid for a whole portfolio, including assets whose OEMs have exited or are unwilling to service. Its addressable market is the entire operating fleet, rather than a single OEM's installed base.
- b) **Cross-fleet learning:** Servicing multiple platforms across many sites and resource regimes generates a comparative performance dataset that enables identification of systematic platform-level failure modes, credible P75/P90 benchmarking and evidence-based warranty and insurance claims an information advantage unavailable to any single-brand servicer and unavailable to the owner in isolation.
- c) **Spares and logistics economies:** Pooled, multi-brand spares inventories, alternative and refurbished sourcing, and shared specialist tooling and crane/access assets reduce both cost and mean time to repair relative to brand-siloed inventories.
- d) **Standardisation and integration:** A common digital platform, tagging convention and reporting standard across a mixed fleet is a prerequisite for portfolio-level optimisation, for InvIT grade reporting and for efficient transaction diligence and is therefore valued by institutional owners at a premium.

- e) **Alignment of interest:** As an independent party with no equipment sales interest, the provider's root-cause analysis and remedial recommendations are commercially neutral, which is precisely what lenders, insurers and financial owners require.
- f) **Adjacency to the integrated model:** For a player that also develops, owns and operates C&I open access capacity, multi-brand O&M capability is self-reinforcing: internal assets provide the initial fleet, data density and reference credentials from which third-party mandates can be won, while third-party fleet data improves the optimisation of owned assets.

6.4.4 Digital and predictive asset management

Predictive maintenance:

Major components such as module, inverters, nacelle, turbine, generator, hydraulics, and electronics require constant monitoring to ensure smooth operability. Traditionally reactive maintenance was carried out when the equipment has shown severe operational faults or complete failure or during scheduled maintenance. The failure of reactive maintenance resulted in longer shutdowns and low availability of the generators.

On the contrary, predictive maintenance accurately forecasts the component failures before they occur based on historical data. This is a critical service in RE generation as it addresses the issues associated with reactive maintenance. Furthermore, proactive maintenance reduces the operational cost by reducing the wear and tear of the equipment in the system. As a result of implementation of predictive maintenance techniques, replacements and major repairs in wind energy have declined considerably over the past years. Therefore, more and more developers are going for predictive maintenance as compared to reactive and scheduled maintenance.

The disadvantage of solely relying on historical data is that some of the assets will be flagged for maintenance despite being functionally sound and performing well. This drives the maintenance cost of the assets into good operational conditions. Hence it needs another layer of technological intervention that segregates, analyses and prioritises the activities. Thus minimise the unwanted spending on maintenance of assets that are in healthy condition. AI and IoT enablement can solve this purpose and lead to cost savings, especially in the case of power plants having assets spread across large areas. For instance, in large solar parks predictive maintenance using AI can lead to significant savings of inspection and maintenance planning of the assets. Moreover, it also aids capacity planning and addition over the years. In the case of distribution networks, precision mapping of the assets leads to proper identification and location of the assets in the network for maintenance and addition of the consumers.

Condition monitoring

A key element of predictive maintenance suite, condition monitoring, is used to monitor the health of the module and turbine and related electrical systems. Its purpose is to predict maintenance issues so site operators can conduct repairs and replacements only when needed to avoid unnecessary and costly up-to-tower jobs. Although the intent is to cut time and cost from O&M tasks, condition monitoring system have become rather detailed in accumulating and analysing data and hence can become expensive. The system can supply the failure rate of the turbine and related equipment for up to 20 years along with simulations to predict future failures thereby reducing the cost of energy considerably by avoiding major replacements. In this context, drones with HD cameras and thermal sensors are becoming a popular solution globally to resolve a number of issues that module and wind equipment managers face by helping them inspect

damage and thermal imaging to detect any internal abnormalities. This mode of airborne inspection is particularly more popular for solar and offshore wind turbine sites.

Impact of technology implementation on asset management services

Technology driven solutions are constantly evolving and are being deployed across the sector. For instance, the implementation of AI-based predictive analytics solutions can help analyse real-time operational data at scale, improve generation forecasting, identify any underperformance, and recognise failures using machine learning. The most advanced implementation of technology across wind energy includes digital twin. Digital twin is a virtual representation of the actual asset for remote interaction by the trained professional. Digital twin minimises the need of human team diagnosing the faults, instead engineers can analyse and identify problems remotely. Thus, saving the efforts of personnel to specific instances. Applications such as demand response management, hybrid energy storage systems, vehicle-to-grid, virtual power plant etc are being tested across the clean energy sector involving implementation of new technologies. Machine learning (ML) and AI can also be applied to recognise patterns, predict faults, and highlight important variables that could otherwise be missed.

6.4.5 Industry participants

Leading wind energy O&M providers in India include Inox Green Energy Services, Renom Energy Services, and Powercon India. These companies specialise in maximising turbine availability, handling balance-of-plant (BoP) infrastructure, and providing multi-brand asset management.

Table 17: Key competitors and their service offerings

Company	O&M Capacity	Key service offerings	Geographical presence
Inox Green Energy Services	13+ GW (10.5 GW Wind and 2.5 GW Solar)	O&M Solutions for WTGs, solar & hybrid; Value-added services, which includes refurbishment, booster sales, carbon credit trading, amongst other offerings	13 States in India
Renom Energy Services	3.2+ GW	O&M of wind, solar and BESS energy Assets, Value Added Services, Asset digitisation & optimisation, Solar inverter services	7 States in India
Vayona Energy	12 GW in India 7 GW in Sri Lanka	Preventive, corrective, and predictive maintenance, revenue based total asset management services	8 States in India
Powercon Ventures	3.5+ GW	Active Operation & Maintenance (O&M), Remote Monitoring, Performance/ KPI Commitment, Asset Management, Yield Optimisation, Asset Longevity	10 States in India
Mitarsh Energy	32 GWp+ of Solar, Wind & BESS	O&M of wind, solar and BESS energy Assets, Preventive Maintenance, Corrective Maintenance, Performance Monitoring & Generation Reporting	12 States in India

Company	O&M Capacity	Key service offerings	Geographical presence
KP Energy OMS Limited	646+ MW	O&M for BOP portion of RE plants, 24X7 monitoring, preventive maintenance, and analytics-driven performance optimisation	4 States in India
Senvion India	1.5 GW	O&M of wind turbines, Value-Added Services	5 States in India

Source: Company websites, Credit Rating Documents, Investor Presentations, Crisil Intelligence

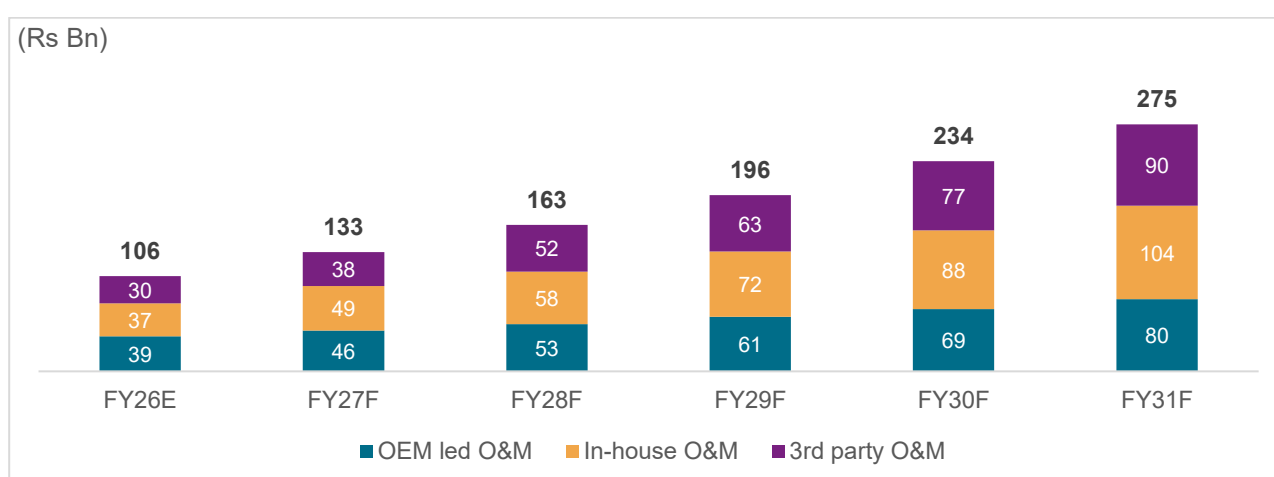
6.4.6 Independent O&M market size FY27–31

The Indian solar and wind O&M market is expected to expand significantly over the forecast period, supported by the continued growth of the installed asset base, increasing project ageing and the growing requirement for reliable plant availability and performance optimisation.

The O&M market comprises a combination of developer-managed, third-party and OEM-led services, with a mix varying by technology. In the solar segment, O&M is predominantly managed by asset owners through in-house teams, while third-party O&M service providers account for approximately 30–35% of the solar O&M market. In the wind segment, OEM-led O&M has a significantly higher prevalence, primarily due to the specialised nature of wind turbine maintenance, availability of proprietary spare parts and technology, and the role of wind turbine manufacturers in providing long-term service arrangements. The BoP and other project-level O&M activities may be undertaken by the asset owner through in-house teams or outsourced to independent third-party service providers.

The operating solar and wind asset base is estimated to surpass 500 GW by fiscal 2031, resulting in an expansion of the addressable O&M market from an estimated Rs 106 billion in fiscal 2026 to approximately Rs. 275 billion by fiscal 2031, representing a CAGR of ~21% over the forecast period. Independent third-party O&M service providers are expected to account for 30-35% of the overall RE O&M market.

Figure 56: India- RE O&M services demand forecast Fiscal 2026 to 2031



E: Estimated; F: Forecast

Source: Industry, Crisil Intelligence

7 Competitive assessment

7.1 About Integrum Energy

Integrum Energy Infrastructure Limited (IEIL or the Company) is an integrated RE solutions partner focused on C&I customers in India, engaged across the RE value chain specialising in execution of multi-technology RE projects to provide a lower delivered cost of energy to customers. Operations of the Company are classified by the stage of service it provides across the customers asset life: energy advisory and structuring; project development; turnkey multi-technology execution; and lifecycle asset services. Since its incorporation in 2021, the Company has progressed through distinct stages of activity and scale, moving from primarily being a solutions advisor to early renewable project execution to being an integrated energy partner with multi-modal offerings across the full asset lifecycle. IEIL's customer base is diversified across multiple industries, including iron and steel, oil and gas, manufacturing, hospital and pharmaceutical, textile, paper, kitchen appliances and renewable energy, and includes both private commercial and industrial customers as well as central public sector undertakings (CPSUs). The Company is configuration-agnostic, and it analyses the customers load profile, tariff position and state regulatory environment, and to recommend and execute a project configuration that minimises the customer's landed cost of energy rather than deploying any single technology on a standalone basis. The company offers solutions under three principal ownership structures, namely CapEx models, IPP models and EaaS models. The selection of the ownership structure for any given engagement is driven by the customer's capital allocation preferences, balance sheet capacity, risk appetite, desired level of operational involvement and the specific energy outcome being targeted.

IEIL's portfolio of executed and ongoing projects spans a wide range of capacities, from 0.08 MW rooftop installations to 70.00 MWp utility-scale hybridised projects, demonstrating the Company's ability to deliver solutions across the full spectrum of customer requirements.

Turnkey scope accounts for 86.5% of O&M capacity of the Company and its customers buy accountability for the outcome, not the supply of equipment. The Company selects states on a defined set of regulatory and economic parameters level of industrialisation / GDP, tariff level, charge stack, banking regime, resource quality and approval timelines and its footprint reflects that selection. According to the Company, its customers are not buying renewable energy - they are buying a lower delivered cost of electricity, and the configuration that achieves it differs by load, by state and by year.

Over FY24–FY26, the Company's business volume has grown materially, with revenue from operations increasing at a CAGR of 72.48% from Rs.2,332.22 million in FY24 to Rs.6,938.39 million in FY26 indicating a significant scale-up in the Company's operating base. During the same period, EBITDA rose at a CAGR of 110.38% from Rs.199.33 million in FY24 to Rs.882.26 million in FY26 outpacing revenue growth and indicating operating leverage, while PAT increased at a CAGR of 103.30% from Rs.152.80 million in FY24 to Rs.631.56 million in FY26 reflecting strong bottom-line growth alongside the Company's scale-up. Orders added by the Company expanded 8.3x over FY24–26, reflecting a significant build-up in contracted execution pipeline. Orders added by the Company grew at a 188.9% CAGR over FY24–26, indicating rapid expansion in the contracted pipeline. Repeat orders of the Company represented 37.21% of the contracted capacity during fiscal 2026, indicating meaningful contribution from existing customers. Hybrid and wind projects constituted 89.12% of the closing order book of the Company as on March 31, 2026, indicating a substantial shift in the contracted pipeline towards non-standalone-solar projects and demonstrating its focus on projects that require multi-technology integration and complex execution capabilities. Turnkey projects

represented 86.49% of the total O&M capacity of the Company as on March 31, 2026, indicating that the majority of the O&M portfolio is linked to projects executed by the Company. Generation under O&M/AMF increased at a CGAR of 43.10% over FY24–26, broadly supporting the expansion in the managed asset portfolio.

Integrum Energy is positioned as a lifecycle RE partner for C&I customers to reduce energy costs, improve RE utilisation, manage regulatory complexity and advance their long-term sustainability objectives. IEIL operates in the C&I renewable energy solutions market, with capabilities spanning energy advisory and structuring, project development, multi-technology EPC execution, O&M and lifecycle asset management and energy optimisation.

7.2 Competitive mapping of players

Competitive mapping covers the details of companies, their products and services within a given market to understand competitive intensity. Some of the key players include Cleanmax Enviro Energy, Oriana Power, KPI Green Energy, ACME Solar Holdings, Waaree RTL and S&W Renewable Energy. These players also have a sizeable quantum of capacity under construction/development. The RE EPC and IPPs face significant competition in the RE industry. The primary competitors include local and international companies which are developers and operators of RE projects. The RE IPPs also compete with traditional energy companies such as Utility companies i.e., generating power from conventional fossil fuels. The mapping also covers the key operational and financial parameters (consolidated figures) of the players.

Table 18: Capability assessment of key players

Capability	Integrum	CleanMax	Oriana Power	KPI Green Energy	Radiance	Sterling & Wilson Renewable Energy	SolarWorld	Serentica	Prozeal	Juniper Green	Suzlon	Inox Wind
Energy advisory/load analysis	✓	✓	✓	NA	NA		✓	✓	✓	✓		
Regulatory/O/A structuring	✓	✓	✓		✓		✓	✓	✓	✓		
Land/project development	✓	✓		✓		✓				✓	✓	✓
Solar EPC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
Wind EPC	✓	✓				✓		✓		✓	✓	✓
Wind solar	✓	✓	✓	✓		✓		✓	✓	✓		
BESS	✓	✓	✓			✓	✓	✓	✓	✓		
Captive/group captive	✓	✓	✓	✓	✓		✓	✓	✓	✓		✓
IPP/group captive	✓	✓	✓	✓	✓		✓	✓	✓	✓		
IPP/OPEX	✓	✓	✓	✓	✓		✓	✓	✓	✓		

Capability	Integrum	CleanMax	Oriana Power	KPI Green Energy	Radiance	Sterling & Wilson Renewable Energy	SolarWorld	Serentica	Prozeal	Juniper Green	Suzlon	Inox Wind
O&M	✓	✓		✓	✓	✓	✓	✓	✓		✓	✓
Asset Management	✓	✓	✓	✓	✓	✓	✓	✓	✓	NA	✓	✓
Digital optimisation	✓	✓	✓	✓								✓
EaaS	✓	✓	NA	NA	✓		NA	NA		NA		
Distributed/VNM	future/emerging	future	future	NA	future	NA	NA	NA	NA	NA	NA	NA

NA – Not Available as not available or reported.

Source: Industry, Company websites, Investor Presentations submitted to SEBI, Crisil Intelligence

Table 19: Operational and under construction/development RE capacity of key players (As of June 2026)

Particulars	Integrum Energy*	KPI Green Energy	Oriana Power	Waaree RTL	Sterling & Wilson Renewable Energy	Inox Wind
Years since incorporation	5	18	13	27	9	17
States	Karnataka, Maharashtra, Gujarat	Gujarat	Uttar Pradesh, Rajasthan and others	Pan-India	Pan-India	Pan-India
Presence across segments (IPP/C&I/Rooftop/EPC/O&M)	C&I, IPP, O&M, EPC	C&I, IPP, Rooftop, EPC, O&M	C&I, IPP, EPC	EPC, O&M	EPC, O&M	EPC, O&M
Operational Capacity (MW)	212.48	1,870.00	575.0+	5,951.60	18,300.00	3,000.00+
Under Construction capacity (MW)	241.00	5,070.00	2,000.0+	2,425.00	10,400.00	NA
Total capacity (MW)	453.48	6,940.00	2,575.0+	8,376.60	28,700.00	NA
O&M Capacity (MW)	93.03	NA	NA	1,154.00	13,500.00	13,400.00

*As of August 2026.

Note: The capacity figures for players such as Waaree RTL, Sterling and Wilson RE and INOX Wind are relatively high as they include significant EPC capacity and therefore may not be comparable with IPP owned operational capacity' NA – Not Available as not available or reported.

Source: Industry, Company websites, Investor Presentations submitted to SEBI, Crisil Intelligence

Table 20: Key operational and financial parameters of selected listed players

Metric	Units	Integrum ¹			Cleanmax			KPI Green		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Operational parameters										
Commissioned Projects	MWp/MW	90.51	34.70	40.50	3,643.94	2,177.99	1,755.21	1,620.00	950.00	445.00
No. of Completed Projects (count)	Nos	7.00	7.00	6.00	NA	NA	NA	NA	NA	NA
Opening Order Book	MWp/MW	88.88	24.90	32.20	NA	NA	NA	NA	NA	NA
Orders Added	MWp/MW	277.10	98.68	33.20	NA	NA	NA	NA	NA	NA
Orders Executed	MWp/MW	90.51	34.70	40.50	1,471.37	422.78	715.07	670.00	505.00	445.00
Closing Order Book	MWp/MW	275.47	88.88	24.90	2,812.05	2,769.66	435.80	4,640.00	2,960.00	1,234.00
Repeat Orders	MWp/MW	103.10	18.18	13.00	NA	NA	NA	NA	NA	NA
Repeat Orders	%	37.21%	18.42%	39.16%	74%	77%	82%	NA	NA	NA
O&M CUF/PLF — Solar	%	18.42%	14.35%	NIL	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Wind	%	26.92%	25.70%	24.29%	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Hybrid	%	30.17%	30.38%	31.31%	NA	NA	NA	NA	NA	NA
Units Generated under O&M/AMF	Mn kWh	294.49	229.99	143.81	NA	NA	NA	410.00	143.50	NA
Grid Availability	%	95.74%	NA	NA	99.24%	99.10%	99.26%	NA	NA	NA
Machine Availability — wind plants	%	96.93%	91.51%	84.56%	35.10%	31.60%	34.52%	NA	NA	NA
Plant Availability	%	95.86%	NA	NA	98.19%	98.17%	98.19%	NA	NA	NA
O&M Capacity under Management — BOP	MW	14.30	14.30	12.20	NA	NA	NA	NA	NA	NA
O&M Capacity under Management — Turnkey	MW	91.51	62.99	46.28	NA	NA	NA	NA	NA	NA

¹ The capacity presented under “Commissioned Projects” represents projects commissioned during the respective fiscal year, however for other players it is the cumulative commissioned capacity as of respective fiscal year.

Metric	Units	Integrum ¹			Cleanmax			KPI Green		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
O&M Capacity under Management — Total	MW	105.81	77.29	58.48	NA	NA	NA	646.00	546.00	523.10
AMF Capacity under Management	MW	162.99	134.39	90.48	NA	NA	NA	NA	NA	NA
Closing Order Book — Hybridisation	MW	101.40	76.51	12.40	NA	NA	NA	NA	NA	NA
Closing Order Book — Standalone: Solar	MW	29.97	10.27	12.50	1,630.35	1,957.26	399.50	NA	NA	NA
Closing Order Book — Standalone: Wind	MW	144.10	2.10	0.00	1,181.70	812.40	36.30	NA	NA	NA
Financial parameters										
Revenue from operations	INR Mn	6,938.39	1,176.66	2,332.22	19,128.73	14,957.01	13,898.37	26,959.10	17,354.54	10,239.00
Growth in revenue from operations	%	489.67%	(49.55%)	284.25%	27.89%	7.62%	49.51%	55.34%	69.49%	59.04%
EBITDA	INR Mn	882.26	171.56	199.33	11,384.74	9,079.83	6,953.35	9,604.92	5,637.68	3,368.43
EBITDA Margin	%	12.72%	14.58%	8.55%	59.52%	60.71%	50.03%	35.63%	32.49%	32.90%
Profit Before Tax	INR Mn	872.41	175.50	199.04	1,349.81	597.47	48.91	6,909.10	4,435.10	2,173.23
Profit After Tax	INR Mn	631.56	126.28	152.80	855.77	194.29	(376.43)	5,092.38	3,252.80	1,616.57
PAT Margin	%	9.10%	10.73%	6.55%	4.47%	1.30%	(2.71%)	18.89%	18.74%	15.79%
Net Debt	INR Mn	(387.36)	(333.84)	(96.45)	101,228.07	67,843.09	51,322.06	39,727.99	6,981.92	6,618.45
Net Debt : Equity ratio	Times	(0.26)	(0.59)	(0.34)	2.18	2.65	2.80	1.36	0.29	0.79
Cost of borrowings	%	3.29%	14.40%	6.66%	7.24%	9.48%	10.34%	6.22%	7.37%	11.93%
Debt service coverage ratio	Times	22.03	8.59	NA	1.54	1.42	1.44	5.46	7.82	4.17
Interest Coverage Ratio	Times	135.49	109.03	1,542.00	0.97	0.92	0.94	4.54	6.33	3.44
ROCE	%	56.23%	32.29%	72.46%	5.14%	6.46%	6.59%	10.78%	13.39%	17.34%
ROE	%	42.11%	22.42%	54.00%	1.85%	0.76%	(2.05%)	17.47%	13.42%	19.34%
Working Capital	INR Mn	1,017.92	300.28	246.86	(29,260.55)	(9,099.85)	(3,318.42)	9,893.74	10,091.45	3,846.52
Receivable days	Days	43.49	142.62	39.62	44.45	53.66	55.30	89.26	105.71	102.29
Inventory days	Days	29.64	134.01	15.95	36.14	41.24	47.36	244.84	160.00	170.75
Payable days	Days	91.51	196.53	42.84	1,029.82	744.02	774.28	114.36	167.34	220.44

Metric	Units	Integrum ¹			Cleanmax			KPI Green		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Cash conversion cycle	Days	(18.38)	80.10	12.74	(949.23)	(649.12)	(671.62)	219.74	98.37	52.60
Current ratio	Times	1.49	1.40	1.68	0.66	0.76	0.78	1.64	2.42	1.49
Quick Ratio	Times	1.35	0.92	1.46	0.66	0.74	0.75	0.94	1.86	1.21
Earning for Debt service	INR Mn	648.40	133.63	156.38	12,514.11	9,823.06	6,882.73	8,244.32	4,652.42	2,880.93
Debt service	INR Mn	29.43	15.56	-	13,508.95	4,794.17	14,501.98	NA	NA	NA

Metric	Units	Juniper Green			Waaree RTL			Solarworld		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Operational parameters										
Commissioned Projects	MWp/MW	1,233.14	854.30	660.20	5,060.00	2,320.00	704.00	348.00	336.17	170.00
No. of Completed Projects (count)	Nos	NA	NA	NA	NA	NA	NA	NA	NA	NA
Opening Order Book	MWp/MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Orders Added	MWp/MW	NA	NA	NA	NA	NA	NA	853.00	376.00	582.00
Orders Executed	MWp/MW	NA	NA	NA	2,727.00	1,524.00	704.00	12.00	24.00	170.00
Closing Order Book	MWp/MW	2,341.16	2,050.00	399.10	2,832.00	3,263.00	2,365.00	1,809.00	NA	NA
Repeat Orders	MWp/MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Repeat Orders	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Solar	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Wind	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Hybrid	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
Units Generated under O&M/AMF	Mn kWh	NA	NA	NA	NA	NA	NA	NA	NA	NA
Grid Availability	%	99.40%	99.31%	99.12%	NA	NA	NA	NA	NA	NA

Metric	Units	Juniper Green			Waaree RTL			Solarworld		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Machine Availability — wind plants	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
Plant Availability	%	99.16%	98.81%	98.57%	NA	NA	NA	NA	NA	NA
O&M Capacity under Management — BOP	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M Capacity under Management — Turnkey	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M Capacity under Management — Total	MW	NA	NA	NA	1,180.00	695.00	587.00	323.00	299.00	119.00
AMF Capacity under Management	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book — Hybridisation	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book — Standalone: Solar	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book — Standalone: Wind	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Financial parameters										
Revenue from operations	INR Mn	7,189.34	5,086.78	3,915.50	33,314.22	15,977.48	8,765.03	13,761.56	5,447.65	5,010.16
Growth in revenue from operations	%	41.33%	29.91%	18.18%	108.51%	82.29%	149.74%	152.61%	8.73%	115.58%
EBITDA	INR Mn	6,061.88	4,245.83	3,379.46	6,410.97	3,068.84	2,071.82	1,470.35	1,067.47	710.93
EBITDA Margin	%	84.32%	83.47%	86.31%	19.24%	19.21%	23.64%	10.68%	19.60%	14.19%
Profit Before Tax	INR Mn	551.94	548.98	574.95	6,396.10	3,004.76	1,984.54	1,610.51	1,066.08	683.71

Metric	Units	Juniper Green			Waaree RTL			Solarworld		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Profit After Tax	INR Mn	404.6	364.8	400.6	4,786.36	2,289.25	1,734.20	1,204.74	770.48	516.9
PAT Margin	%	5.63%	7.17%	10.23%	14.37%	14.33%	19.79%	8.75%	14.14%	10.32%
Net Debt	INR Mn	95,630.44	28,500.19	18,141.27	(2,769.22)	(161.81)	(804.08)	(3,165.13)	(125.66)	206.00
Net Debt : Equity Ratio	Times	2.79	0.85	1.05	(0.30)	(0.04)	(0.33)	(0.37)	(0.04)	0.28
Cost of borrowings	%	4.15%	6.35%	7.86%	8.05%	15.99%	14.21%	9.21%	6.11%	9.75%
Debt service coverage ratio	Times	1.59	1.64	1.79	146.02	56.58	36.70	8.63	19.91	11.59
Interest Coverage Ratio	Times	0.92	0.98	1.13	47.60	20.25	29.72	6.87	17.09	10.42
ROCE	%	2.79%	3.61%	9.31%	62.54%	61.94%	66.72%	16.56%	26.95%	56.22%
ROE	%	1.18%	1.09%	2.31%	51.23%	50.27%	70.34%	14.21%	24.93%	70.24%
Working Capital	INR Mn	(13,894.90)	14,845.63	3,880.65	3,889.46	1,717.73	1,215.15	6,598.26	2,001.67	579.90
Receivable days	Days	46.79	48.42	29.17	91.47	99.41	90.34	65.24	58.44	18.48
Inventory days	Days	NA	NA	NA	8.63	5.25	18.59	17.66	2.77	1.08
Payable days	Days	NA	NA	NA	70.47	78.35	105.31	82.89	48.27	11.25
Cash conversion cycle	Days	NA	NA	NA	29.63	26.30	3.62	0.01	12.94	8.30
Current ratio	Times	0.97	2.79	1.92	1.35	1.32	1.31	1.88	1.94	2.20
Quick Ratio	Times	NA	NA	NA	1.26	1.31	1.24	1.74	1.94	2.17
Earning for Debt service	INR Mn	6,774.54	4,672.65	3,534.12	5,005.48	2,501.35	1,863.02	1,463.68	835.07	588.99
Debt service	INR Mn	5,195.30	4,022.04	2,664.69	NA	279.46	NA	NA	NA	NA

Metric	Units	Sterling and Wilson RE			INOX Wind ²			Suzlon		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Operational parameters										
Commissioned Projects	MWp/MW	17,200.00	NA	NA	3,400.00	3000.00+	NA	NA	NA	NA
No. of Completed Projects (count)	Nos	NA	NA	NA	NA	NA	NA	NA	NA	NA
Opening Order Book	MWp/MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Orders Added	MWp/MW	10,100.00	NA	NA	NA	NA	NA	NA	NA	NA
Orders Executed	MWp/MW	4,500.00	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book	MWp/MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Repeat Orders	MWp/MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Repeat Orders	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Solar	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Wind	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M CUF/PLF — Hybrid	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
Units Generated under O&M/AMF	Mn kWh	NA	NA	NA	NA	NA	NA	NA	NA	NA
Grid Availability	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
Machine Availability — wind plants	%	NA	NA	NA	96.50%	96.30%	96.00%	NA	NA	NA
Plant Availability	%	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M Capacity under	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA

² Inox Wind undertakes EPC activities through its subsidiary, Inox Renewables and O&M activities through Inox Green.

Metric	Units	Sterling and Wilson RE			INOX Wind ²			Suzlon		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Management — BOP										
O&M Capacity under Management — Turnkey	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
O&M Capacity under Management — Total	MW	13,500.00	8,700.00	7,670.00	13,000.00+	5,076.00	3,350.00	NA	NA	NA
AMF Capacity under Management	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book — Hybridisation	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book — Standalone: Solar	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Closing Order Book — Standalone: Wind	MW	NA	NA	NA	NA	NA	NA	NA	NA	NA
Financial parameters										
Revenue from operations	INR Mn	75,480.50	63,018.60	30,353.70	43,975.18	35,571.48	17,463.04	166,791.10	108,513.20	64,968.40
Growth in revenue from operations	%	19.77%	107.61%	50.64%	23.62%	103.70%	138.23%	53.71%	67.02%	9.25%
EBITDA	INR Mn	(3,088.50)	2,467.30	(225.70)	8,914.21	7,437.01	2,482.14	30,396.50	18,188.10	9,427.40
EBITDA Margin	%	(4.09%)	3.92%	(0.74%)	19.51%	20.09%	13.73%	18.22%	16.76%	14.51%
Profit Before Tax	INR Mn	(2,702.40)	1,625.40	(1,723.20)	6,591.21	5,368.10	(426.94)	24,217.20	14,466.30	6,594.90
Profit After Tax	INR Mn	(2,957.90)	855.50	(1,723.20)	4,491.0	4,376.3	(481.5)	31,633.9	20,716.3	6,603.5
PAT Margin	%	(3.92%)	1.36%	(5.68%)	10.21%	12.30%	(2.76%)	18.97%	19.09%	10.16%
Net Debt	INR Mn	6,298.90	1,905.30	1,374.70	9,397.46	12,272.58	20,127.51	(9,816.90)	(8,295.00)	(3,168.90)

Metric	Units	Sterling and Wilson RE			INOX Wind ²			Suzlon		
		FY26	FY25	FY24	FY26	FY25	FY24	FY26	FY25	FY24
		(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
Net Debt : Equity Ratio	Times	0.97	0.19	0.14	0.15	0.28	0.72	(0.10)	(0.14)	(0.08)
Cost of borrowings	%	13.85%	13.65%	16.31%	7.79%	8.88%	8.23%	81.59%	53.74%	10.49%
Debt service coverage ratio	Times	(2.16)	2.62	(0.11)	7.65	4.78	1.35	13.61	17.21	8.92
Interest Coverage Ratio	Times	(2.08)	2.12	(0.18)	3.44	3.32	0.56	5.89	6.12	4.58
ROCE	%	(6.28%)	14.16%	3.09%	9.56%	10.99%	3.98%	34.61%	29.62%	20.46%
ROE	%	(45.52%)	8.50%	(17.78%)	7.04%	9.92%	(1.71%)	33.43%	33.93%	16.84%
Working Capital	INR Mn	3,454.50	5,646.70	7,797.30	36,925.07	14,835.50	1,210.49	44,661.20	23,930.90	20,209.20
Receivable days	Days	73.82	60.40	97.52	282.14	189.10	205.29	110.90	95.80	84.27
Inventory days	Days	0.09	0.05	0.04	245.87	218.26	417.57	123.63	141.26	187.08
Payable days	Days	204.44	170.37	202.81	184.65	148.41	214.83	128.07	120.94	122.18
Cash conversion cycle	Days	(130.53)	(109.93)	(105.25)	343.37	258.96	408.04	106.46	116.12	149.17
Current ratio	Times	1.16	1.28	1.35	2.01	1.55	1.05	1.65	1.56	1.76
Quick Ratio	Times	1.16	1.28	1.35	1.54	1.15	0.62	1.10	1.02	1.00
Earning for Debt service	INR Mn	(1,307.30)	2,093.40	628.50	8,532.41	7,889.18	3,044.78	39,439.90	25,856.20	10,142.70
Debt service	INR Mn	5,125.10	3,371.80	26,067.00	NA	6,585.61	NA	2,148.90	1,425.40	19,335.80

Note: Financial ratios across companies may not be directly comparable due to differences in revenue mix, asset ownership, EPC exposure, IPP operations and accounting treatment; (A): Audited; NA – Not Available as not available or reported for the particular period.

Source: Company websites, Annual Reports, Investors Presentations as available on stock exchanges, RHP Filings, Crisil Intelligence

Formulae used:

- **Commissioned Projects (MWp/MW):** Aggregate capacity of projects commissioned during the period. MWp (DC) for solar, MW (AC) for wind; combined for hybrid.
- **No. of Completed Projects (count):** Distinct projects achieving commissioning and final customer handover during the period; phased projects counted separately.
- **Opening Order Book (MWp/MW):** Unexecuted value of secured contracts at period start (= prior period's closing order book).

- **Orders Added (MWp/MW):** New orders secured during the period. Private sector: on execution of binding agreement + advance; PSU/Government: on date of LoA.
- **Orders Executed (MWp/MW):** Capacity of projects commissioned and handed over during the period.
- **Closing Order Book (MWp/MW):** Opening order book + Orders added – Orders executed.
- **Repeat Orders (MW):** Capacity contracted from customers with at least one prior completed/contracted project (incl. group/associate companies).
- **Repeat Orders (%):** Repeat order capacity as a percentage of total capacity contracted in the period.
- **O&M CUF/PLF — Solar (%):** Actual generation vs. rated-capacity generation for solar plants under O&M scope.
- **O&M CUF/PLF — Wind (%):** Actual generation vs. rated-capacity generation for wind plants under O&M scope.
- **O&M CUF/PLF — Hybrid (%):** Actual generation of a hybrid plant (combined wind and solar) as a percentage of evacuation capacity.
- **Units Generated under O&M/AMF (mn kWh):** Electricity generated by plants under the O&M and asset-management scope of our Company.
- **Grid Availability (%):** Proportion of the period the grid/evacuation infrastructure was available, up to the GSS interconnection point.
- **Machine Availability (%) — wind plants:** Proportion of the period generating equipment was available, excluding grid/force-majeure/customer-caused downtime; computed as Grid Availability + Plant Availability, for wind plants only.
- **Plant Availability (%):** Proportion of the period the plant was capable of exporting power (equipment and grid simultaneously available).
- **O&M Capacity under Management — BOP (MW):** Capacity under active O&M contracts, Balance-of-Plant scope only.
- **O&M Capacity under Management — Turnkey (MW):** Capacity under active O&M contracts, full Turnkey scope.
- **O&M Capacity under Management — Total (MW):** Aggregate capacity under active O&M contracts as at period end (BOP + Turnkey).
- **AMF Capacity under Management (MW):** Aggregate capacity under active asset-management agreements as at period end.
- **Closing Order Book — Hybridisation (MW):** Order book for hybridisation projects (shared generation infrastructure across wind/solar as defined).
- **Closing Order Book — Standalone: Solar (MW):** Standalone solar order book component.
- **Closing Order Book — Standalone: Wind (MW):** Standalone wind order book component.
- **Revenue from Operations:** Revenue from sales, other operating revenues and government grants.
- **Growth in Revenue from Operations (%):** Growth in Revenue from Operations is calculated as a percentage of Revenue from Operations of the relevant year minus Revenue from Operations of the preceding year, divided by Revenue from Operations of the preceding year.
- **EBITDA:** Profit before finance cost, tax, depreciation and amortisation excluding other income and exceptional items
- **EBITDA Margin (%):** EBITDA divided by Revenue from Operations.
- **Profit before tax:** Profit after operating and non-operating income and expenses but before income tax
- **Profit after tax:** Profit after tax
- **PAT Margin (%):** PAT divided by Revenue from Operations.
- **Net Debt:** Net Debt is calculated by subtracting a company's total cash and cash equivalents, bank balances from its total borrowing as at the end of the financial year/period.
- **Net Debt : Equity:** Net Debt divided by Total Equity attributable to owners of the Company as at the end of the financial year/period.

- **Cost of Borrowings (%):** Total interest expense divided by the average of sum borrowings expressed as a percentage.
- **Debt Service Coverage Ratio:** EBITDA divided by total of Interest cost and principal and lease payments.
- **Interest Coverage Ratio:** EBIT divided by total finance cost.
- **Return on Capital Employed ROCE (%):** ROCE is Profit before Interest and Taxes as a % of Capital Employed. Capital Employed is the sum of Total Equity minus deferred tax assets plus Total Debt plus deferred tax liabilities. Profit before Interest and Taxes is calculated as Profit before Tax plus finance costs
- **Return on Equity (%) ROE (%):** PAT divided by Total closing Equity attributable to owners of the company, multiplied by 100.
- **Working Capital:** Current assets minus Cash and cash equivalents minus Investments minus Current Liabilities
- **Receivable Days:** $(\text{Average Trade Receivables} \div \text{Revenue from Operations}) \times 365$.
- **Inventory Days:** $(\text{Average Inventory} \div \text{Cost of Materials Consumed} + \text{Direct Cost}) \times 365$.
- **Payable Days:** $(\text{Average Trade Payables} \div \text{Cost of Materials Consumed and Direct Cost}) \times 365$.
- **Cash Conversion Cycle:** Receivable Days plus Inventory Days minus Payable Days.
- **Current Ratio:** Current ratio is calculated by dividing the current assets by current liabilities
- **Quick Ratio:** $(\text{Current Assets} - \text{Inventories})$ divided by Current liabilities.
- **Earnings for Debt Service:** Net Profit after tax plus non-cash operating expenses (depreciation, other amortisation) plus Interest.
- **Debt Service:** Interest & lease payments plus principal repayments.

8 Risk and challenges impacting RE Sector

RE solution partners for C&I in India must manage a complex matrix of regulatory, operational, and financial risks. Success depends on robust risk assessment, diversified portfolios, strong due diligence, and agile execution capabilities.

1. Regulatory and Policy

Any adverse shift in government policies, including reductions in incentives or changes in energy regulations, can significantly impact revenue and profitability of integrated solution partner. Frequent changes in central and state RE policies, open access regulations, and tariff orders can impact project viability and long-term planning.

Lengthy timelines in obtaining regulatory approvals, especially for open access and net metering, can delay project execution. Grid operators may curtail RE generation, especially during periods of oversupply or grid constraints, affecting returns.

2. Operational

Variability in RE generation (solar/wind) can lead to grid instability, requiring advanced forecasting, scheduling, and balancing solutions. Dependence on weather leads to unpredictable generation, impacting power supply reliability for C&I customers. Ensuring consistent performance and uptime of RE assets across multiple locations is operationally intensive.

3. Financial

Delays in payments from C&I customers can lead to cash flow mismatches. High perceived risks (regulatory, offtake, counterparty) can increase the cost of capital and limit access to long-term financing.

4. Execution and Project Management

Managing multiple vendors, EPC contractors, and supply chains increases execution risk. Delays in project commissioning due to regulatory, land, or equipment issues can affect returns and relationships. Ensuring consistent build quality across dispersed sites is also challenging. During monsoon/rainy season, integrated solution partners may not be able to execute projects due to site accessibility, waterlogging etc. which can affect their revenue during these periods.

5. Geographic Concentration

Over-concentration in specific regions (e.g., high solar irradiation or wind-rich states) may expose providers to local regulatory or climatic risks, adequate labour, geo-political risks, disasters etc. State-level policy changes can disproportionately impact projects concentrated in certain geographies.

6. Land Acquisition

Land acquisition for renewable energy projects in India is frequently a protracted and challenging process. Delays commonly stem from fragmented land records, the need for multiple bureaucratic approvals, and resistance from local communities. Large-scale renewable energy parks, in particular, often encounter opposition from farmers and residents concerned about displacement, loss of agricultural land, or insufficient compensation. The absence of a centralised land policy means that each state enforces its own regulations, further complicating the process and increasing costs especially with the need to establish dedicated regional land acquisition teams. Rising land prices and legal disputes can inflate project costs and delay execution.

7. Connectivity Arrangement

Obtaining timely grid connectivity and transmission/wheeling approvals is a major bottleneck, especially for interstate open access projects. Insufficient or delayed grid infrastructure can limit the ability to evacuate generated power. The sluggish expansion of transmission infrastructure poses a significant obstacle to renewable energy development in India. Many RE rich remote areas suffer from insufficient grid connectivity, resulting in bottlenecks that hinder capacity growth. Establishing new transmission lines usually takes more than three years and is often delayed due to right-of-way conflicts, difficult terrain, and challenges in obtaining environmental clearances.

8. Customer Credit

C&I offtakers may face financial stress, leading to delayed or defaulted payments. Enforcing long-term PPAs or lease agreements can be challenging in case of disputes.

9. Working Capital Management

Payment delays from consumers can strain working capital and necessitate higher short-term borrowing. Managing spare parts and consumables across multiple O&M sites adds to working capital requirements.

10. Operations & Maintenance (O&M)

Ensuring timely and efficient O&M for RE assets, especially remotely located can be logistically complex. Rising labor and input costs can impact O&M margins. Meeting guaranteed generation or uptime levels can be challenging, especially in adverse weather or grid conditions.

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